SINGLE CRYSTAL NMR OF HIGH TEMPERATURE SUPERCONDUCTORS

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The author reports Cu NMR studies of the high temperature superconductor $YBa_2Cu_3O_{7-\delta}$ (T_c=90K) using a 1mg single crystal.

Three types of measurements have been made for both the "plane" and "chain" Cu crystallographic sites and with the static magnetic field placed succesively along each of the three crystal axes a, b, and c: NMR lineshapes, spin-lattice relaxation, and transverse relaxation.

From these measurements four NMR tensors are deduced: the magnetic shift tensor, the electric field gradient tensor, the spin-lattice relaxation rate tensor, and the transverse relaxation rate tensor.

As a preliminary matter the author uses the symmetries of the magnetic shift and electric field gradient tensors to assign the two families of NMR lines to their respective crystallographic sites.

From an analysis of the electric field gradient tensors it is shown that both the plane and chain Cu have valences Cu^{2+}, giving a single hole in the Cu atomic d-shell.

The author then analyzes the spin-lattice relaxation rate and magnetic shift of the plane and chain Cu; included are both analyses in which only normal state data are considered, and in which superconducting state measurements of magnetic shifts are considered as well. These analyses show that the plane Cu atom behaves very nearly as a local moment. The chain Cu is probably also in the local moment limit; however, its NMR properties are largely determined by its interaction with oxygen holes in the chain.

From transverse relaxation measurements a strong spin-spin coupling of plane Cu nuclei is observed; this coupling results from an indirect, electron-mediated mechanism involving exchange coupling between neighboring plane Cu local moments. The exchange coupling energy is found to be J_{eff}=1700cm^{-1}.

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In my case an extended "Acknowledgements" section is in order, because I have had the privilege to work with many great people.

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Throughout my years as a graduate student I have worked very closely with Dr. Dale Durand, both in his capacity as a graduate student and as a post-doctoral investigator. He has shown an enormous amount of concern not only for the projects in which he was directly involved, but also for the concerns and the education of every student in our lab. When I first joined the Slichter laboratory group Dale's patient tutelage helped me to quickly become an effective member. As my thesis work has progressed we have spent many hours working together and discussing various problems. He has given freely his time and ideas, many of which were key ingredients in important experiments in this thesis.

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In recent months Sean Barrett has made valuable new measurements on high temperature superconductors. In this thesis I have in

places used some of the results of Sean's measurements in analysis of my data. I thank him for allowing me to do so prior to publication.

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Chapter 1: Introduction

The discovery by Bednorz and Muller of high-temperature copper-oxide superconductors in 1986 [Bednorz et al, 1986] spawned one of the most exciting problems in solid state physics. Bednorz and Muller found that the compound $La_{2-x}Ba_xCuO_4$ displayed superconductivity at the world-record transition temperature of approximately 30K. Bednorz and Muller's discovery was soon confirmed, and work was begun worldwide to vary the chemical composition with the goal of finding still higher transition temperatures. In 1987 Wu et. al [1987] found that a compound later determined to have the chemical formula $YBa_2Cu_3O_7$ had a transition temperature of approximately 90K, thus exceeding the boiling point of liquid nitrogen (77K), and suggesting the possibility of myriad applications of superconductivity which could become practical if liquid nitrogen, an inexpensive and readily available cryogenic fluid, could be used to maintain the material's superconductivity. Bednorz and Muller received the Nobel Prize in 1987 for their discovery.

The study of high-temperature superconductors is naturally divided into two broad categories. First, an enormous effort has been made to understand and to manipulate the materials' properties in order to make them practical for application. In this thesis we focus on the second broad category of study, which is of the microscopic physics of these materials. The questions faced in this category include the following: In what way, if any, are the normal state and superconducting state properties of these materials different from conventional (low transition temperature) superconductors? Is there a "pairing" of electrons as in the Bardeen Cooper Schrieffer theory, and if so, is this pairing the conventional mechanism involving the electron-phonon interaction, or, is it some new exotic mechanism, perhaps involving magnetism?

To understand superconductivity in these materials will require an understanding of the special properties of the normal state. It is the purpose of this thesis to elucidate these properties. Specifically, we have studied the material $YBa_2Cu_3O_7$, and obtained valuable information on the copper atoms and the charge carriers in this material.

We have used the technique of copper nuclear magnetic resonance (NMR). Our work is unique in that our measurements are made on a tiny (1mg) single-crystal of $YBa_2Cu_3O_7$. By using a single crystal, and orienting the static magnetic field along principal axes we have precisely measured crucial anisotropic properties of the material.

The material has two crystallographic copper atom sites, the so-called "plane" and "chain" copper sites. As a preliminary matter we have resolved conclusively a controversial issue-the assignment of the two copper nuclear magnetic resonances to their respective crystallographic sites.

We have determined the valence of each of the two Cu sites to be Cu^{2+}, giving a single hole in the Cu 3d shell.

We have determined that in the normal state this hole behaves very much as a localized, rather than as an itinerant hole, and has a permanent magnetic moment, much like the Cu^{2+} ions in copper sulfate, familiar to workers in electron spin resonance. We have, in fact, demonstrated the existence of exchange coupling between plane Cu atoms. The localized behaviour of this d-shell hole excludes the possibility that a simple band theory picture will describe the physics of the system. Conversely, a great body of nuclear magnetic resonance data indicate that Cu d-shell holes must "pair" in the superconducting state--behaviour much more commonly associated with itinerant electrons. We believe that the resolution of this paradox will be an important step in the process of understanding high temperature superconductivity.

In addition to the localized holes in the plane and chain Cu 3d shells one also expects from simple valence arguments that one extra hole per unit cell will be distributed in some way on the oxygen atoms. Some of our data suggests that in the normal state these holes, in sharp contrast to the holes in the Cu 3d shells, are well described by a simple band picture with only modest electron-electron interaction effects.

Chapter 2: High Temperature Superconductivity

2.1 Introduction

2.11 Superconductivity

2.111 Phenomenology

Superconductivity was discovered in 1911 by Kamerlingh Onnes [1911]. Onnes' development of a technique to liquify helium and thus refrigerate to a few degrees Kelvin was crucial to the discovery, since for most simple materials, such as lead, tin, and mercury superconductivity appears only at temperatures less than a few degrees Kelvin.

Onnes observed that at low temperatures the electrical resistance of these materials vanishes. Perfect conductivity is the first experimental manifestation of superconductivity.

The second manifestation was discovered in 1933 by Meissner and Ochsenfeld [1933], who found that superconductors exhibit perfect diamagnetism. Of course, if one initially has a superconductor with zero applied field and then turns the field on, then perfect conductivity alone guarantees that magnetic flux will not enter the sample. The phenomenon observed by Meissner and Oschenfeld is more subtle. They began their experiment with a sample in an applied magnetic field, and with the sample temperature above the superconducting transition temperature T_c, and thus in the normal state. They then cooled the sample into the superconducting state and found that currents spontaneously began to flow in the sample to expel the already-present magnetic flux. Perfect diamagnetism is thus the second experimental manifestation of superconductivity.

There are two types of superconductors, known as Type I and Type II. Type I superconductors behave as described above. They superconduct, exhibiting both perfect conductivity and perfect diamagnetism, when at temperatures less than T_c and at magnetic fields less than H_c. Type II superconductors are characterized by a transition temperature T_c and two critical magnetic fields H_{c1} and H_{c2}, where $H_{c1} < H_{c2}$. For temperatures less than T_c and magnetic fields less than H_{c1} they exhibit both perfect conductivity and perfect diamagnetism. For magnetic fields larger than H_{c2} they revert to the normal state. For magnetic fields between H_{c1} and H_{c2} Type II superconductors exhibit perfect conductivity, but

incomplete flux exclusion, or diamagnetism. All known high T_c superconductors are Type II. For a more complete review of superconductivity see Tinkham [Chapter 1, 1975].

2.112 Phenomenological Theories

An early (1935) phenomenological description of superconductivity was given by F. London and H. London [1935], to describe the newly discovered Meissner effect. The basic postulate of the London theory is that the time derivative of current is proportional to electric field. The postulate is a reasonable one, since unscattered charge carriers should accelerate with the force applied, which is proportional to the electric field. Combining their postulate with Maxwell's Equations leads one to predict both perfect conductivity and perfect diamagnetism--applied magnetic fields fall to zero in a superconductor with the characteristic London penetration depth λ, which is contained in the proportionality constant between current rate of change and electric field.

A more complete and very useful phenomenological theory was given by Ginzburg and Landau [1950], the so-called "Landau-Ginzburg" theory. Their premise is that the superconducting charge carriers can be described by a macroscopic complex wave function Ψ. The number of carriers per unit volume is given by $|\Psi|^2$. One writes the energy of a system in terms of $|\Psi|^2$ and in terms of electromagnetic fields present (see Tinkham, Chapter 4 [1975]). Landau-Ginzburg theory has two length scales. First, the so-called "coherence length" ξ is the length scale over which Ψ may vary substantially. Second is the penetration depth λ, the characteristic length for the spatial decay of applied magnetic fields. Landau-Ginzburg theory has been derived from the microscopic theory of Bardeen-Cooper-Schrieffer in a limiting case [Gor'kov, 1959], and has been succesful in predicting numerous experimental results, including the Josephson effect, on which most of the technology of superconductor devices is based.

In Landau-Ginzburg theory one may classify superconductors as Type I or Type II according to the ratio $\kappa=\lambda/\xi$. A value of κ greater than $1/\sqrt{2}$ indicates a Type II superconductor. High T_c superconductors are in the extreme Type II limit, with λ and ξ on the order of 1000 and 10 Angstroms, respectively, and κ on the order of 100.

2.12 Bardeen-Cooper-Schrieffer Theory

In 1957 Bardeen, Cooper, and Schrieffer [1957] presented a microscopic theory of superconductivity, for which they received the Nobel Prize in 1972 (see Tinkham Chapter 2 for review [Tinkham, 1975]). The theory has seen extensive experimental verification and is widely accepted as accounting for the important physics of superconductivity.

A key concept in the BCS theory is the Cooper pair [Cooper, 1956]. Cooper showed that electrons at the edge of a Fermi sea which experience an attractive interaction may form a bound pair, even if the attractive interaction is vanishingly small. The existence of the Fermi sea with a sharp cut-off in electron state occupation number at the Fermi energy is crucial to the existence of the bound state. One can think of electrons at the edge of the Fermi sea forming many bound pairs. The pairing continues until the Fermi sea is distorted to make pairing no longer favorable. The BCS theory does not use this over-simplified approach, but instead treats a many-electron wavefunction with an attractive interaction between electrons.

The BCS theory not only explains superconductivity in terms of pairing and an attractive electron-electron interaction but also specifies the origin of the attractive, "pairing" interaction. An important clue as to the source of the pairing lies in measurements of the so-called "isotope effect." In an isotope effect experiment one prepares samples of two different isotopes of the same element. In many instances [Maxwell, 1950; Reynolds et. al., 1950] it is found that T_c varies as $M^{-1/2}$ where M is the mass of the isotope. The experiment strongly suggests that the frequencies of the lattice vibrations (phonons) are important. BCS theory states that the pairing interaction between electrons is mediated by lattice vibrations, or phonons. In a simple picture, one can imagine electron A passing through the lattice. As it moves, its negative charge may polarize the lattice, leaving behind a trail of positive charge. Electron B is then attracted to this path.

Thus, there are two major principles of BCS theory. First, superconductivity results from an electron-electron attractive interaction and the resulting formation of Cooper pairs. Second, this pairing interaction is mediated by the electron-phonon interaction. It is widely believed that pairing is involved in high temperature superconductivity, and that the BCS

formalism for the treatment of the many electron wavefunction with pairing is correct. It seems very possible, however, that the actual pairing mechanism in high T_c superconductors is not the conventional electron-phonon interaction. There are two reasons for this. The first is simply that the transition temperature is so high. Theoretical calculations [McMillan, 1968; Cohen, et. al., 1982] place an upper limit to T_c for phonon pairing at approximately 30K. That is consistent with the $La_{2-x}Ba_xCuO_4$ compound, but not with $YBa_2Cu_3O_{7-\delta}$ (T_c = 90K). Second, the T_c=90K material displays little or no isotope effect [Batlogg, et al [a], 1987].

2.13 Discovery of High T_c

The field of high temperature superconductivity began with the discovery by Bednorz and Muller [1986] that the compound $La_{2-x}Ba_xCuO_4$ (x=.1-.2) superconducts up to approximately 30K. Investigators rushed to make variations in the chemical formula, and Wu et. al. [1987] found a T_c of 90K in a compound later determined [Cava et. al. 1987] to have the chemical formula $YBa_2Cu_3O_{7-\delta}$. Subsequent progress has been made at a somewhat slower pace, with the discovery of the T_c=110K compound Bi-Sr-Ca-Cu-O [Maeda, 1988] and the T_c=118K compound Tl-Ba-Ca-Cu-O [Sheng, 1988] (values given for T_c correspond to the onset of diamagnetic shielding).

High T_c superconductors to date have an important feature in common; they all contain Cu and oxygen. In fact, they all contain so-called CuO_2 "planes." Crystallographic structures are given for four high T_c superconductors in Figure 2.1a-d, with the CuO_2 planes indicated. A compound without Cu, $Ba_{1-x}K_xO_{3-y}$, has been found by Mattheiss et al [1988] to have a transition temperature of approximately 30K. At present it is still a matter of great debate as to whether this compound should be classified with the others as a high temperature superconductor, or whether, on the other hand, it can be shown to be more like the conventional BCS phonon-pairing superconductors.

2.2 Experimental Facts

Thorough reviews of the experimental properties of high temperature superconductors exist [e.g. Ginsberg, 1989]. The most unusual properties are the very high transition temperatures, short coherence lengths, and the

strong anisotropies of many quantities. Coherence lengths are on the order of 10 Angstroms, with $\zeta_\perp/\zeta_{||}$, the ratio of the coherence length perpendicular to the CuO_2 planes to that parallel typically 5 to 1. Welp et al [1989] find $\zeta_\perp$=3 Angstroms and $\zeta_{||}$=16 Angstroms $YBa_2Cu_3O_{7-\delta}$, with a transition temperature 90K; these coherence lengths correspond to upper critical fields of 674 Tesla and 122 Tesla for fields in the plane and perpendicular to the plane respectively.

Each high temperature superconducting compound is closely related to a compound displaying antiferromagnetism; for example oxygen deficient $YBa_2Cu_3O_{7-\delta}$ (with δ=1) is an antiferromagnet.

An appreciable isotope effect has been observed in $La_{1.8}Sr_{.2}CuO_4$ [Battlogg, et al [b],1987], but only a very small isotope effect is seen in $YBa_2Cu_3O_{7-\delta}$ [Battlogg, a,1987].

2.3 Theoretical Picture

The theoretical picture of the mechanism of high temperature superconductivity is not yet clear. Most theories retain the BCS principle of electron pairing, but differ from conventional BCS in that the pairing is not from the electron-phonon mechanism. Theorists generally agree that the phonon mechanism cannot account for superconductivity at temperatures higher than approximately 30K [McMillan, 1968; Cohen, et al,1982]; thus unconventional pairing theories are common [see for example Wolf and Kresin, 1987]. The existence of antiferromagnetic compounds which are closely related to high temperature superconductors has lead many theorists to propose pairing mechanisms in which magnetism is important

2.4 $YBa_2Cu_3O_{7-\delta}$

2.41 Crystal Structure

The crystal structure of $YBa_2Cu_3O_7$ is shown in Figure 2.1b. The orientation of the crystal axes is given. The lattice parameters at room temperature are given by: a=3.8282, b=3.8897, c=11.6944 Angstroms [Greedan et al, 1987]. We note that the structure contains two different types of Cu atoms (shown in figure). The "plane Cu," or Cu(2) has an environment which is approximately, though not exactly, four-fold symmetric about the crystal c-axis. Four oxygen atoms lie near the plane Cu just out of the plane formed by the plane Cu atoms. In addition, one finds above the plane Cu

(along c) the "bridging" oxygen, giving the Cu a total of five very nearby oxygens. The "chain" Cu is part of the so-called Cu-O "chains" which run in the b direction. The chain is a repeated Cu-O unit. Two "chain" oxygens are very near the chain Cu. In addition, two "bridge" oxygens lie above and below the chain Cu along c. These four oxygens give the chain Cu an environment that is crudely four-fold symmetric about the crystal a-axis. We later use these crystallographic symmetries, together with the symmetries of the measured NMR data, to assign the two Cu resonances to their respective crystallographic sites.

2.42 Valence Picture

The complex "molecule" $YBa_2Cu_3O_7$ presents some difficulties in the assignment of a valence charge to each atom. The valence for Y and Ba are rather uncontroversial. Charges of 3+ and 2+ respectively give these atoms a closed shell. The simplest formal valence picture is the following: we assign the oxygen atoms a charge of 2-. This leaves a total charge of +7 that must go on the Cu atoms. Since there are two plane Cu and one chain Cu per cell the sensible assignment is to assign the two plane Cu's a valence of 2+, and the single chain Cu a valence of 3+.

The measurements in this thesis indicate that the formal valence picture is not a good description of the actual atomic charges. We find that in fact the chain Cu, as well as the plane Cu, has a charge of 2+, not 3+. The extra hole most likely is found on the oxygen atoms, giving at least some of the oxygen atoms a valence greater than (less negative than) 2-.

2.43 Magnetism picture

The compound $YBa_2Cu_3O_6$ is an insulating antiferromagnet. The spins are located on the plane Cu atoms and point in the plane [Tranquada, 1988].

$YBa_2Cu_3O_7$ is not antiferromagnetic, but it is superconducting. The magnetism in this material is of special interest since many theoretical models of invoke magnetism as an element in a pairing mechanism. Magnetism in $YBa_2Cu_3O_7$ is a major subject of this thesis.

2.5 NMR and High T_c Superconductors

We discuss briefly results of important NMR experiments on high temperature superconductors. A review of NMR of high temperature superconductors is given by Pennington and Slichter [1989].

NMR has played an extremely important role in the characterization of conventional superconductors. The classic work of Hebel and Slichter [1959] and of Anderson and Redfield [1959] verified BCS predictions of NMR relaxation times as a function of temperature in the superconducting state and provided, along with ultrasonic attenuation measurements, the first confirmation that electrons are paired off in the superconducting state

2.51 $La_{2-x}Ba_xCuO_4$

To date very little work has been reported on $La_{2-x}Ba_xCuO_4$, the high temperature superconductor discovered by Bednorz and Muller. We have measured the Nuclear Quadrupole Resonance Lineshape of this material (Figure 2.2). We obtain, however sharply different lineshapes for two samples which were prepared in an identical fashion. Additionally we have found that the spin-lattice relaxation of the Cu nuclei is from an electric quadrupolar mechanism, rather than from a magnetic mechanism. At present we do not have a complete theory to explain our results on this compound.

2.52 $YBa_2Cu_3O_{7-\delta}$

2.521 Normal State Results

Many groups [Lutgemeier, et al, 1987; Mali et al, 1987; Furo et al, 1987; Warren, Walstedt et al, 1987; Riesemeier, et al, 1987; Kohara et al, 1987] have observed the Nuclear Quadrupole Resonance (NQR) lines of $YBa_2Cu_3O_{7-\delta}$. Two NQR resonance lines, corresponding to the two Cu crystallographic sites in $YBa_2Cu_3O_{7-\delta}$, have been observed, one at 22MHz and another at 31.5MHz. Warren and Walstedt initially assigned the 22MHz line to the plane Cu. Subsequent to the measurements reported in this thesis they have reversed this assignment. We shall, thus, refer to measurements on the 31.5MHz line as being simply from the plane Cu.

We have measured the NQR spin-lattice relaxation rate as a function of temperature in powder samples (Figure 2.3)

The plane Cu spin-lattice relaxation rate in the normal state does not obey the well known Korringa law; i.e. it is not linear in temperature.

The chain Cu spin-lattice relaxation rate in the normal state follows the Korringa law, although at high temperatures (above 200K) the temperature dependence becomes faster than linear, suggesting perhaps the activation of a new mechanism.

Walstedt et al [1988] have measured the spin-lattice relaxation anisotropy (i.e., the dependence of relaxation rate upon magnetic field orientation) of the plane Cu on powder samples. They find an anisotropy that is somewhat lower than we find, but with error bars that overlap those of our data. They show that the anisotropy is not consistent with the predictions of a simple band theory.

2.522 Superconducting State Results

Various groups have measured spin-lattice relaxation rates in the superconducting state, using the zero-field NQR approach [Mali et al, 1987; Furo et al, 1987; Warren, Walstedt et al, 1987; Riesemeier, et al, 1987; Kohara et al, 1987] . The spin-lattice relaxation of both the plane and chain Cu both drop sharply as one cools into the superconducting state.

In conventional BCS superconductors the spin-lattice relaxation typically increases as one moves just below the transition temperature. The spin-lattice relaxation in a conventional mechanism is given by:

$$\frac{1}{T_1} \alpha \int_0^\infty |V|^2 \rho^2(\varepsilon) f(\varepsilon)(1-f(\varepsilon))d\varepsilon \qquad (2.1)$$

where V is the scattering matrix element, ρ is the density of states, and f is the Fermi-Dirac distribution. There are two reasons for the usual increase in the relaxation rate as one moves into the superconducting state. The first is that as one forms a gap the energy states become crowded, making the integral over ρ^2 larger. Second, the pairing of electrons causes interference in the scattering matrix elements for states with equal and opposite **k** vectors which enhances NMR relaxation (and conversely decreases ultrasonic attenuation). The enhancement of NMR relaxation just below T_c, known as the "coherence peak," may be smeared or not present if one has substantial anisotropy in the energy gap. At lower temperatures the width of the function $f(\varepsilon)(1-f(\varepsilon))$, which is proportional to temperature, becomes substantially less

than the energy gap, and the relaxation decreases exponentially as $\exp(-2\Delta/T)$.

The "coherence peak" is not observed in either the plane or chain Cu.

By measuring the low temperature relaxation Warren et al [1987] found a decay of the relaxation corresponding to a gap energy of $8k_BT_c$ for the plane Cu, and $2.4k_BT_c$ for the chain. Using the BCS relation that the gap energy is given by $3.5k_BT_c$ these energies correspond to 200K and 60K respectively. The plane Cu relaxation, then decays much more steeply with decreasing temperature than the chain.

In addition to the BCS predictions of spin-lattice relaxation, BCS theory also predicts the behaviour of the NMR "Knight shift" in the superconducting state. The Knight shift is proportional to the Pauli electron spin susceptiblity. BCS describes a pairing of electrons in which spin susceptibility vanishes. The Knight shift in the superconducting state has been measured by Takigawa et al [1989 (a,b)] and by Barrett et al [1989] in our laboratory. The simplest interpretation of these data indicates that, as in conventional BCS theory, the Knight shift in the superconducting state vanishes in $YBa_2Cu_3O_{7-\delta}$. We will at certain points make use of these data in our analysis.

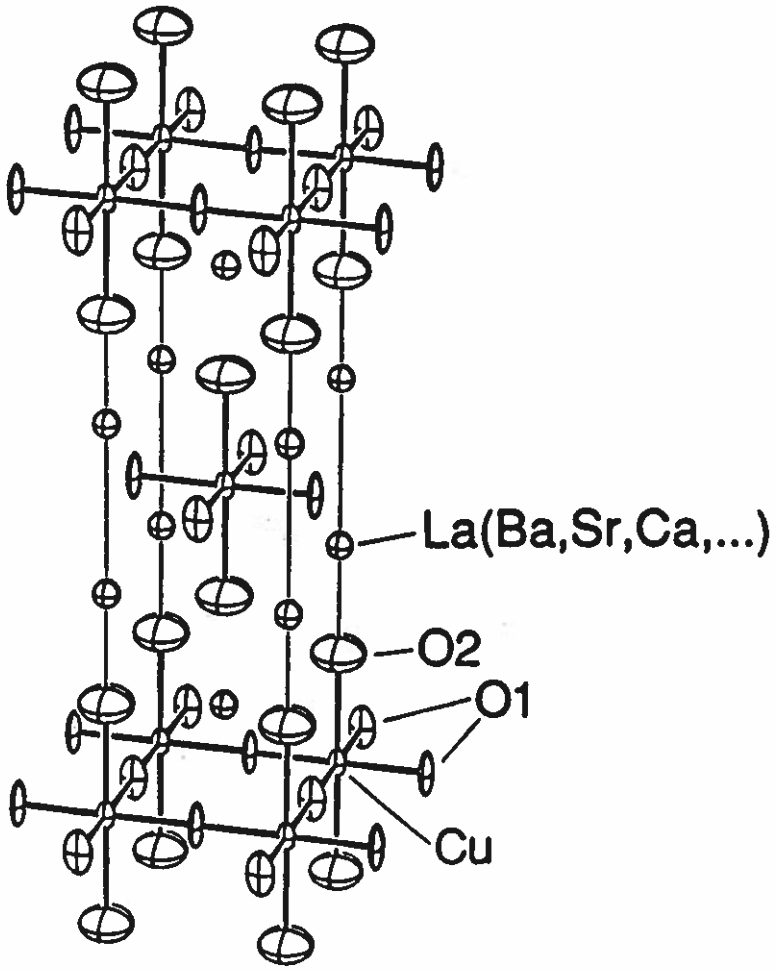


Figure 2.1a: The structure of $La_{2-x}A_xCuO_4$, where A=Ba, Sr, Ca. Figure supplied by J.D. Jorgenson.

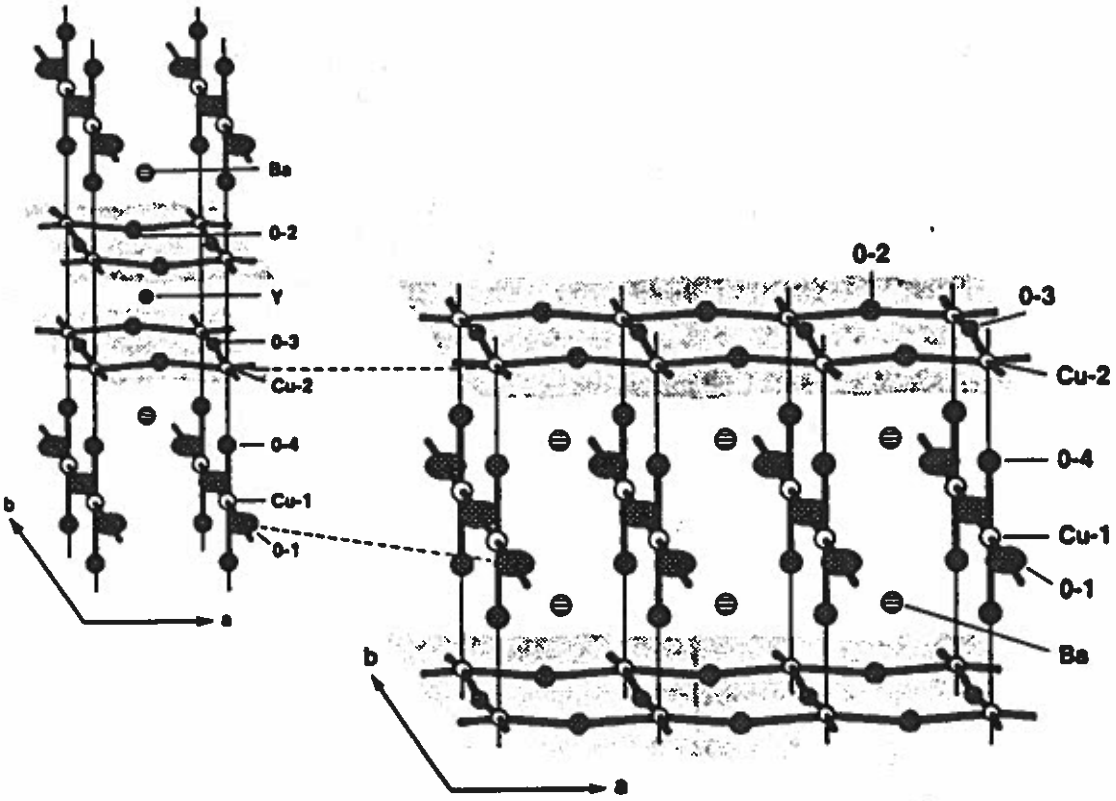


Figure 2.1b The structure of $YBa_2Cu_3O_7$. The CuO_2 "planes" are shown shaded. Taken from Beno et al [1987].

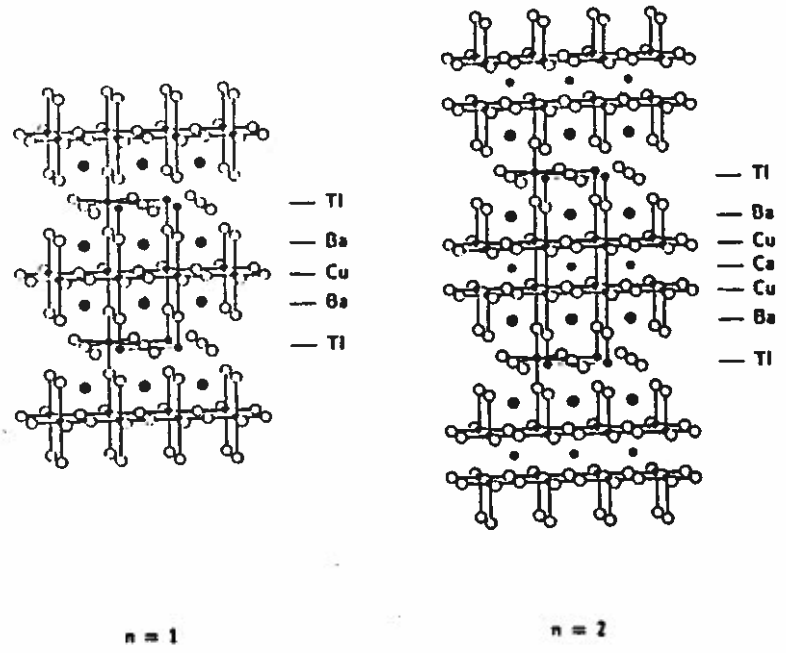


Figure 2.1c The structure of $TlBa_2Ca_{n-1}Cu_nO_{2n-3}$. The corresponding bismuth compound has the same structure, with Tl replaced by Bi. Figure supplied by A.W. Sleight and C.C. Torardi.

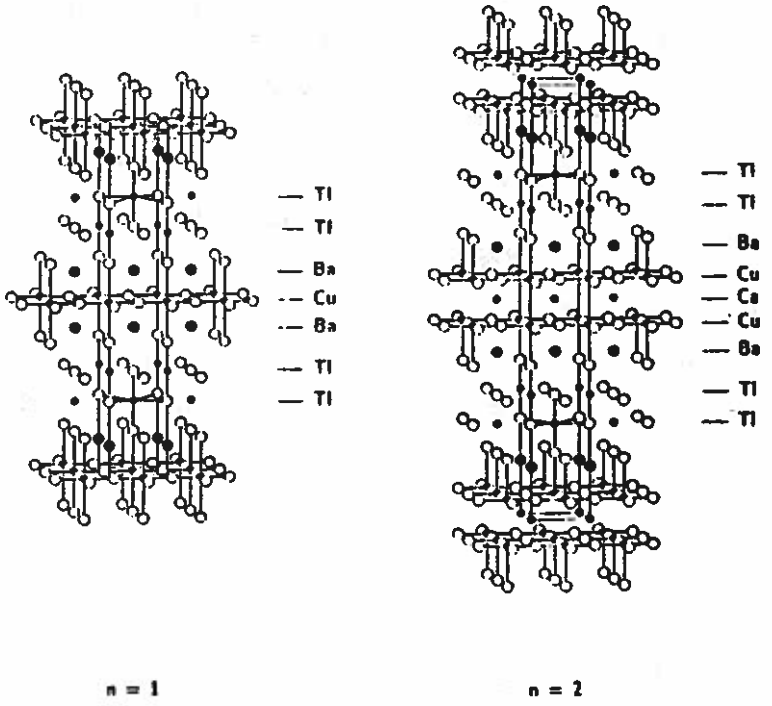


Figure 2.1d The structure of $Tl_2Ba_2Ca_{n-1}Cu_nO_{2n+4}$. The corresponding bismuth compound has the same structure, with Tl replaced by Bi. Figure supplied by A.W. Sleight and C.C. Torardi.

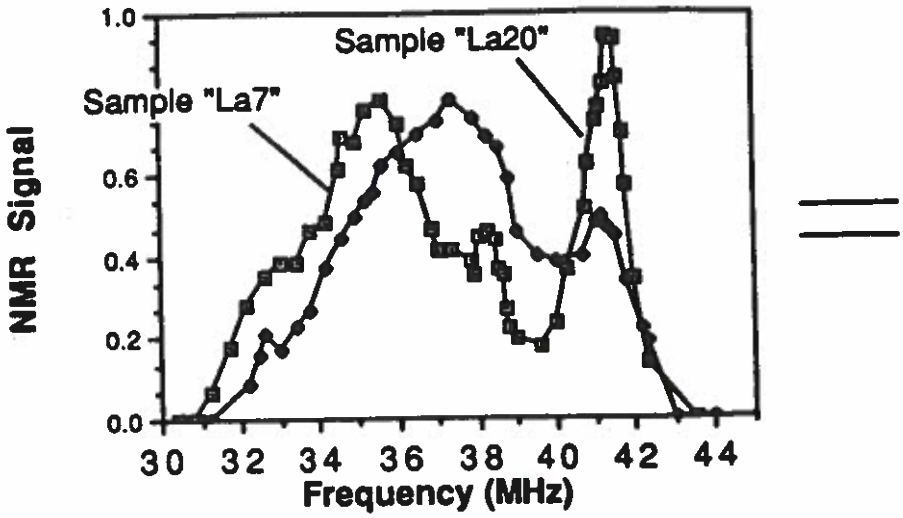


Figure 2.2 The nuclear quadrupole resonance lineshapes for two samples, sample "La7" and sample "La20." The samples have nominally identical composition $La_{1.85}Ba_{.15}CuO_4$. They were prepared with identical procedures. Yet, one finds substantially different lineshapes.

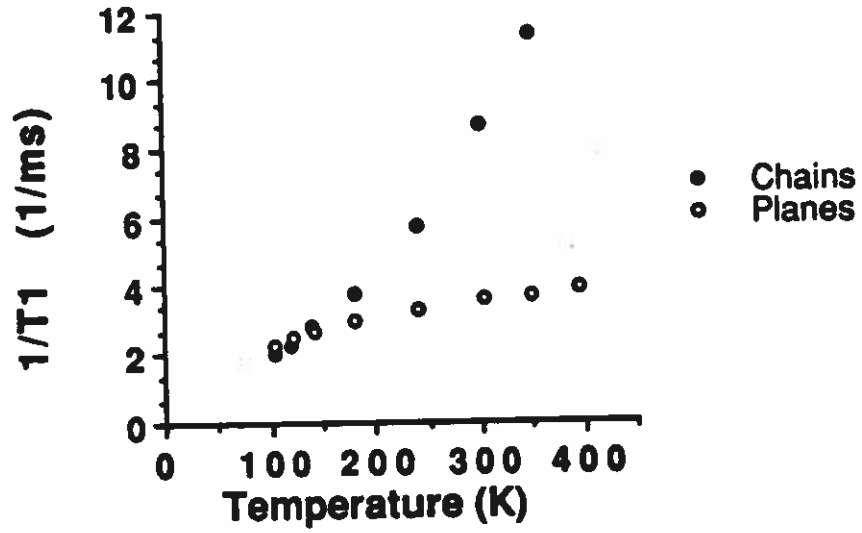


Figure 2.3 Powder Sample NQR Measurement of Spin-Lattice Relaxation Rate of the Plane and Chain Cu in $YBa_2Cu_3O_{7-\delta}$.

Chapter 3: Nuclear Magnetic Resonance

Nuclear Magnetic Resonance (NMR) is the experimental technique used in this thesis. We have done three types of NMR measurements; each type has been performed with the static field H_0 placed succesively along each of the three crystal axes, a, b, and c (see Figure 2.1b) and for both the plane and the chain Cu. First, we have made measurements of the NMR lineshapes and line postions. These measurements enable us to calculate two NMR tensors--the electric field gradient tensor and the magnetic shift tensor. Second, we have measured the spin-lattice relaxation rate, giving the spin-lattice relaxation rate tensor. Finally, we have measured the time dependence and functional form of the transverse relaxation.

We now discuss the four types of NMR tensors that have been measured in this thesis and the major contributions to each of these tensors .

The richness of the $YBa_2Cu_3O_{7-\delta}$ system necessitates a somewhat complex analysis of NMR data.

3.1 NMR Tensors

We measure the diagonal components of four different tensors in the principle axis system for each of the two Cu crystallographic sites: the electric field gradient tensor, the magnetic shift tensor, the spin-lattice relaxation rate tensor, and the transverse relaxation rate tensor.

3.11 The Electric Field Gradient Tensor

Cu has two isotopes, Cu^{63} and Cu^{65}, occuring in abundances of 69 and 31 percent respectively. Each isotope has, in addition to a nuclear magnetic dipole moment, an electric quadrupole moment Q. Electric charges in a material generate electric field gradients at the site of the Cu nucleus and interact with the nuclear quadrupole moment. If we denote the spin as I then we can express the Hamiltonian representing the interaction between the Cu quadrupole moment and the local electric field gradient in terms of Q, the electric field gradient tensor, and products of the components of I. This same vector I, of course, appears in the Zeeman Hamiltonian representing the interaction of the nuclear spin with the magnetic field. The net result is that electric field gradients shift NMR resonance lines. Observation of these shifts allows one to measure the electric field gradient

at the site of the Cu nucleus. The complete Cu nuclear spin Hamiltonian and its solutions are discussed in Appendix A.

We have used NMR to measure the electric field gradient at both the plane and chain Cu sites. The electric field gradient is the sum of the electric field gradients from two principle sources: first, the electric field gradient from the charges in electron orbitals on the Cu atom, and second, the electric field gradient from other ions in the lattice, which we treat as point charges. The contribution from the lattice ions should in general be enhanced by the so-called Sternheimer factor, γ_s, representing polarization of core electron orbitals.

3.12 The Magnetic Shift Tensor

The magnetic shift tensor is the shift in the NMR resonance frequency away from the shift of a reference compound, and specifically only the part of the shift that is proportional to the applied magnetic field; one does not include the part of the shift due to electric field gradient effects. The magnetic shift tensor is often a combination of magnetic shifts from several sources. We list and briefly describe the shifts that are important in our measurements:

3.121 The Orbital (Chemical) Shift

The orbital (chemical) shift results primarily from "unquenching" of orbital angular momentum. For example, the atomic state x^2-y^2 consists of equal combinations of $m_l = +2$ and $m_l = -2$, giving no net current circulation and hence no magnetic field at the nuclear site. An applied field along z, however, mixes the state $x^2 - y^2$ with a small fraction of :'y. The resultant state has some small circulating current which creates an extra magnetic field at the nucleus. A detailed treatment of this kind of orbital shift is given in Appendix B.

3.122 The Knight Shift

The spin shift due to the susceptibility of conduction electrons is called the "Knight Shift." (Note: In our earlier published work [Pennington (a,b,c,d), 1988; 1989] we loosely used the term "Knight shift" to describe a shift due to the susceptibility of localized electrons; in this thesis, however, the term "Knight shift" will be reserved to describe shifts from conduction electrons.) The Knight shift will be proportional to the Pauli electron spin

susceptibility and to the electron-nuclear hyperfine coupling. In $YBa_2Cu_3O_7$ the Cu Knight shift is present in a form of "transferred hyperfine coupling."

3.1221 Transferred Hyperfine Coupling I

Transferred hyperfine coupling I is the interaction of the Cu nuclear spin with the spin of a hole in an oxygen p-orbital band. If an oxygen has a valence of 2-, of course, then the band is filled, giving no susceptibility and no shift. Holes in the band, however, give the famous Pauli paramagnetism. The oxygen p-orbitals may be hybridized with the Cu 4s orbital. An oxygen hole then, has some probability density on the Cu nuclear site. The hole couples with the Cu nucleus through the so-called "Fermi contact interaction" [see Slichter Chapter 4.7, 1980]. The coupling is isotropic. The shift in the resonance frequency will be proportional to the coupling strength, which we call B_h ("h" stands for "hole"), and the hole spin susceptibility χ_h. We have:

$$K_h = B_h \chi_h \qquad (3.1)$$

where K is the fractional Knight shift from this mechanism.

This hyperfine coupling mechanism, in addition to producing a "Knight shift," also contributes a spin-lattice relaxation rate that is proportional to temperature (see section 3.131).

3.123 The (Localized) Spin Susceptibility Shift

In addition to the Knight shift, one also finds in $YBa_2Cu_3O_{7-\delta}$ an NMR shift from nuclear spin interactions with localized moments.

In normal state $YBa_2Cu_3O_{7-\delta}$ we find that plane Cu atoms have a permanent spin-1/2 local moment. This local moment is hyperfine coupled to the nuclear spin. The local moment fluctuates rapidly, on time scales of about 10^{-15} seconds, much faster than the NMR resonance frequency. Thus, the NMR line is shifted by an amount determined by the time-averaged magnitude and direction of polarization of the localized spin in the strong magnetic field--an amount proportional to the susceptibility of the localized spin.

There are two sources of hyperfine coupling to the localized spin--direct and transferred.

3.1231 Direct Hyperfine Coupling

Direct hyperfine coupling exists between a Cu nuclear spin and the spin of the electron localized on the Cu atom. The coupling is from two sources: the magnetic dipole-dipole interaction of the nuclear and electron spins, and the so-called "core polarization"--the electron spin in the 3d orbital distorts the core electron orbitals, and distorts the up and down spin states in a slightly different way. The sum of these two effects is represented by the coupling tensor $A_{\alpha\alpha}$. The shift for a magnetic field along c is given by:

$$K_{cc} = \frac{A_{cc}}{\gamma_n \gamma_e \hbar^2} \chi_s \qquad (3.2)$$

3.1232 Transferred Hyperfine Coupling II

An oxygen atom with a filled shell which separates two Cu atoms can mediate a transferred hyperfine interaction between a Cu nuclear spin and the spin of the electron on the Cu neighbor. Mila et al [1989] first pointed out the importance of this kind of transferred hyperfine coupling in $YBa_2Cu_3O_{7-\delta}$; the effect is well known in concentrated magnetic systems [Huang et al, 1967]. We describe the coupling between the electron spin in a Cu A 3d-orbital and the nuclear spin on Cu B. We take the Cu A electron spin to be up. We have between these atoms an oxygen atom with a closed shell. Now, we suppose that an oxygen electron virtually spends time in the Cu A 3d-orbital. Its spin must in that event be down. The spin of the remaining oxygen electron is then up. The remaining electron orbital, however, is hybridized with the Cu 4s orbital. Thus, it hyperfine couples to the Cu B nucleus through the contact interaction. Of course in $YBa_2Cu_3O_7$ a Cu atom will be coupled in this way to four other Cu atoms. If the coupling parameter for one Cu neighbor is B_1 then the total Knight shift contribution will be given by:

$$K = \frac{4B_1}{\gamma_n \gamma_e \hbar^2} \chi_s \qquad (3.3)$$

where χ_s is the susceptibility of the Cu spins.

3.13 The Spin-Lattice Relaxation Rate Tensor

The spin-lattice relaxation rate is the rate at which nuclear spins, disturbed from thermal equilibrium, relax back to thermal equilibrium. It is often characterized by a single relaxation time called "T_1." Cu, however, is a spin-3/2 nucleus, giving a four-level system in a magnetic field. Its relaxation

may have as many as three exponential components, depending on the detailed non-equilibrium state. The functional forms of relaxation for various pulse sequences is given in Appendix D. We parametrize our relaxation with a rate constant W_1, which is defined in Appendix C.

There are two major sources of Cu spin-lattice relaxation in $YBa_2Cu_3O_7$, both of which involve electron nuclear hyperfine coupling. The sources are the relaxation from interaction with localized spins, and "Korringa" relaxation, relaxation from interaction with conduction electrons.

3.131 Korringa Relaxation (Transferred Hyperfine Coupling I)

Korringa relaxation results from the same electron-nuclear couplings that give the Knight shift; it results from interactions with conduction electrons. In $YBa_2Cu_3O_{7-\delta}$ the electron-nuclear coupling is through the Transferred Hyperfine Coupling I (Section 3.1221) mechanism.

The relaxation rate contribution from this mechanism is proportional to temperature. When electron-electron interactions can be neglected one has the "Korringa relation" which relates the spin lattice relaxation W_1 to the Knight shift K_h:

$$\frac{3T}{2W_1}(K_h)^2 = \left(\frac{\hbar}{4\pi k_B}\right)\frac{\gamma_e^2}{\gamma_n^2} \qquad (3.4)$$

If this relation is obeyed then it is a strong indication that one has relaxation from a conduction band in which electron-electron interaction effects give only modest corrections to a simple band theory.

By combining measurements described in this thesis with measurements by Barrett et al [1989] in our laboratory of the Knight shift in the superconducting state we have found that this mechanism dominates the relaxation of the chain Cu in $YBa_2Cu_3O_{7-\delta}$.

3.132 Relaxation from Interactions with Localized Spins

We have found that the spin-lattice relaxation of the plane Cu is dominated by hyperfine interactions of the nuclear spin with the spin of a localized electron moment on the plane Cu atom. The moment fluctuates rapidly and produces rapidly changing hyperfine fields at the nucleus which relax the nuclear spin. We treat this case quantitatively in Chapter 7.

3.14 The Transverse Relaxation Tensor

In a spin-echo experiment one applies a 90° pulse followed after a delay time t/2 by a 180° pulse. A spin-echo then appears at time t. The transverse relaxation refers to the decay of the size of the spin-echo signal as a function of the delay time. "Exponential decay" refers to decay following the form $S(t) = \exp(-t/T_2)$ where the decay is parametrized by a time T_2.

In our experiments we have in all cases been able to fit our transverse relaxation data with a functional form combining an exponential and a Gaussian:

$$S(t) = \exp(-t/T_2)\exp(-t^2/2\tau^2) \qquad (3.5)$$

In all cases a T_2 contribution is present which results from the same mechanisms giving the T_1. This we denote "T_{2R}" where "R" refers to Redfield, who developed the theory to relate this quantity to the measured spin-lattice relaxation [1957; see Slichter, Chapter 5 for review (1980)]. In each of our measurements we have also found some additional relaxation, which in some cases is exponential and in others Gaussian. For the plane Cu we show that the additional relaxation is due to an electron mediated nuclear spin-spin coupling. For the chain Cu the source of the extra relaxation is not as clear.

Chapter 4: Experimental Techniques

4.1 Apparatus

4.11 Magnet and Sample Dewar

The magnet used in the NMR measurements is a superconducting solenoid operating in persistent current mode with a field of 81.2kG, giving a Cu NMR frequency of 91.628MHz in CuCl, a diamagnetic salt used as our reference compound.

The measurements reported are at 100K. The Oxford magnet has a vertical, cylindrical, room-temperature bore, with the field along its axis. The sample Dewar [Ross, 1986] has a long cylindrical tail which extends into the region of strongest field. The axis of the cylindrical tail, thus, is parallel to the field H_0.

The Dewar has an inner chamber for the probe. The inner chamber extends into the strong field region of the magnet. An outer chamber, with a reservoir for liquid nitrogen is located outside of the magnet bore. Thermally anchored to the outer chamber is a Cu radiation shield extending into the bore and surrounding the inner chamber. With liquid nitrogen in the reservoir the Cu shield serves to cool the probe. Temperature is measured with a copper-constantan thermocouple in thermal contact with the brass probe can. The heating element is a wire wrapped around the probe can in a manner that minimizes stray magnetic fields.

The inner chamber of the sample Dewar is sealed with an atmosphere of nitrogen gas. It is necessary to eliminate O_2 gas from the region, as it may condense to form liquid oxygen, which is highly combustible.

4.12 Coil and Resonant Circuit

The Resonant Circuit used is shown in Figure 4.1a. The sample is placed within the inductor-coil. In an NMR "pulse" a large (typically 100-300V peak to peak) rf voltage is applied to the circuit. The voltage across the coil is increased by a factor Q, resulting in currents typically of the order of 100 A and alternating magnetic fields of 10-100G.

The coil is not only used to apply pulses to the spins, but also to detect their precession in the strong, static magnetic field. The precessing nuclear magnetic dipoles induce an emf which is detected and measured.

When working with a small sample as we are it is helpful to also have a tiny NMR coil, so that the sample comes as close as possible to filling the entire coil volume. We have worked with a coil having a volume of .2cc, about 1/10 the volume of coils used typically in our lab.

In addition we have used silver wire [A. D. MacKay Co.] rather than copper in order to avoid spurious copper signals.

4.13 NMR Spectrometer

The NMR spectrometer used is basically that described by Shore [1986]. The only major changes to the spectrometer are the replacement of the Pulse Sequence Generator with a more powerful one of my design and the addition of a quadrature detection system designed by Chris Klug and Dale Durand.

The spectrometer is controlled by the user with an IBM PC/XT. The powerful and user-friendly software which interfaces with the user and with the apparatus has been written by Dale Durand [1989].

The Pulse Sequence Generator is a device used for the rapid time-control of experiments. It exerts control by means of TTL outputs. The new Pulse Sequence Generator allows pulse sequences of almost arbitrary length and complexity. A description of the new pulse sequence generator is given in Appendix E. Dale Durand has greatly assisted in the construction of the Pulse Sequence Generator.

The quadrature detection system of Durand and Klug has also benefitted our lab greatly, reducing data acquisition time in nearly every experiment by a factor of 2. In an NMR experiment the signals detected are at rf frequencies. They are, however, mixed to audio frequency at the final detection stage. In this process one loses information unless one mixes the NMR signal with rf of two orthogonal phases. In the past we have at times mixed first with one phase and then repeated the experiment, mixing at a second phase. The new quadrature detection system allows us to measure with both phases simultaneously.

4.2 Single Crystal Samples

The samples used for NMR measurements in this thesis are single crystals of $YBa_2Cu_3O_{7-\delta}$. The crystals were prepared in the laboratory of

Prof. Don Ginsberg by Joe Rice and Ed Bukowski. The preparation of the samples is described by Rice et al [1988, 1989].

The typical crystal has dimensions of 1mm x 1mm x .15mm. The shortest dimension is along the crystal c-axis. The crystals, then, are flat, as shown in Figure 4.1b, with the c-axis perpendicular to the flat plane. The crystal lattice constants a and b are very nearly equal, and in all crystals used crystal "twinning" was present; if we imagine Cartesian axes in the crystal ab plane, then some crystal domains have the crystal a(b)-axis along x(y), but in neighboring domains the b(a)-axis is along x(y).

In our first reported measurements on single crystal $YBa_2Cu_3O_{7-\delta}$ [Pennington et al (a, 1988)] we have used a multiple-crystal sample, consisting of approximately 30 single crystals having a total mass of 2mg. In subsequent reported work [Pennington et al (b,c,d, 1989;)], and in measurements reported here, we use a single crystal with a mass of 1mg.

All samples used show a sharp transition to the superconducting state at approximately 90K. Figure 4.2 demonstrates the Meissner effect in our 1mg single crystal. Shown is the sample susceptibility versus temperature, measured with an external magnetic field of 16 Oersted applied perpendicular to the crystal c-axis. The susceptibility is shown for both the "field cooled" and "zero-field" cooled procedures. In the "field cooled" procedure the magnetic field is first applied with the sample in the normal state, at a temperature greater than T_c. As the sample is cooled into the superconducting state currents flow which expel the magnetic field. In the "zero-field cooled" procedure the sample is cooled into the superconducting state with zero applied field. The field is then turned on, and the sample warmed into the normal state. In a perfect demonstration of the Meissner effect the quantity $-4\pi\chi$ will equal 1 at $T << T_c$. In general, however, some flux-trapping will occur when one uses the "field-cooled" procedure, giving a value of $-4\pi\chi$ somewhat less than 1. We obtain low temperature values of $-4\pi\chi$ of .59 and .95 for the "field-cooled" and "zero-field cooled" procedures respectively. These values suggest a sample that is 100% superconducting. In addition these values may be slightly underestimated, since upon mounting the crystal in the NMR probe a small piece of the crystal was

accidentally chipped off. The mass of the crystal used in the calculation of χ is thus slightly overestimated.

4.3 Crystal Alignment

We have made NMR measurements on a single crystal of $YBa_2Cu_3O_{7-\delta}$, with the field applied along each of the three principal axes of the crystal.

4.31 Crystal Alignment for H_0 Along c

The NMR probe for mounting $YBa_2Cu_3O_{7-\delta}$ crystals with H_0 along c was designed and constructed by Dale Durand. The probe top-piece fits at the top of the sample Dewar. A rigid stainless steel rod extends through the center of the probe top-piece and into the region of strong field. The stainless steel rod is soldered into the top of the probe can. The diameter of the probe can is chosen to be very slightly smaller than the inner diameter of the inner chamber of the sample dewar. This restricts the rigid rod to point parallel to the field. Gripping the rigid rod in a lathe, the probe can top is faced-off to have a surface perpendicular to the rigid rod. A plexiglas piece is milled to form a sample support with a face parallel to the probe can top face.

We then take advantage of the crystal shape. The crystals have flat faces perpendicular to their c-axes (Figure 4.1b). The flat face is mounted on the plexiglas piece. We spray the mounted crystals with "Crystal Clear," an aerosol used as an insulating coat in electronics. The spray holds the crystals on the support at all temperatures between 77 and 300K. The single crystal probe configuration is sketched in Figure 4.1c.

We have made measurements on both a single crystal sample [Pennington et al [b,c,d] 1989] and on a sample with about 30 crystals mounted [Pennington et al [a], 1988]. The linewidths of these measurements should be very sensitive to crystal misalignment. We find identical linewidths and line positions in these samples, indicating that our alignment method is very accurate.

4.32 Crystal Alignment Along a/b

Due to crystal twinning, we effectively align part of the crystal with H_0 along the a-axis, and part along b.

The alignment is a two step procedure. The first step is to use probe design that guarantees that H_0 is perpendicular to c. The procedure given above for the H_0 ll c alignment is used, with the exception that the plexiglas piece is milled to give a face perpendicular to the probe can top rather than parallel. The flat surface of the crystal is mounted on that face, guaranteeing H_0 perpendicular to c.

Given H_0 perpendicular to c there is a remaining angular degree of freedom. To adjust this degree of freedom we use a microscope with a cross polarizer. Polarized light is shined upon the crystal. The light reflects of the crystal and through a polarizer at a right angle to the initial polarization. The light is observed after this "cross polarizer." If the polarization vector of the light is not rotated upon reflection off the crystal surface, as occurs when the polarization vector is along a principal axis of the crystal (a or b), then the cross polarizer will transmit zero intensity and the crystal will look black. If the initial polarization of the light is not along the crystal principal axis, then the polarization will be rotated and the crystal will not appear black. We obtain precision of approximately 1° using this technique.

4.4 Pulse Sequences

Various NMR pulse sequences have been used in order to overcome difficulties in the NMR of this material. The NMR spin-echo and the concept of a pulse sequence is given in Slichter [1980, p.38 and Chapter 8].

Novel pulse sequences are used to accomplish various objectives. An important problem in the study of Cu in $YBa_2Cu_3O_{7-\delta}$ is the very rapid decay of the spin-echo envelope. In a spin-echo experiment 90° and 180° degree pulses are applied with a time spacing τ. The spin-echo occurs at a time τ following the 180° pulse. Typically the signal will decay with increasing τ, so one chooses a short τ. Following a pulse, however, there is a period of time in which the rf amplifiers are saturated. At later times still the rf amplifier is not saturated but is picking up signals from the ringdown of the coil. This ringdown is closely related to the phase of the pulse applied; the sign of the ringdown will change if the phase of the pulse changes by 180°. One can design pulse sequences in which the NMR signals add coherently but the ringdowns cancel.

4.41 The Spin-Echo

The spin-echo is used in every pulse sequence described in this thesis. For an explanation of the spin-echo see Slichter [1980, p.38]. The pulse sequence is given by:

$90_x - \tau - 180_x - \tau$ - echo

The notation "90_x" denotes a 90° pulse about the x-axis in the rotating frame. The pulse is in fact a rotation of the magnetization vector, initially along z, by 90° about the x-axis in the rotating frame. The 90° pulse is followed by a waiting period τ and then a 180° pulse, also a rotation about the x-axis. Following a time τ the echo appears and is measured.

In this sequence the recovery from both the 90° and 180° pulses is present at the echo position.

4.42 Add-Subtract Spin-Echo

The Add-Subtract Spin-Echo is used to cancel the ringdown voltage resulting from the 180° pulse. The sequence is:

$90_x - \tau - 180_x - \tau$ - echo	ADD
$90_{-x} - \tau - 180_x - \tau$ - echo	SUBTRACT

By alternating between 90_x and 90_{-x} pulses one alternates the sign of the resulting echo. Adding the first echo and subtracting the second gives contructive addition. The phase of the ringdown of the 180° pulse does not alternate, thus it cancels in the add-subtract procedure. Note, however, that the ringdown from the 90° pulses, though much smaller than that of the 180° pulses, does not cancel and will be present at the time of the echo.

4.43 Add-Subtract, Four-Phase

This sequence is given by:

$90_x - \tau - 180_x - \tau$ - echo	ADD
$90_{-x} - \tau - 180_x - \tau$ - echo	SUBTRACT
$90_{-x} - \tau - 180_y - \tau$ - echo	ADD
$90_x - \tau - 180_y - \tau$ - echo	SUBTRACT

In this sequence it is clear that the ringdowns of both the 90° and 180° pulses cancel after the add-subtract procedure. In addition the echoes add constructively. Note, for example, that the first step in the sequence results in a final magnetization vector along y. The fourth results in magnetization

along -y, thus having the opposite sign. These steps are added and subtracted, respecively, giving constructive addition of the signals.

The effectiveness of this sequence in eliminating ringdown problems is demonstrated in Figure 4.3.

4.44 Inversion Recovery

The inversion recovery sequence is used in this thesis to measure spin-lattice relaxation ("T_1"). In this sequence one first inverts the magnetization by applying a 180° pulse. One then waits a parameter time t, as the magnetization "relaxes" to again point along the +z axis. After the relaxation period one measures the magnetization with a spin-echo sequence. Of course, it is advantageous to use spin-echo techniques such as add-subtract and add-subtract, four-phase. In the latter case one uses the procedure described in 4.43 but in addition precedes each 90° pulse by a time t with a 180_x pulse. One holds τ fixed and varies t to find the spin-lattice relaxation.

A difficulty with this sequence is that in addition to the spin-echo which one wishes to measure, four additional echoes, called "stimulated echoes," which result from having a third pulse, are present [Hahn, 1950]. These stimulated echoes may obscure the measurement.

4.45 Alternating Phase Inversion Recovery

Stimulated echoes result from the slight error in the 180° pulse length. By alternating the phase of the 180° pulse, for example by alternating between 180_x and 180_{-x} one cancels the stimulated echoes. We prove this cancellation in Appendix F.

This technique is very important in the measurment of very fast spin-lattice relaxation. In this thesis it was used in the measurement of W_1 for the plane Cu with H_0 perpendicular to the crystal c axis.

4.46 Spin-Echo Double Resonance (SEDOR)

The SEDOR double resonance sequence is used to measure spin-spin coupling between nuclei having very different resonance frequencies. The sequence is given by:

Nucleus A: 90 - τ - 180 - τ - echo

Nucleus B: - t - 180 -

The sequence consists of a spin-echo sequence applied to Nucleus A. At a time t following the 90° pulse on Nucleus A a 180° pulse is applied to Nucleus B. The delay time τ for the Nucleus A spin-echo is held fixed. The spin-echo signal of Nucleus A is measured as a function of t, the time of the 180° pulse applied to Nucleus B. The 180° pulse on Nucleus B will have no effect on Nucleus A unless the two nuclei are coupled to one another. Thus, one can demonstrate unambiguously the existence of spin-spin coupling and measure its strength.

We have applied the SEDOR sequence to measure spin-spin coupling between plane Cu nuclei. There are two different Cu isotopes, Cu^{63} and Cu^{65}. In our experiment Cu^{65} played the role of Nucleus A and Cu^{63} that of Nucleus B.

4.47 Elimination of Drift

The newly designed Pulse Sequence Generator makes possible pulse sequences of very great length. This has allowed us to cut down on any possible effects of "drift" in the apparatus. The problem is as follows: one wishes to make an NMR measurement as a function of some time parameter, for example τ in a spin-echo experiment. It may require many hours and extensive signal averaging to measure the signal versus, say, eight values of τ. If one averages with one value of τ, stores the results, changes τ, etc., one has the possibility that something in the apparatus may "drift" in some way, for example by systematically giving smaller signals for the last times measured. We eliminate this possibility by using a pulse sequence which varies τ from sweep to sweep. Data is stored in eight separate arrays. If the experiment takes many hours then one knows that each value of τ experiences the same drift.

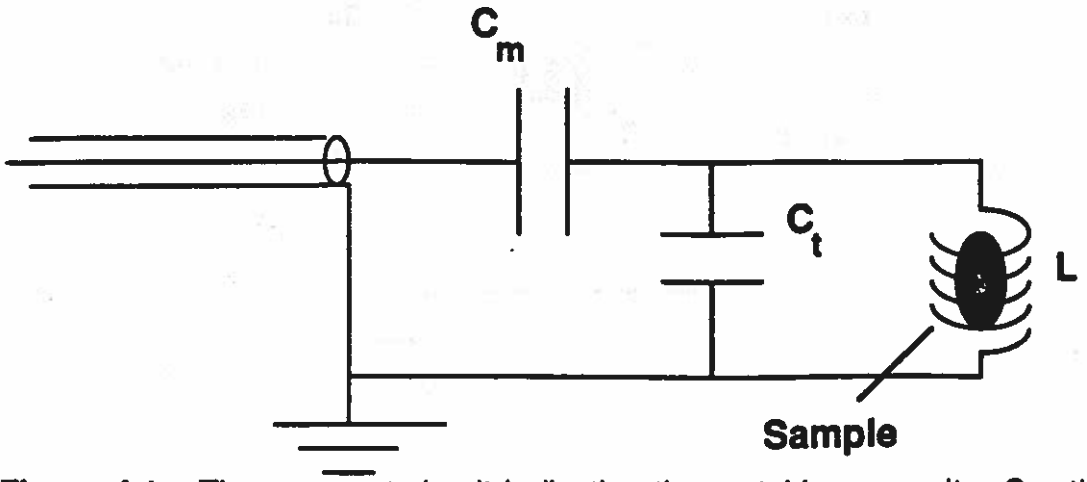


Figure 4.1a. The resonant circuit indicating the matching capacitor C_m, the tuning capacitor C_t, and the inductor coil. The circuit is matched to 50 ohms.

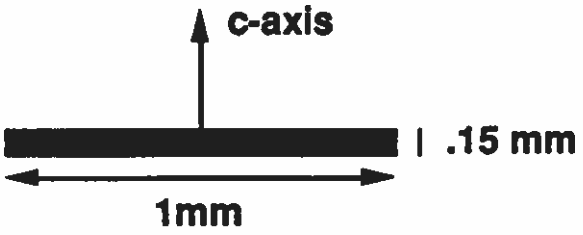


Figure 4.1b The flat shape of the single crystal.

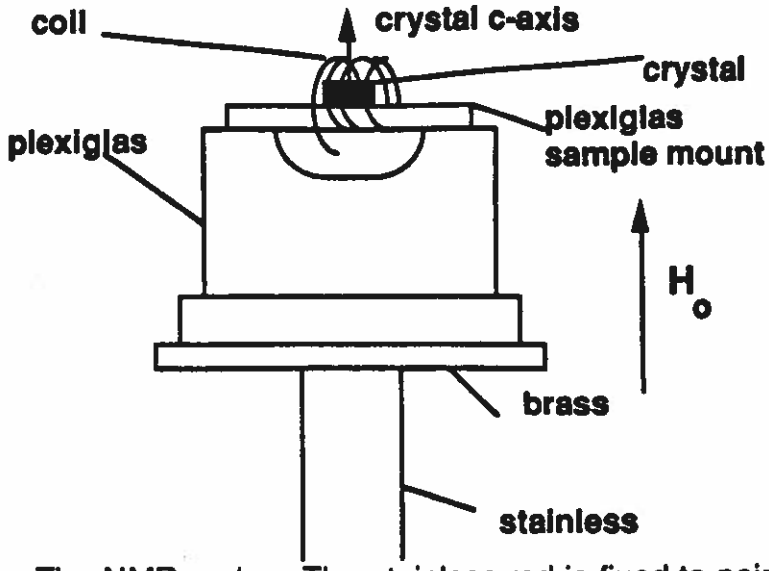


Figure 4.1c. The NMR probe. The stainless rod is fixed to point parallel to H_0. The brass face of the probe can is lathed to be perpendicular to the stainless rod.

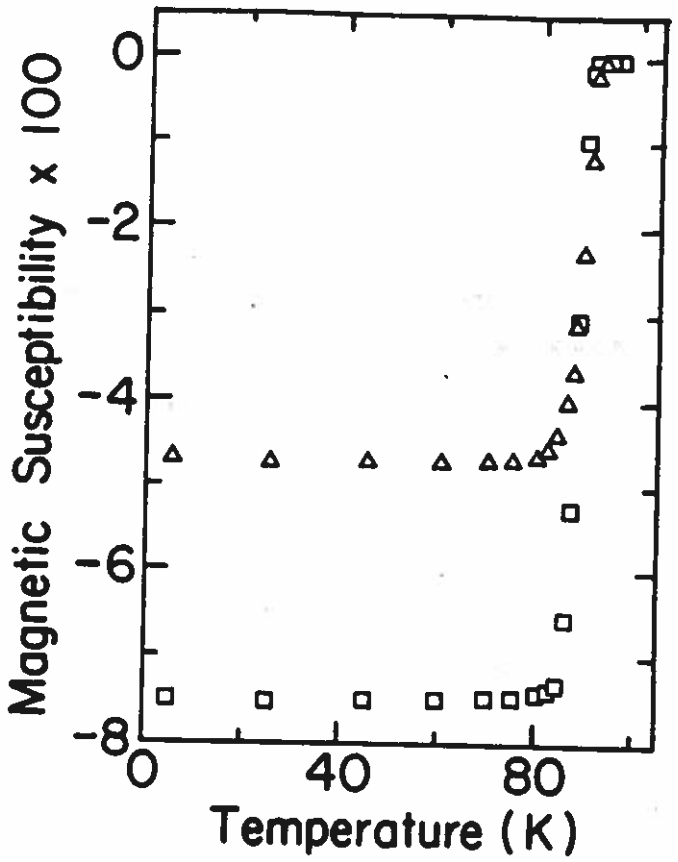


Figure 4.2 SQUID susceptibility measurement showing the Meissner effect in our 1mg crystal, and indicating a sharp superconducting transition at 90K. The magnetic field was applied in the plane of the crystal, perpendicular to the c-axis. Measurement performed by Joe Rice.

Figure 4.3:

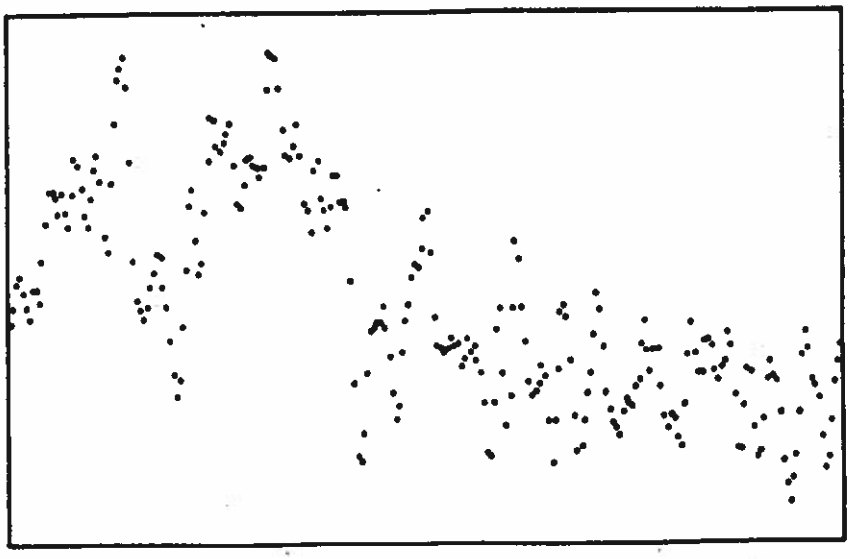


Figure 4.3a. The NMR signal shown above uses the Add-Subtract (Section 4.42) technique. The echo signal contains a very large recovery in addition to a somewhat smaller signal. The delay time between the 90 and 180 degree pulses is 45μs. The time interval shown is 128μs.

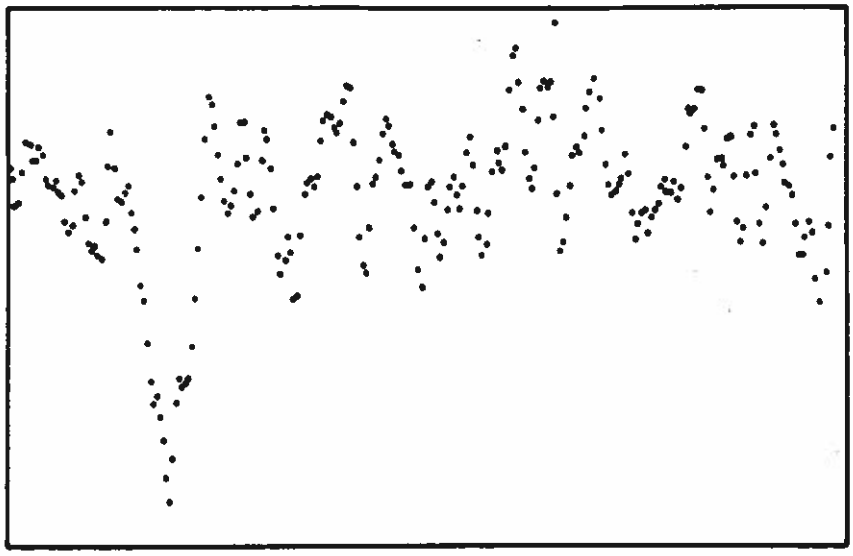


Figure 4.3b The signal shown above uses the Add-Subtract Four-Phase Method, in which the recoveries of both the 90 and 180 degree pulses are cancelled. The same number of sweeps is averaged as in Figure 4.3a. The times used are identical as well. The recovery glitch is much smaller, and the echo more easily detected.

Chapter 5: Experimental Results

5.1 Measurements Made

We have observed the nuclear magnetic resonance of both the plane Cu and the chain Cu in $YBa_2Cu_3O_{7-\delta}$. The measurements reported are at an applied magnetic field of 81.2kG, giving a Cu^{63} NMR frequency of 91.628MHz in our reference compound, the diamagnetic salt CuCl. Many measurements have been performed on Cu^{63} only, since its abundance (69%) is highest, but in some cases we have also observed Cu^{65} (abundance 31%) for comparison. Both Cu^{63} and Cu^{65} are spin-3/2 nuclei, giving a four-level system corresponding to the four values of m_I where I is nuclear spin. In an NMR experiment one observes coherence between states differing by $\Delta m_I = \pm 1$, thus we may observe the (3/2,1/2), (1/2,-1/2), or the (-1/2,-3/2) transitions. We have measured the Cu^{63} NMR lineshapes corresponding to each of these transitions, for the static field H_0 along the crystal a, b, and c axes, and for both the plane and chain Cu^{63}, for a total of 18 lineshapes. In addition we have measured the Cu^{65} (1/2,-1/2) transition for H_0 along a, b, and c.

The energy splittings of the transitions are determined by the applied field, the magnetic shift tensor, and the electric field gradient at the Cu nucleus. The Hamiltonian for the spin-3/2 system is discussed in detail in Appendix A. From analysis of the line positions we determine the magnetic shift tensor and the electric field gradient tensor.

In addition to measuring the NMR lineshapes for each transition we have also measured the spin-lattice relaxation and the transverse relaxation at temperatures of 100K and 200K. The spin-lattice and transverse relaxation are commonly associated with the familiar "T_1" and "T_2" relaxation times respectively.

For a four-level spin-3/2 system, however, the spin-lattice relaxation consists of a sum of as many as three exponentials. For relaxation from an interaction between a rapidly fluctuating magnetic field and the nuclear magnetic moment this multiple exponential relaxation can be characterized with a single rate parameter which we denote W_1. W_1 is the transition rate from the m=3/2 to the m=1/2 state. The functional forms of spin-lattice

relaxation are discussed in detail in Appendix D. Calculation of W_1 from a fluctuating magnetic field is given in Appendix C.

We have found that the spin-lattice relaxation for both the plane and chain Cu is due to the interaction of the nuclear magnetic moment with a rapidly fluctuating magnetic field; the plane Cu nucleus, for example, experiences an effective magnetic field from its hyperfine interaction with the rapidly fluctuating spin of the Cu d-shell hole. The rapidly fluctuating field will also contribute in a known way to the NMR transverse relaxation. It gives a single exponential transverse decay; thus we may speak meaningfully of a "T_2." The contribution of this mechanism to T_2 is calculated from Redfield Theory [Redfield, 1957; see also Slichter, Chapter 5 (1980)] in Appendix C. We have found, however, that for many transitions we have an extra transverse relaxation mechanism. For H_0 parallel to c, for example, the (1/2,-1/2) transition for the plane Cu has extra relaxation which has a Gaussian form, rather than exponential. We analyze this extra relaxation in detail.

5.2 Lineshapes

Figures 5.1a-m show the NMR lineshapes.

All lineshapes were taken using spin-echoes, specifically the "Add-Subtract, Four Phase" method described in 4.43.

The Cu^{63} and Cu^{65} lineshape for the plane Cu with H_0 along c was taken by pulsing at frequencies 92.8MHz and 99.4MHz respectively. The pulse length for the 90° pulse is 2.5μs, effectively pulsing a frequency width of 200kHz. The linewidth is 25kHz full width at half maximum; thus we easily excite the entire line. The lineshape is then measured by taking the Fourier transform of the spin-echo. Several million echoes were averaged for each isotope. The repetition rate was one echo every 2ms, and the delay time between the 90 and 180° pulses 65μs.

All of the remaining lineshapes are substantially broader, and were taken "point-by-point ; " one pulses at the frequency of the point measured and integrates the spin-echo. Typically three to four hundred thousand echoes were averaged. The echo repetition rate was again one per 2ms. The delay time used was not the same for each lineshape but in all cases was between 30 and 50μs.

The positions of the resonance lines are given in Table 5.1.

5.21 Electric Field Gradient, Magnetic Shift Tensors

From the frequencies in Table 5.1 we have obtained the electric field gradient tensor and the magnetic shift tensor for the plane and chain Cu. We use the Cu nuclear spin Hamiltonian described in Appendix A. The frequencies of the nine Cu^{63} transitions--three transitions for each of the three field orientations, specifies unique values for the tensors. We have found these values by the following procedure: we know the values approximately from analysis using second order perturbation theory. Thus, we try values, diagonalize the Hamiltonian, look at the frequency spectrum, and compare it with that observed experimentally. An analytical diagonalization of the Hamiltonian has been performed by Durand and Barrett [1989]; it is given in Appendix A.

We obtain the values given in Table 5.2, with the magnetic shift given in percent, and the electric field gradient $\nu_{\alpha\alpha} = \frac{eQ}{2h} V_{\alpha\alpha}$ where $V_{\alpha\alpha}$ is the second partial derivative of the potential with respect to the α coordinate.

5.22 Site Assignment of Nuclear Quadrupole Resonance Lines

Prior to the publication of our data a great controversy existed over the assignment of the two Cu Nuclear Quadrupole Resonance Lines. Nuclear Quadrupole Resonance is done in essentially the same way as NMR, except that there is no applied field. The energy levels are split by the interaction of the electric field gradient with the nuclear quadrupole moment. Clearly, with no applied field to give a preferred direction one cannot obtain information about anisotropy from a Nuclear Quadrupole Resonance experiment. Several groups (see 2.53) had reported spin-lattice relaxation times for two Cu NQR lines, one at 31.5MHz, the other at 22.0MHz.

Some groups assigned the 31.5MHz NQR line to the Cu in the chain site, while others, including Warren and Walstedt [1987] assigned it to the plane. The great importance of this issue is that the two sites have very different spin-lattice relaxation behaviour both in the superconducting and the normal states. The 31.5MHz NQR relaxation decreases sharply out as one goes down in temperature, while the 22.0MHz line decreases somewhat less sharply. Warren et al [1987] characterized the freezing out of

the relaxation by effective gaps at the two sites. The 31.5MHz site had a gap corresponding in the BCS theory to a transition temperature of 200K (with $2\Delta/3.5k_BT_c$=200K), while the 22MHz site gap was equivalent to 60K.

The magnetic shift tensor and electric field gradient tensor make the site assignment obvious. First we note that the crystallographic site of Cu in the plane has approximately a four-fold symmetry about the crystal c direction. The chain Cu site also has an approximate four-fold symmetry, but it is a symmetry about the crystal a-axis; not c. Now, we note that the family of NMR lines which we have identified as being from the "plane" site has axial symmetry about c for both its magnetic shift and electric field gradient tensors. Conversely the family of lines which we say are from the chain have a magnetic shift tensor which is approximately four-fold symmetric about the crystal a-axis. (Its electric field gradient tensor is not symmetric about any axis, but its symmetry can be understood in a detailed analysis. See Chapter 6.) Our site assignment of the NMR lineshapes is thus very clear. Then, to associate the NMR resonances with the corresponding NQR resonance we simply calculate the NQR frequency of the two families of NMR lines. The nuclear quadrupole resonance frequency may be calculated from the electric field gradient tensor from the relation:

$$\nu_Q^2 = \nu_{cc}^2 + \frac{(\nu_{aa} - \nu_{bb})^2}{3} \qquad (5.1)$$

(Note that permutation of the indices results in the same result since $\nu_{aa}+\nu_{bb}+\nu_{cc}=0$). This gives an NQR frequency of 31.5MHz for the site we have identified by symmetry to be the plane site. Similarly it gives 22.0MHz for the site identified as the chain site.

5.3 Spin-Lattice Relaxation

We have used the Alternating Phase Inversion Recovery (4.45) pulse sequence to measure the spin-lattice relaxation at 100K and 200K. We invert the population of a transition, for example, the (1/2,-1/2) transition, by applying a 180° pulse. We then wait a time t and measure the population difference of the transition using a spin-echo. We compare this population difference with the value obtained when we use a very long value of t. The difference, $M(t)-M(t=\infty)$ is the deviation of the system from thermal equilibrium. In all measurements we have used a time for $t=\infty$ that is long compared to the longest characteristic relaxation times. For the plane Cu

with H_0 parallel to c, for example, the longest time component is approximately 1.5ms; in that instance we used a $t=\infty$ time of 12ms, giving a completely negligible error. We have typically signal averaged for about 12 or more hours for each relaxation curve, giving five hundred thousand to one million echoes for each point on the curve. In order to eliminate the possibility of a systematic error from "drift" of the apparatus we have averaged each data point throughout the entire averaging period as described in (4.47).

The time dependence of $M(t)-M(\infty)$ for relaxation from a rapidly fluctuating magnetic field is calculated in Appendix D in terms of W_1, the transition probability per unit time from the m=3/2 to the m=1/2 state. We find for the (1/2,-1/2) transition:

$$M(t)-M(\infty) \propto .9e^{-4W1t} + .1e^{-(2/3)W1t} \qquad (5.2)$$

and for the (3/2,1/2) and (-1/2,-3/2) transitions:

$$M(t)-M(\infty) \propto .5e^{-2W1t} + .4e^{-4W1t} + .1e^{-(2/3)W1t} \qquad (5.3)$$

Figure 5.2 shows unequivocally that we have this kind of relaxation behavior. Plotted is the relaxation curve at 100K for the (1/2,-1/2) transition at 92.8MHz, the (3/2,1/2) transition at 124.3MHz, and the (-3/2,-1/2) transition at 61.3MHz, all for the plane Cu with H_0 parallel to c. Each curve is fitted to the appropriate functional form given above with the **same** value of W_1. The fact that we can fit the relaxation in this manner demonstrates two things. First, the relaxation is due to a "magnetic" rather than an "electric quadrupolar" mechanism; that is, the relaxation is due to the interaction of the nuclear magnetic moment with a fluctuating magnetic field, and not due to the interaction of a fluctuating electric field gradient with the nuclear electric quadrupole moment. Second, the magnetic fields giving the relaxation fluctuate on a time scale that is very fast compared to the NMR resonance frequencies (which range from 61 to 124 MHz). Were this not the case then the relaxation of the 61MHz and 124 MHz transitions would be substantially different from one another.

Shown in Figures 5.3a-j are the results of this experiment for the (1/2,-1/2) transitions for the plane and chain Cu measured at 100 and 200K. For each of the T=100K curves shown we have also measured the (3/2,1/2)

and (-1/2,-3/2) transitions as well and found that, as in Figure 5.2, we could fit all three curves with the same W_1 to within experimental error.

From these fits we have extracted values of W_1. They are given in Table 5.3.

In addition to listing values of W_1 we also find it convenient to characterize the relaxation in terms of the magnitude and correlation time of the rapidly fluctuating magnetic field causing the relaxation. We assume a field h with an exponential correlation function with time constant τ_0. Then we show in Appendix C that for H_0 along the z axis W_1 is given by:

$$W_{1z} = (\frac{3}{2})\gamma_n^2[\overline{h_x^2} + \overline{h_y^2}]\tau_0 \qquad (5.4)$$

where $\overline{h_x^2}$ is the mean square value of $h_x(t)$ and γ_n the nuclear gyromagnetic ratio. Measuring $W_{1\alpha}$ for the three orientations α of the magnetic field enables one to deduce the three quantities $\overline{h_\alpha^2}\,\tau_0$ for α=x,y,z. These values as well are given in Table 5.3.

We note that the spin-lattice relaxation for the plane Cu is very anisotropic, with $\overline{h_c^2} / \overline{h_a^2} = 7.9$ at 100K. The relaxation at the chain Cu, although perhaps slightly anisotropic, is isotropic to within experimental error.

5.4 Transverse Relaxation

We have used the Four Phase Add-Subtract Echo Sequence (4.43) to measure transverse relaxation. In the experiment we measure the integral of the spin-echo versus two times the delay time τ. As in the spin-lattice relaxation measurements we have used the technique described in (4.47) to eliminate the effects of "drift" of the apparatus. We have used echo repetition rates that are somewhat slower than those used in the lineshape measurements. In each case we have used repetition rates to insure that at least 99% of equilibrium magnetization has recovered. For the plane Cu, with H_0 along c, for example, we have used a repetition rate of one echo every 5ms.

We report measurements on the (1/2,-1/2) transition for the plane and chain Cu and for the static field along each of the three crystal axes. In each case we have also measured transverse relaxation on the (3/2,1/2) and

(-3/2,-1/2) transitions and found results that are consistent to within experimental error.

Ordinarily in NMR the transverse relaxation arises from one or both of two mechanisms: 1) nuclear spin-spin coupling (via the magnetic dipolar or indirect interaction) or 2) spin-lattice relaxation (the so-called "lifetime broadening"). The first mechanism leads to relaxation curves which have a wide variety of time dependences for which in general it is not possible to find the exact theoretical form. Under certain circumstances the decay has a simple time dependence such as an exponential or a Gaussian. The second mechanism leads to an exponential decay with a time constant related to the spin-lattice relaxation by Redfield theory [1959, see also Slichter, 1980]. We will denote this time constant T_{2R}.

We have seen from our study of the spin-lattice relaxation that the relaxation times could be described in terms of fluctuating fields $h_\alpha(t)$ at the nuclei whose power spectrum was independent of frequency from 61 to 124 MHz (Figure 5.2). Applying Redfield theory to the data and assuming that the frequency independence extends to zero we can deduce the contribution of the h_α to the transverse relaxation. This calculation is performed in Appendix C.

The deduced values of T_{2R} are given in Table 5.4.

We have found in each measurement that in addition to the calculated T_{2R} an additional transverse relaxation is present. In some cases the extra relaxation is also exponential, characterized by a T_2'. In other cases the extra relaxation is Gaussian with a time constant τ. In each case the relaxation can be described by the relation:

$$S(t) = \exp(-t / T_{2R})\exp(-t / T_2')\exp[-(t^2 / 2\tau^2)] \quad (5.5)$$

where S(t) is the NMR signal with t equal to two times the delay time between the 90 and 180° pulses. The measured values of T_2' and τ at temperatures 100K and 200K are given in Table 5.4.

We have performed experiments with the goal of determining the source of the transverse relaxation not accounted for by Redfield theory. We focus first on the plane Cu.

For H_0 parallel to the c-axis the plane Cu relaxation in excess of T_{2R} is Gaussian, characterized by τ=120±7μs at T=100K.

A candidate source for the extra relaxation in this case is a strong coupling of "like" Cu nuclei, where two nuclei are "like" nuclei in an NMR experiment if they are both excited by the rf pulse. In practice two nuclei are "like" if they are separated in resonance frequency by less than the nuclear gyromagnetic ratio multiplied by the amplitude of the rf magnetic field H_1. Typical values of this frequency cutoff in our experiment are γH_1=50-200kHz. The linewidth of the plane Cu^{63} signal is approximately 25kHz; thus, plane Cu nuclei are considered "like." The chain Cu are separated from the plane Cu by a frequency difference of approximately 700kHz; thus, they are "unlike" the plane Cu. Let us briefly explain the concept of coupling of "like" nuclei. If nucleus A and B are "unlike," and we perform a spin echo experiment on nucleus A in which nucleus B is never pulsed, then any coupling between the nuclei will be "refocused" and not observed at the echo position; in terms of the spin-echo on nucleus A the coupling with B appears to be merely a static field, and the classic effect of the spin-echo [Hahn, 1950] is to eliminate the effects of static fields. (Of course, if nucleus B does not maintain a static orientation, then its coupling will be observed, and its effects on the transverse relaxation are in general very difficult to analyze. Such is the case when B nuclei undergo so-called "mutual spin flips" in which a pair of B nuclei, one initially parallel and the other antiparallel to H_0 simultaneously flip, in an energy conserving process.) If, however, "like" nuclei are coupled, then the rf pulses flip both spins, and therefore the coupling of spin B to spin A is not static, and thus is not refocused in a spin-echo.

To test for the presence of spin-spin coupling of "like" plane Cu nuclei we have measured the spin-echo envelope of both Cu^{63} and Cu^{65}. Cu^{63} has an abundance that is 2.2 times greater than that of Cu^{65}. These two isotopes are separated in frequency by about 7MHz--a large separation so that they appear as "unlike" nuclei. The plane Cu^{63} nuclei, then, will only couple to other plane Cu^{63} nuclei. The Cu^{65}, in turn, will only couple to other Cu^{65}'s. Since there are more Cu^{63} nuclei one would expect its echo to decay more rapidly than Cu^{65}. In fact, if one has a Gaussian decay of the echo from this process, then in the limit of many neighbors the Gaussian τ's will be related by:

$$\tau^{65}=\tau^{63}(\gamma^{63}/\gamma^{65})^2\sqrt{P_{63}/P_{65}} \qquad (5.6)$$

where γ is the nuclear gyromagnetic ratio and P is the isotopic abundance. (The abundance dependence can be understood if one notes that the second moment of the NMR lineshape for n equivalent neighbor spins is proportional to n [Slichter, Section 3.3,3.4 1980]. In Chapter 7 we treat our present situation in detail; one has four nearest neighbor spins each with a certain probability, determined by isotopic abundances, to be a "like" spin. The detailed analysis show that the above relation holds to within ten percent).

In Figure 5.4 is shown a fit to the plane Cu^{63} and Cu^{65} spin-echo envelopes in which the τ's have been scaled by the factor given above. This scaling demonstrates convincingly that the extra transverse relaxation results from spin-spin coupling.

We have performed an additional experiment to verify directly the spin-spin coupling. In Figure 5.5 is given the results of a SEDOR experiment (Section 4.46). In the inset is shown the pulse sequence. We perform a spin-echo experiment on Cu^{65}, holding the delay between the 90 and 180 degree pulses fixed. At a time t following the Cu^{65} 90 degree pulse we apply a 180 degree pulse to Cu^{63}. Clearly, if there is no coupling then the Cu^{63} 180 degree pulse will have no effect on the Cu^{65} echo. If, on the other hand, the Gaussian component of the Cu^{63} spin-echo is due to coupling of "like" Cu^{63}'s, then one predicts that our SEDOR experiment will have the time dependence:

$$S(t)=\text{const. x } \exp(-(2t)^2/2\tau^2) \qquad (5.7)$$

τ should be given by $\tau=\tau_{63}$ x $(\gamma^{65}/\gamma^{63})$ and τ_{63} is given in Table 5.4; this results in a value τ=112μs. We observe τ=130μs, in reasonable agreement with experiment.

We conclude then that the extra Gaussian relaxation for the plane Cu with H_0 along c results from a strong nuclear spin-spin coupling. The coupling is an order of magnitude larger than what one obtains from the magnetic dipole-dipole interaction of the nuclei. In the following chapter we interpret this result in terms of the so-called Suhl-Nakamura coupling, in which nuclear spin-spin coupling is mediated by interaction with J-coupled electrons.

In all other cases of extra transverse relaxation we have found that the relaxation does not scale as does a coupling of like spins, with abundance an important factor, but instead scales only with the gyromagnetic ratio. For example for the chain Cu the Gaussian relaxation for H_0 along c scales as:

$$\tau_{65}=(\gamma_{63}/\gamma_{65})^2\tau_{63}. \qquad (5.8)$$

In these cases spin-spin coupling may be important; however, if it is important then mutual spin flips must be occuring which result in this isotopic dependence. In this case there is no exact solution to the relaxation process; thus, with the known presence of the conventional magnetic dipole-dipole coupling, one cannot determine whether or not indirect nuclear spin-spin coupling is present. An exception occurs for the plane Cu with H_0 along a and b. In this case our detailed model of the mechanism of the spin-spin coupling predicts the behaviour that is observed.

TABLE 5.1: Line Positions (MHz)

PLANE Cu H_0 along:	$(1/2,-1/2)_{63}$	$(1/2,-1/2)_{65}$	$(3/2,1/2)_{63}$	$(-3/2,-1/2)_{63}$
a	94.17±.02	100.32±.02	76.37±.05	108.0±.05
b	94.17±.02	100.32±.02	76.77±.05	107.57±.05
c	92.789±.001	99.40±.005	61.3±.05	124.3±.05
CHAIN Cu H_0 along:				
a	93.15±.05	99.7±.05	73.9±.1	111.9±.1
b	92.45±.05	98.95±.05	73.05±.1	111.3±.1
c	93.475±.03	99.775±.03	92.05±.05	92.35±.05

TABLE 5.2 Magnetic Shift and Electric Field Gradient Tensors

	Plane Cu	Chain Cu
K_{aa}(%)	.59±.04	1.38±.07
K_{bb}	.59±.04	.55±.07
K_{cc}	1.267±.001	.60±.04
ν_{aa}(MHz)	15.94±.05**	-19.03±.07*
ν_{bb}	15.56±.05**	19.17±.07*
ν_{cc}	-31.50±.05	-.16±.03

TABLE 5.3 Spin-Lattice Relaxation Rates
($W_{1\alpha}$, α denotes field orientation)

	T=100K	
	Plane Cu	Chain Cu
$W_{1a}(ms^{-1})$	4.7±.3	1.1±.1
W_{1b}	4.7±.3	.9±.1
W_{1c}	1.05±.1	.8±.1
$(\frac{3}{2})\gamma_n^2 \overline{h_a^2} \tau_0(ms^{-1})$	.53±.05	.35±.09
$(\frac{3}{2})\gamma_n^2 \overline{h_b^2} \tau_0$	.53±.05	.50±.09
$(\frac{3}{2})\gamma_n^2 \overline{h_c^2} \tau_0$	4.2±.3	.60±.09
	T=200K	
	Plane Cu	Chain Cu
$W_{1a}(ms^{-1})$	5.6±.3	1.9±.5
W_{1b}	5.6±.3	1.8±.4
W_{1c}	1.4±.2	1.9±.3
$(\frac{3}{2})\gamma_n^2 \overline{h_a^2} \tau_0(ms^{-1})$	.7±.1	.9±.4
$(\frac{3}{2})\gamma_n^2 \overline{h_b^2} \tau_0$	.7±.1	1.0±.4
$(\frac{3}{2})\gamma_n^2 \overline{h_c^2} \tau_0$	4.9±.3	.9±.4

TABLE 5.4 Transverse Relaxation Times Given in μs

Temperature 100K

Case:	Cu^{63} T_{2R}	$Cu^{63}T_2'$	$Cu^{65}T_2'$	τ^{63}	τ^{65}
Cu(2)H_0llc	190±10	a	a	120±7	147±8
Cu(2)H_0lla,b	87±5	550±200	450±300	b	b
Cu(1)H_0llc	440±40	a	a	370±20	300±20
Cu(1)H_0lla	360±30	700±150	700±200	b	b
Cu(1)H_0llb	410±40	a	----------	290±20	---------

Temperature 200K

Case:	Cu^{63} T_{2R}	$Cu^{63}T_2'$	$Cu^{65}T_2'$	τ^{63}	τ^{65}
Cu(2)H_0llc	170±30	a	----------	130±10	---------
Cu(2)H_0lla,b	74±4	----------	----------	---------	---------
Cu(1)H_0llc	210±50	a	a	340±60	---------
Cu(1)H_0lla	210±50	600±300	----------	b	---------
Cu(1)H_0llb	200±50	a	----------	330±90	---------

[a] In these cases the additional relaxation is Gaussian so no T_2' is needed.

[b] In these cases the additional relaxation is exponential so no τ is needed.

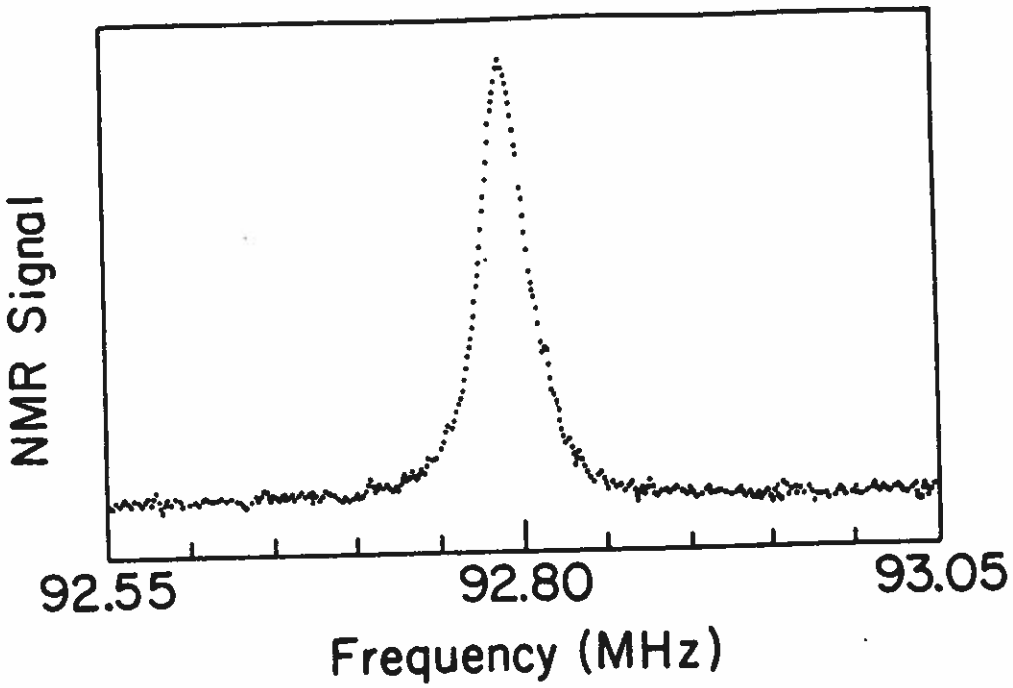


Figure 5.1a: Cu^{63} NMR lineshape of the (1/2,-1/2) transition of the plane Cu with H_0 along the crystal c-axis.

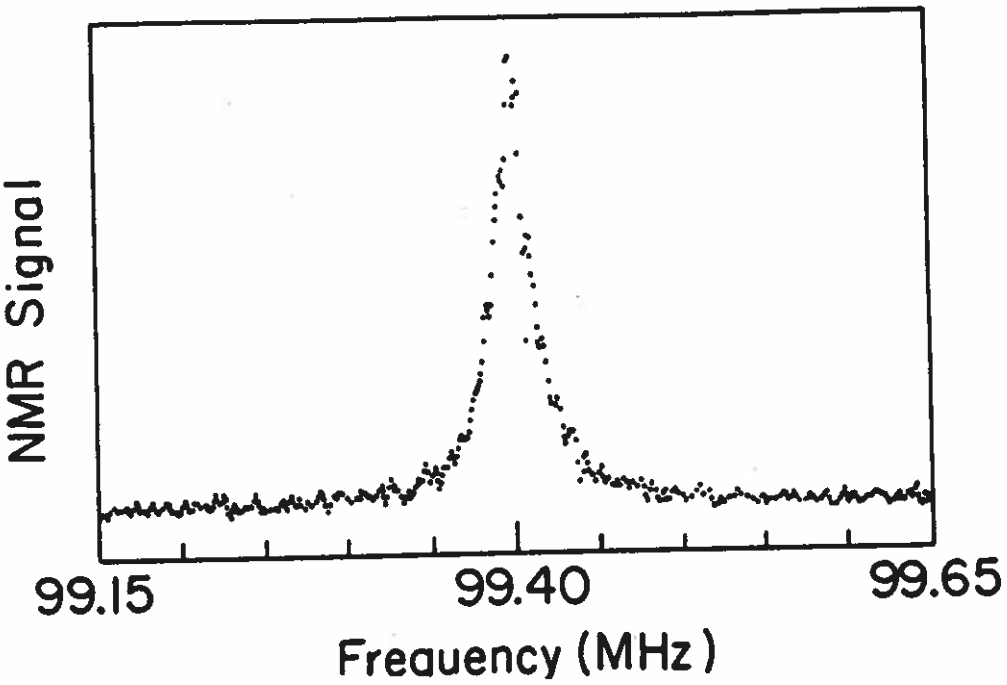


Figure 5.1b: Cu^{65} NMR lineshape of the (1/2,-1/2) transition of the plane Cu with H_0 along the crystal c-axis.

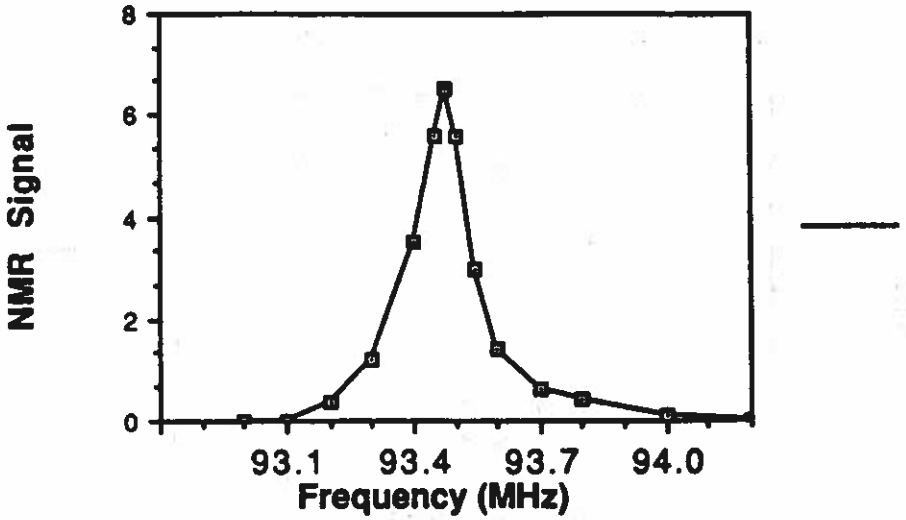


Figure 5.1c Cu^{63} NMR lineshape of the (1/2,-1/2) transition of the chain Cu with H_0 along the crystal c-axis.

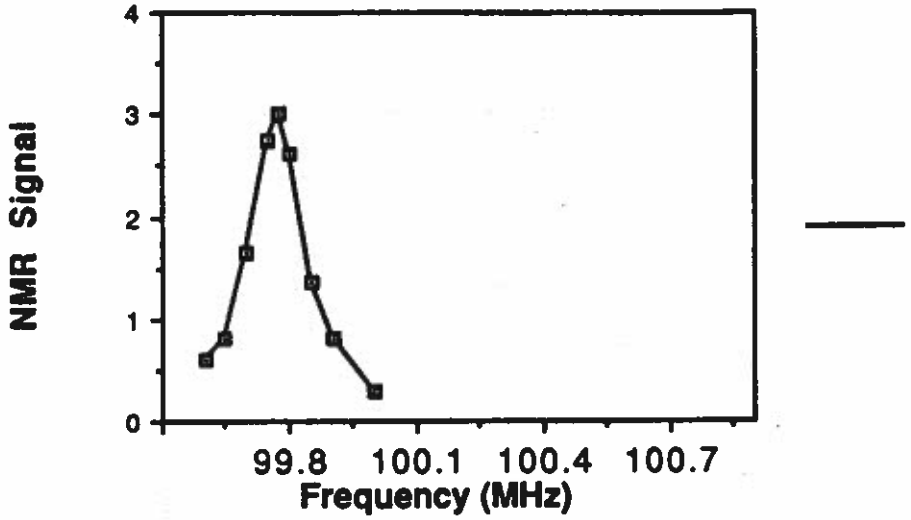


Figure 5.1d Cu^{65} NMR lineshape of the (1/2,-1/2) transition of the chain Cu with H_0 along the crystal c-axis.

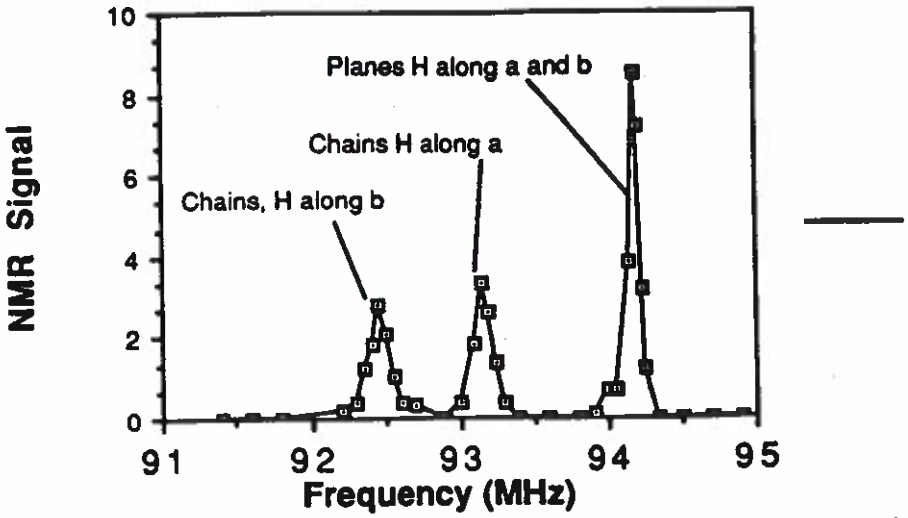


Figure 5.1e Cu^{63} NMR lineshape of the (1/2,-1/2) transitions of the plane and chain Cu for H_0 along the crystal a and b axes. Due to crystal twinning of the a and b axes part of the crystal has H_0 along a and part along b.

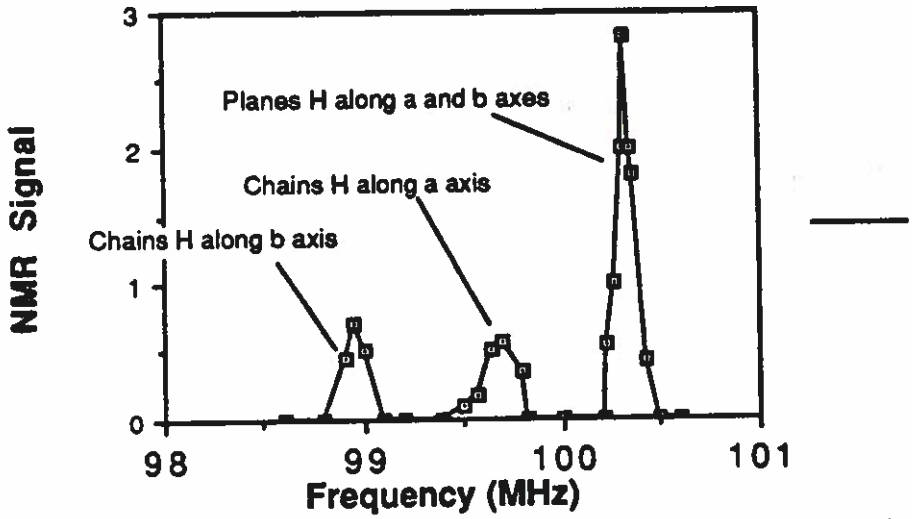


Figure 5.1f Cu^{65} NMR lineshape of the (1/2,-1/2) transitions of the plane and chain Cu for H_0 along the crystal a and b axes. Due to crystal twinning of the a and b axes part of the crystal has H_0 along a and part along b.

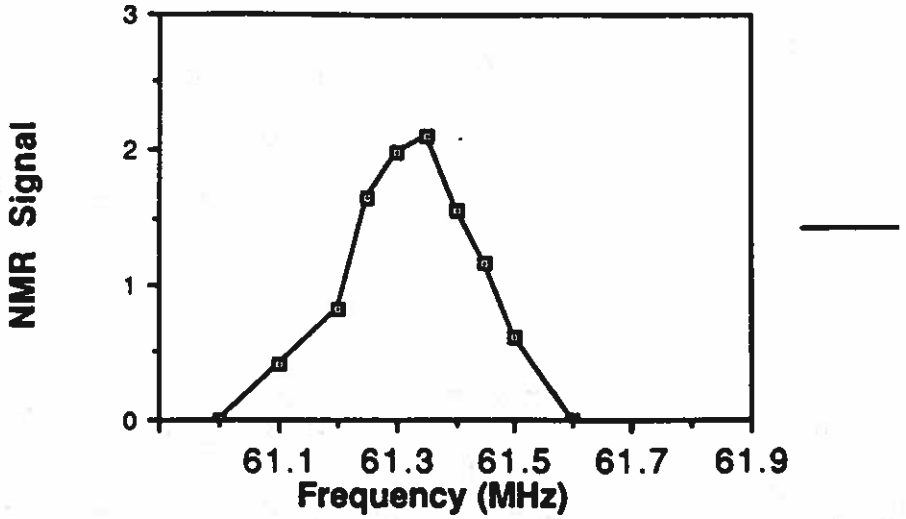


Figure 5.1g Cu^{63} lineshape of the (-1/2,-3/2) transition of the plane Cu with H_0 along the crystal c-axis.

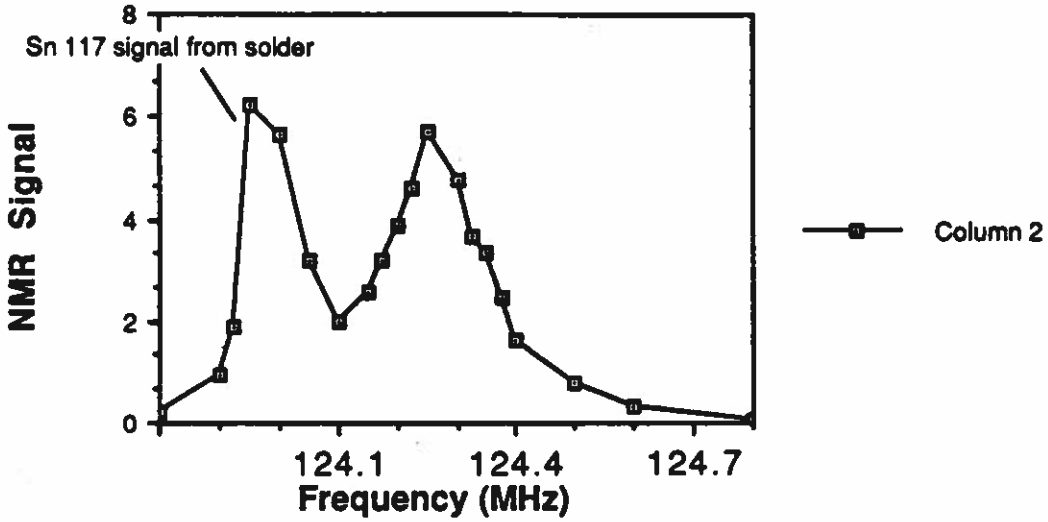


Figure 5.1h Cu^{63} lineshape of the (1/2,3/2) transition of the plane Cu with H_0 along the crystal c-axis. A spurious signal from tin nuclei in solder is also shown.

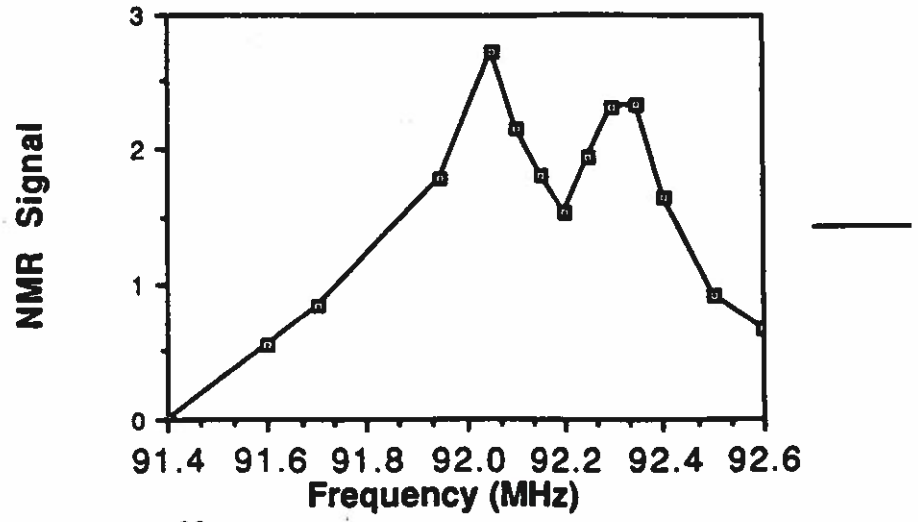


Figure 5.1l Cu^{63} lineshape of the (3/2,1/2) and (-3/2,-1/2) transitions of the chain Cu with H_0 along the crystal c-axis.

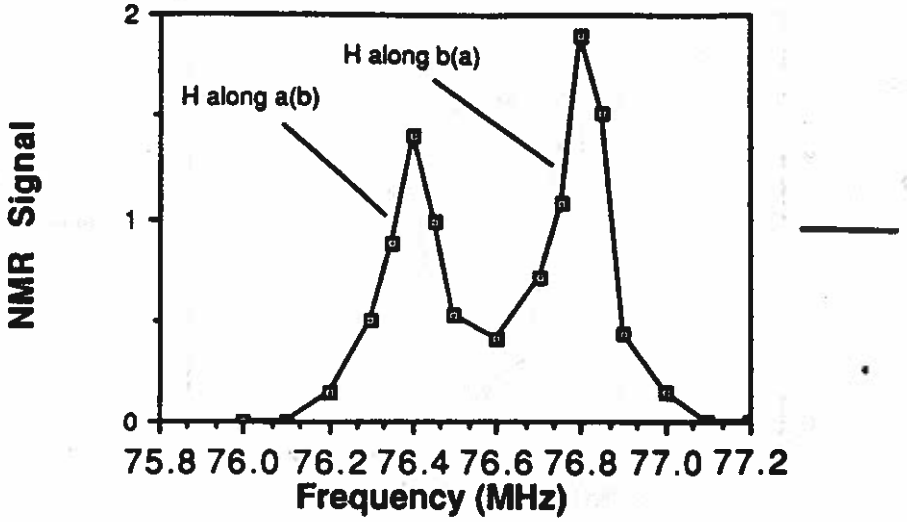


Figure 5.1j Cu^{63} lineshape of the (-3/2,-1/2) transition of the plane Cu with H_0 along the crystal a and b axes. The assignment of field orientation (along a or b) for the two peaks could be reversed. The assignment shown corresponds to that used in our tabulation of the NMR tensors.

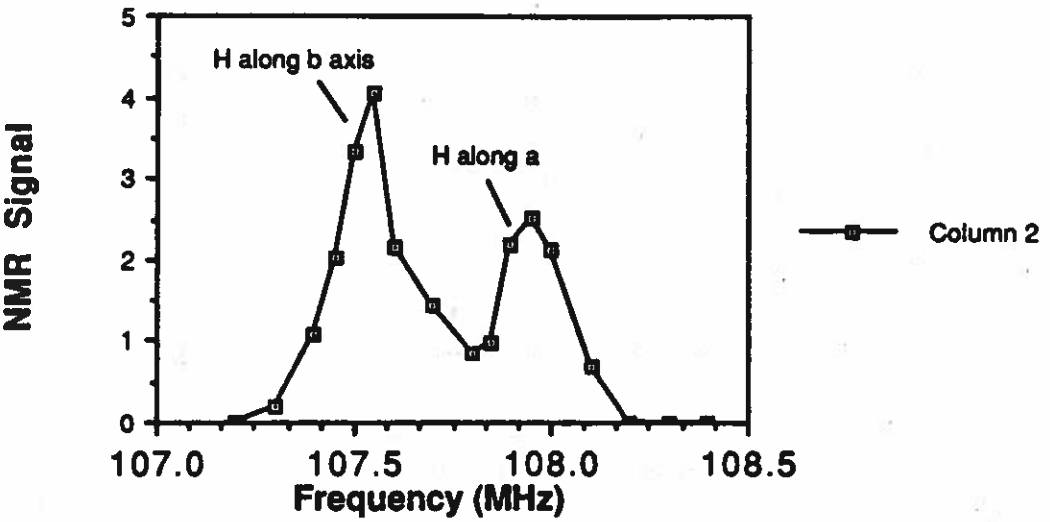


Figure 5.1k Cu^{63} lineshape of the (3/2,1/2) transition of the plane Cu with H_0 along the crystal a and b axes. The assignment of field orientation (along a or b) for the two peaks could be reversed. The assignment shown corresponds to that used in our tabulation of the NMR tensors.

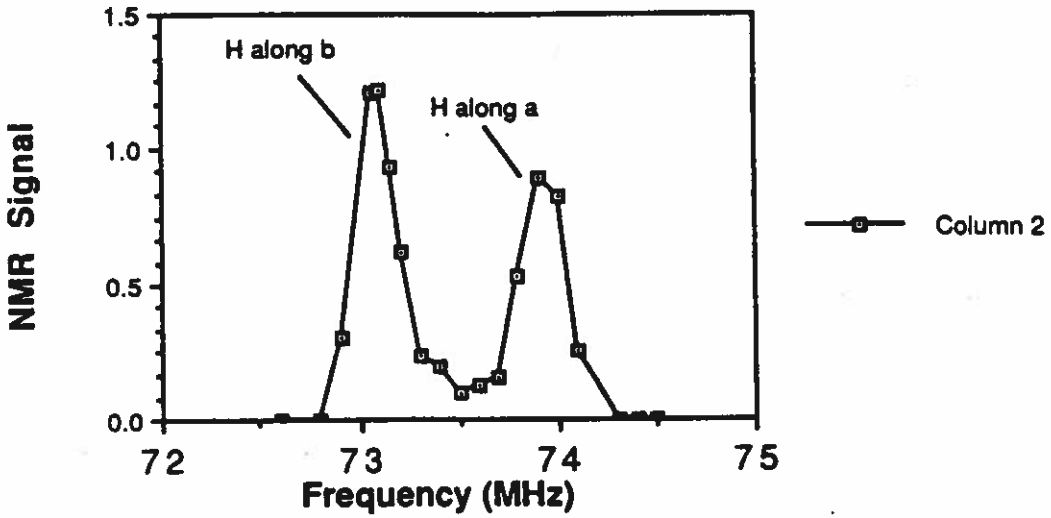


Figure 5.1l Cu^{63} lineshape of the (-3/2,-1/2) transition of the chain Cu with H_0 along the crystal a and b axes. The assignment of field orientation to the two peaks is based upon the symmetry of the magnetic shift tensor.

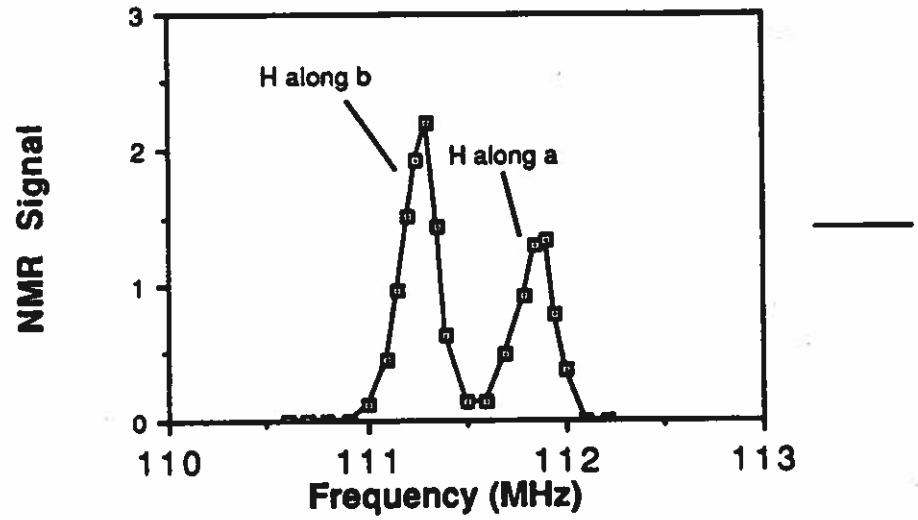


Figure 5.1m Cu^{63} lineshape of the (3/2,1/2) transition of the chain Cu with H_0 along the crystal a and b axes. The assignment of field orientation to the two peaks is based upon the symmetry of the magnetic shift tensor.

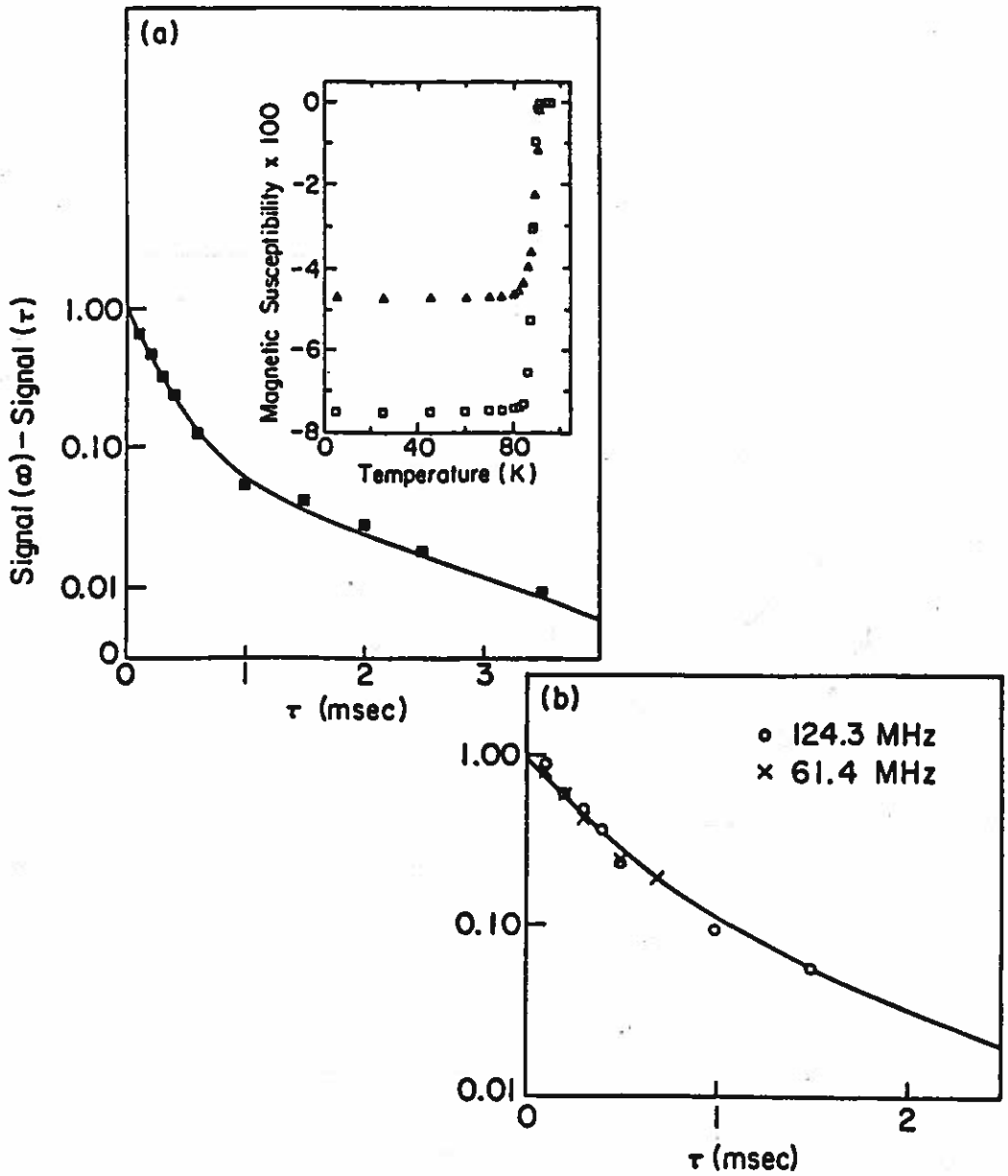


Figure 5.2 inset: the magnetic susceptibility of the crystal vs. temperature showing a sharp superconducting transition at 90K and a complete Meissner effect. Main figure: The spin-lattice relaxation of the plane Cu^{63} for the transitions (a) (1/2,-1/2) at 92.8MHz, (b) (3/2,1/2) and (-3/2,-1/2) at 124 and 61MHz. The points are the data plotted as $\log_{10}[\text{signal}(\infty) - \text{signal}(\tau)]$, where τ is the time after the inversion pulse. The solid line in (a) is a theoretical fit to the data assuming a mangetic relaxation mechanism independent of frequency (see text) where the fundamental rate W_1 has been adjusted to yield the best agreement with the data (W_1=1ms^{-1}). The solid line in (b) is the theoretical prediction for these transitions using the same W_1 as deduced in (a).

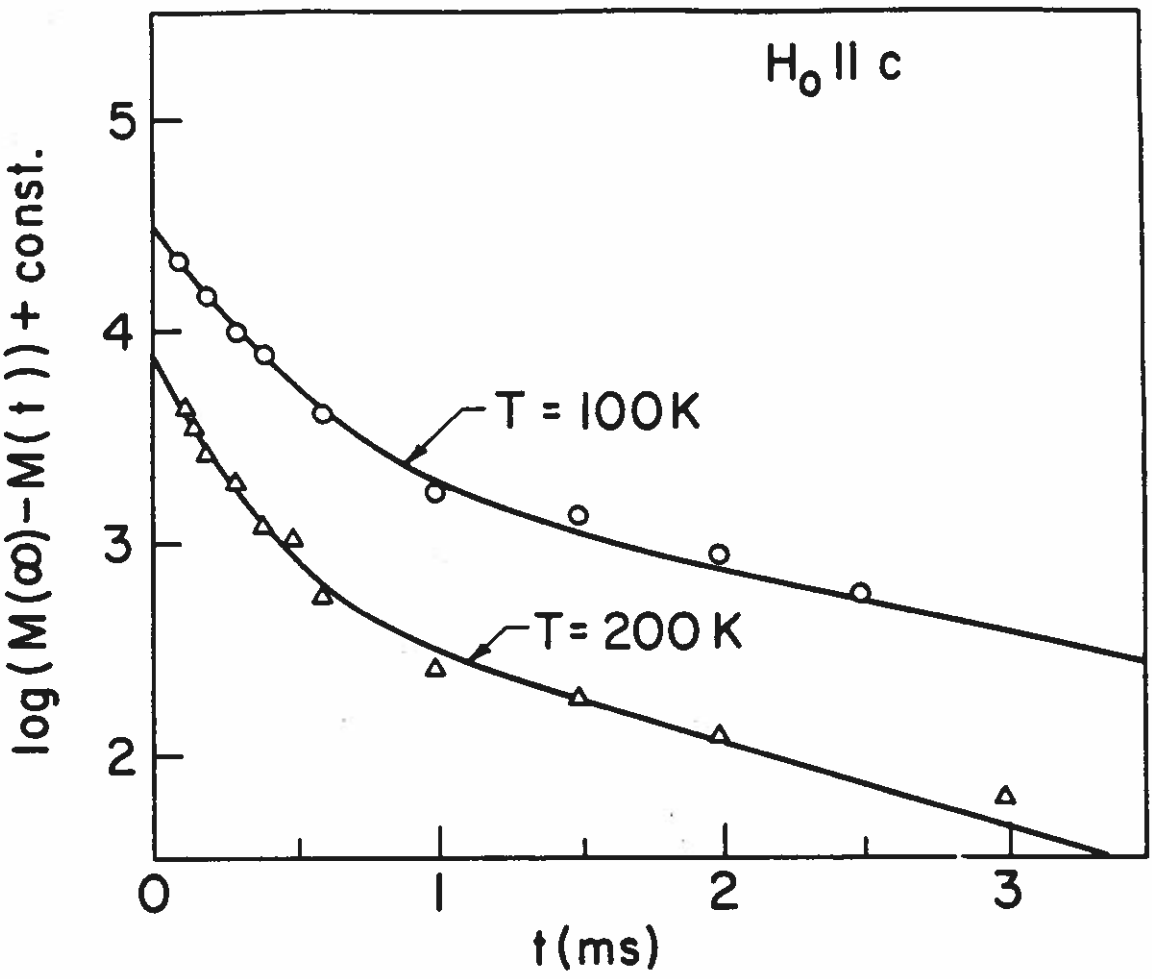


Figure 5.3a The spin-lattice relaxation at temperatures 100K and 200K of the plane Cu^{63} for H_0 along the crystal c-axis. The points are the data plotted as $\log_{10}(\text{signal}(\infty) - \text{signal}(\tau))$ where τ is the time after an inversion pulse. The solid lines are theoretical fits to the data assuming a magnetic relaxation mechanism as in Figure 5.2a. The parameters W_1 used are those given in Table 5.3.

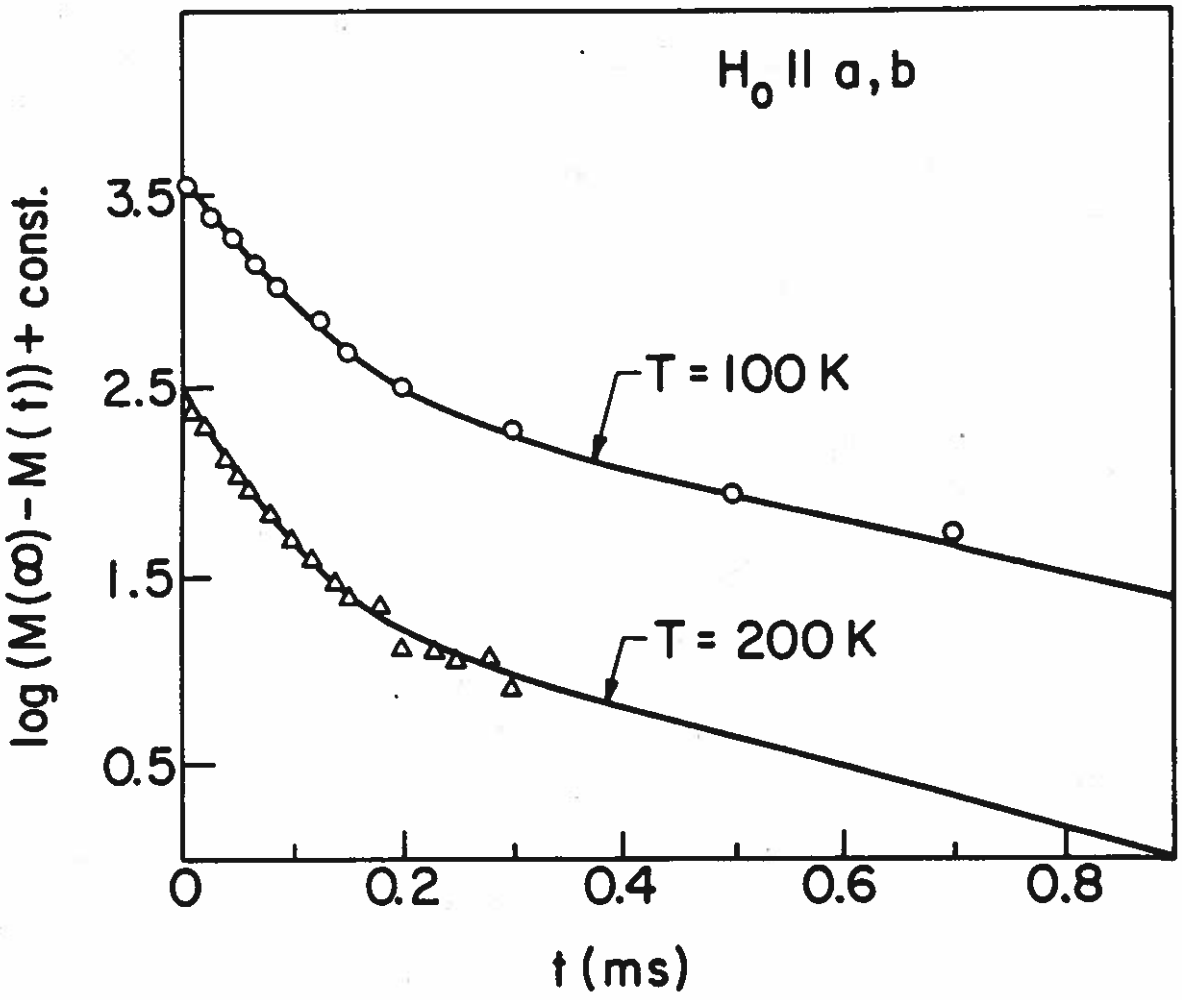


Figure 5.3b The spin-lattice relaxation at temperatures 100K and 200K os the plane Cu^{63} for H_0 along the crystal a/b axis (the NMR lines overlap for these two orientations). The points are the data plotted as $\log_{10}(\text{signal}(\infty) - \text{signal}(\tau))$ where τ is the time after an inversion pulse. The solid lines are theoretical fits to the data assuming a magnetic relaxation mechanism as in Figure 5.2a. The parameters W_1 used are those given in Table 5.3.

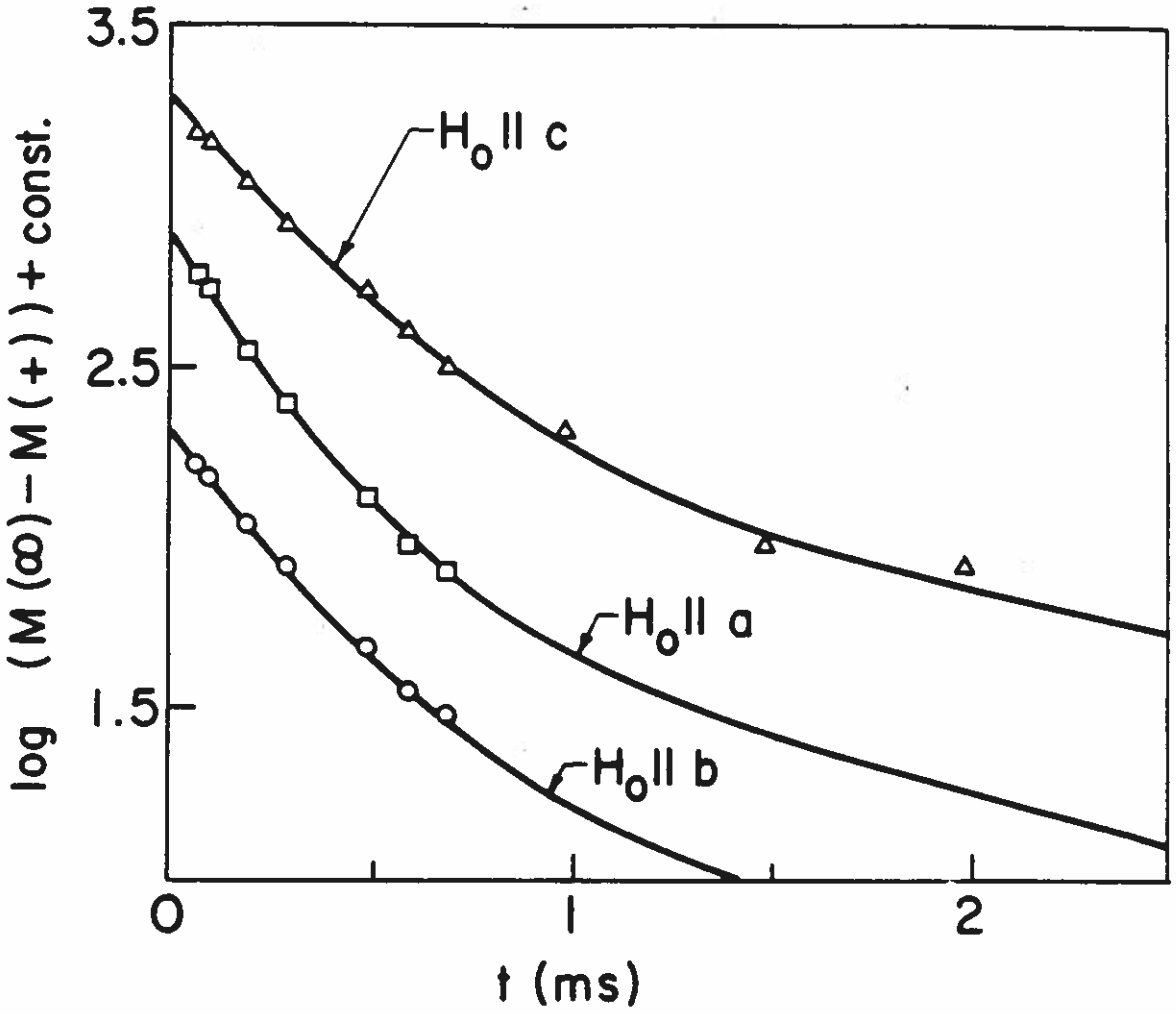


Figure 5.3c The spin-lattice relaxation at temperature 100K of the chain Cu^{63} for H_0 along the a (squares), b (circles), and c (triangles) axes. The points are the data plotted as $\log_{10}(\text{signal}(\infty) - \text{signal}(\tau))$ where τ is the time after an inversion pulse. The solid lines are theoretical fits to the data assuming a magnetic relaxation mechanism as in Figure 5.2a. The parameters W_1 used are those given in Table 5.3.

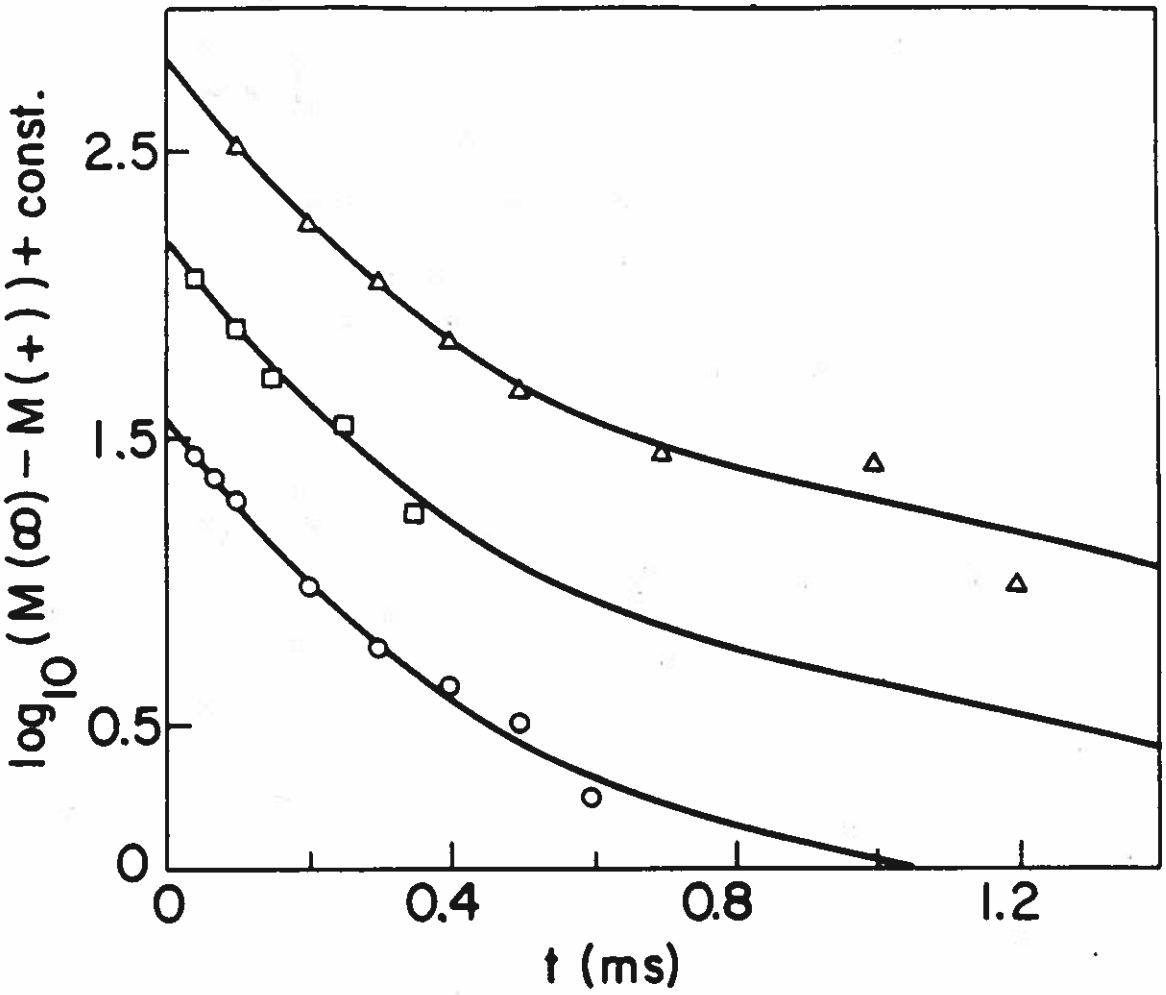


Figure 5.3d The spin-lattice relaxation at temperature 200K of the chain Cu^{63} for H_0 along the a (squares), b (circles), and c (triangles) axes. The points are the data plotted as $\log_{10}(\text{signal}(\infty) - \text{signal}(\tau))$ where τ is the time after an inversion pulse. The solid lines are theoretical fits to the data assuming a magnetic relaxation mechanism as in Figure 5.2a. The parameters W_1 used are those given in Table 5.3.

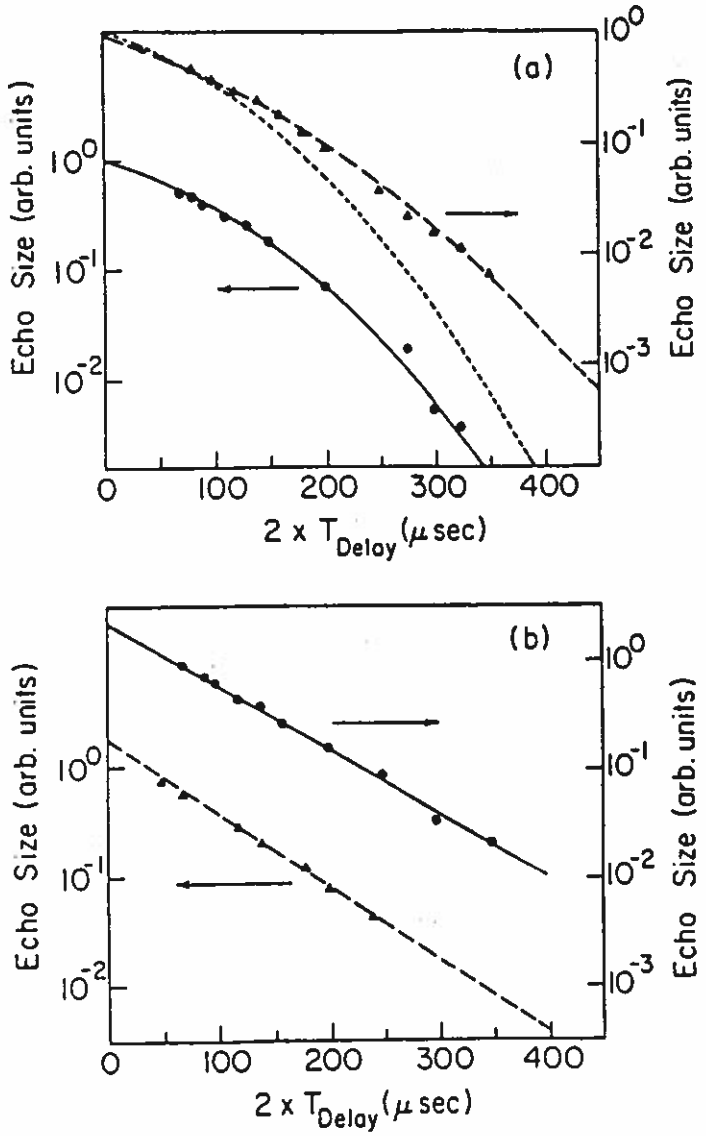


Figure 5.4 The transverse NMR relaxation for both plane Cu^{63} (circles)and Cu^{65} (triangles) for (a) H_0 along c and (b) H_0 along a. Plotted is the size of the spin echo vs $2T_{delay}$, where T_{delay} is the time between the 90 and 180 degree pulses. The solid (dashed) lines are fits to the Cu^{63} (Cu^{65}) data using parameters given in Table 5.4. The time constants used in (b) are related by $T_{2,65}=(\gamma_{63}/\gamma_{65})^2T_{2,63}$. The τ's used in (a) are related by $\tau_{65}=(\gamma_{63}/\gamma_{65})^2(P_{63}/P_{65})^{1/2}\tau_{63}$, where P is the isotopic abundance. For comparison the dotted line in (a) is an attempt to fit the Cu^{65} data by the scaling $\tau_{65}=(\gamma_{63}/\gamma_{65})^2\tau_{63}$.

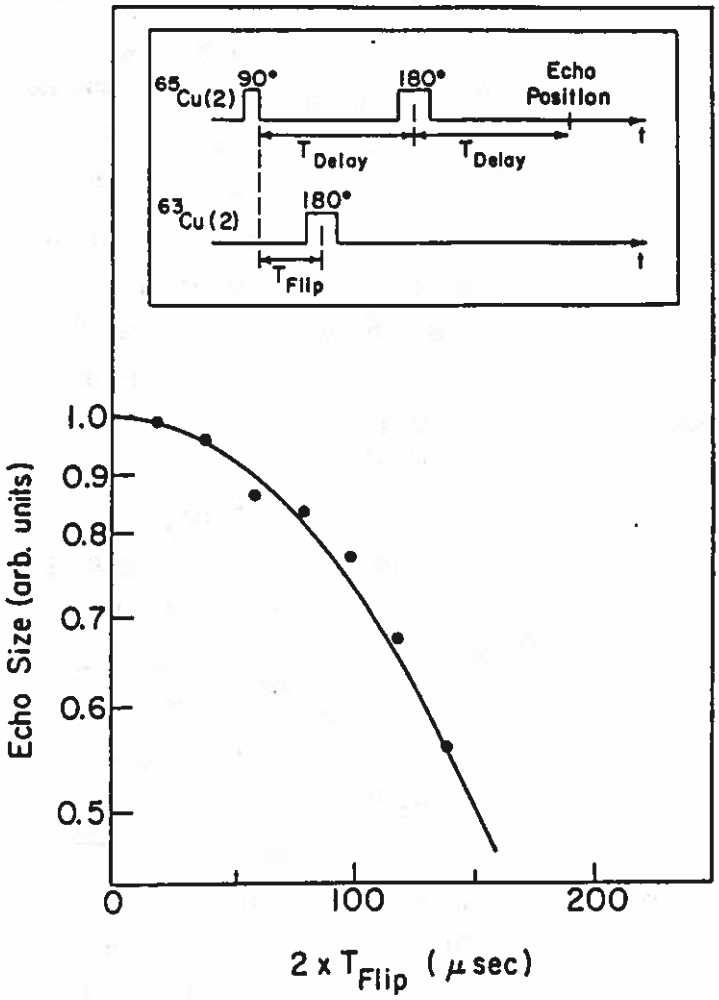


Figure 5.5 The plane Cu^{63}-Cu^{65} SEDOR pulse sequence (inset) and data (main figure). Plotted is the echo size vs. $2T_{flip}$. The solid line is a Gaussian with $\tau_{SEDOR} = 130\mu s$.

Chapter 6: The Valence Charges of the Cu Atoms

An important starting question in the study of high T_c superconductors is the assignment of valence charges to the atoms. In $YBa_2Cu_3O_7$ the assignment of valence charges of +3 and +2 to Y and Ba respectively is noncontroversial; the debate focuses on the valences of the Cu and O atoms. In a formal valence picture one assigns a charge of 2- to the oxygen atoms. This leaves a remaining charge of +7 to be distributed among the two plane Cu's and the one chain Cu. A sensible formal valence picture is to assign a charge of 3+ to the chain Cu and 2+ to the plane Cu. It is known, however, that it is very costly in energy to ionize a Cu^{2+} to Cu^{3+}. Thus, we ask if in fact both plane Cu and chain Cu are Cu^{2+}.

We find that we can explain the electric field gradient tensor of both the plane and chain Cu with a charge assignment of 2+ for both. Additionally we determine the orbital occupied by the single hole in the Cu d-shell at both the plane and chain Cu sites. A similar analysis by Adrian [1988] of our early data [Pennington, (a), 1988] has also appeared in print. Using a slightly different approach he arrives at similar conclusions.

Our approach exploits the detailed understanding of the Cu^{2+} ion which has been obtained in the classic Electron Spin Resonance Studies of Bleaney et al [1955]. There are two important contributions to the electric field gradient. The first is the contribution from lattice ions. Their contribution is enhanced by the so-called "Sternheimer factor" γ_s which takes into account effects of slight distortion of Cu core electron states in the presence of the electric field of the lattice ions; the Sternheimer factor is known to be approximately 10-20 for the Cu^{2+} ion. The second contribution is from any holes in the Cu d-shell orbitals. Of course, a filled d-shell has a spherically symmetric charge distribution, and thus contributes no electric field gradient. Bleaney et al find that a single hole in the Cu d-shell $x^2 - y^2$ orbital gives an electric field gradient (ν_{xx},ν_{yy},ν_{zz}) equal to approximately (-35,-35,70)MHz. We refer to the ν_{zz} contribution as ν_{axial}. A hole in $3z^2-r^2$ gives the same contribution but with the opposite sign.

We calculate the lattice contribution apart from the Sternheimer factor using a computer sum over the lattice. There is not complete consensus on the value of the Cu electric quadrupole moment Q. We use the value of the

quadrupole moment of Cu^{63} given by Bleaney of -.16 (in units of $e \times 10^{-24} cm^2$); this value is in disagreement with that used by many groups (-.211). The latter value is based upon a theoretical calculation by Sternheimer [1967]. We have used the value based on ESR measurements because we feel that this procedure is most likely more reliable than the theoretical approach, and in order to maintain self-consistency with other parameters measured by ESR. We assume that the plane oxygens are O^{2-}, but that there is one extra hole, resulting from the conversion of the chain Cu^{3+} to Cu^{2+}, which resides on the chain and bridge oxygens making them each $O^{-5/3}$. Using this charge configuration we find that the total lattice contribution to the electric field gradient (neglecting the Sternheimer enhancement) is:

$$(\nu_{aa},\nu_{bb},\nu_{cc})_{chains}=(-5.2091,1.7649,3.4442)MHz$$
$$(\nu_{aa},\nu_{bb},\nu_{cc})_{planes}=(2.1897,1.6756,-3.8653)MHz$$

We then take the parameters ν_{axial} and γ_s as unknowns. We take the plane Cu to have a hole in the x^2-y^2 orbital and the chain in y^2-z^2, where we have associated x,y, and z with the crystal axes a,b, and c respectively. We then choose values of ν_{axial} and γ_s to fit the experimental value of ν_{cc} for both the planes and the chains. We find ν_{axial}=71.7MHz and γ_s=10.4, giving for $(\nu_{aa},\nu_{bb},\nu_{cc})$ the following results:

	Calculated	Experimental
Chain Cu	(17.6,-17.6,0)	(19.03,-19.07,0.16)
Plane Cu	(-13.1,-18.4,31.5)	(-15.94,-15.56,31.50)

The calculated results agree reasonably well with experiment. We note that a different valence or orbital would result in a drastically different electric field gradient, since ν_{axial} is typically such a large number. Thus, we conclude that both the plane and chain Cu are Cu^{2+}. The plane(chain) Cu has a single hole in the $x^2-y^2(y^2-z^2)$ orbital.

We have in addition tried other charge configurations for the oxygen ions. For several configurations we give the lattice contribution (without Sternheimer enhancement), the extracted electric field gradient tensor, and the parameters, ν_{axial} and γ_s.

Oxygen:	Charge	γ_s	ν_{axial}(MHz)	EFG's (MHz)
Configuration 1:				
Chain	-1	4.0	46.3	**Lattice:**
Bridge	-2			Chain
(-4.9825,-.8152,5.7977)				
Plane	-2			
	Plane(2.1160,1.5896,-3.7056)			
				Total:
				Chain (26.4,-26.4,0)
				Plane(-14.7,-16.8,31.5)
Configuration 2:				
Chain	-2	53.4	242.3	**Lattice:**
Bridge	-1.5			
	Chain(-5.3224,3.0550,2.2674)			
Plane	-2			
	Plane(2.2266,1.7186,-3.9452)			
				Total:
				Chain (-42.1,42.1,0)
				Plane (-2.2,-29.3,31.5)
Configuration 3:				
Chain	-2	6.0	47.3	**Lattice:**
Bridge	-2			
	Chain(-6.1579,2.2122,3.9457)			
Plane	-1.75			Plane
(1.5396,1.0917,-2.6313)				
				Total:
				Chain (10.4,-10.4,0)
				Plane(-14.4,-17.1,31.5)

Configuration 4:

Chain	-13/7	7.3	54.6	**Lattice:**
Bridge	-13/7			
	Chain(-5.7512,2.0205,3.7308)			
Plane	-13/7			
	Plane(1.8182,1.3419,-3.1601)			

Total:
Chain (12.5,-12.5,0)
Plane (-14.0,-17.5,31.5)

None of these configurations give results that are as reasonable as those obtained by putting the hole evenly on the chain and bridge oxygens. Configuration 1 gives a Sternheimer factor that is too low and a poor fit to the chain electric field gradient. Configuration 2 gives an unreasonable value of ν_{axial}, γ_s, and a poor fit. Configurations 3 and 4 each involve having holes on the plane oxygens. They give values of ν_{axial} and γ_s that are rather low, and additionally they do not fit the chain electric field gradient very well. They do, however, give a good fit for the plane electric field gradient. If we consider the imperfections in this type of calculation--such as the the use of the point-charge model for the lattice contribution, then we probably can not rule out configurations 3 and 4.

As a final check on our model we note that we can use the **same** values of ν_{axial} and γ_s to fit the Nuclear Quadrupole Resonance frequency of $La_{1.85}Ba_{.15}CuO_4$, discussed briefly in 2.61. If we suppose in this material that we have the usual valence assignments for La (3+) and Ba (2+), that the Cu is Cu^{2+}, and that the extra holes are spread evenly on each of the oxygen atoms, then we calculate a nuclear quadrupole resonance frequency of 35MHz for a Cu^{63} that is far from a Ba site. This frequency is near the center of the broad lineshape observed.

Chapter 7: Local Moments and Charge Carriers in $YBa_2Cu_3O_{7-\delta}$

7.1 Introduction

We have seen from our electric field gradient data that plane and chain Cu each have valence Cu^{2+}, with a single hole in the Cu 3d shell. In the formal valence picture, however, the chain Cu has a valence Cu^{3+}. In converting the chain Cu from 3+ to 2+ we are left with an extra hole per unit cell which is associated with oxygen 2p orbitals. It is these two types of holes--those associated with Cu atoms and those associated primarily with oxygen atoms--which dominate the magnetic shift, the spin-lattice relaxation, and the transverse relaxation of the plane and chain Cu. We now present a detailed analysis of the NMR data in order to better understand the nature of these holes.

A previous analysis of our data has appeared in print [Pennington et al (b,c,d), 1989]; Pennington et al analyzed the normal state spin-lattice relaxation and magnetic shift and showed that for the plane Cu the data could be explained using reasonable parameters with the plane Cu atom thought of as being in the permanent moment limit; in their analysis they neglect contributions of oxygen holes to the normal state plane Cu magnetic shift and spin-lattice relaxation. They use parameters obtained from electron spin resonance studies of the Cu^{2+} ion by Bleaney et al [1955]. They further postulated that the relaxation of the chain Cu was due to a transferred hyperfine interaction with oxygen holes in the chain. Takigawa et al [1989] have reported measurements of magnetic shifts in the superconducting state which they suggest demonstrate the presence of a substantial isotropic oxygen hole contribution to the normal state magnetic shift of the plane Cu. Mila et al [1989] suggested that the source of the isotropic shift is a transferred hyperfine coupling of the Cu nuclear spin to the spin of the localized electron moment on neighboring Cu atoms (Section 3.1232). Monien, Pines, and Slichter (MPS) [1989] then presented an analysis of the measurements of Pennington et al and of Takigawa et al. They have made a thorough exploration of the limits of uncertainty that can be placed on various calculated quantities, where the uncertainties are determined from uncertainties in experimental data and by the non-zero range of reasonable values for input parameters determined from ESR studies of the Cu^{2+} ion.

They have utilized NMR data obtained from antiferromagnetic $YBa_2Cu_3O_{6+x}$ and combined it with data on magnetic shifts and spin-lattice relaxation rate anisotropy to show that one can determine hyperfine coupling constants with quite good precision. MPS find that to explain the plane Cu data of Pennington et al and of Takigawa et al requires the presence of a small oxygen hole contribution to the magnetic shift of the plane Cu atoms. They found, however, inconsistencies in their analysis of the chain Cu data. Recent measurements of Barrett et al [1989] in our laboratory have removed these inconsistencies. Barrett has repeated the measurements of Takigawa et al using a more precise technique to determine the magnetic fields present in the mixed state of $YBa_2Cu_3O_{7-\delta}$, and thus to more precisely measure magnetic shifts. The present analysis follows the treatment of MPS. We have, however, extended their analysis by using a computerized iteration procedure to demand self-consistency of all parameters calculated. We are grateful to Hartmut Monien for his help in explaining his approach.

Our analysis will proceed as follows. We first look briefly at ways to describe the Cu 3d shell holes. We consider the two contrasting possibilities of a band theory limit with equal populations of up and down spin and no spin density, and conversely the local moment limit, with each hole localized on a Cu site and having a permanent spin moment.

We then analyze the plane Cu NMR data in a manner somewhat similar to that of our previous analysis, but with the inclusion of the effects of transferred hyperfine coupling and making use of data obtained on the antiferromagnetic system; in this preliminary analysis we will neglect the influence of the oxygen holes on the Cu relaxation and shift, as we have done in our previous analysis. The analysis is self-consistent and demonstrates that the hole in the plane Cu d-shell behaves very much like a localized magnetic moment, rather than moving into an itinerant state best described by band theory. We next incorporate the effects of the oxygen holes into our analysis of the plane Cu relaxation, using the measurements of Barrett et al [1989]; we find that these effects are small--negligible in terms of spin-lattice relaxation, and small but not negligible in terms of the magnetic shift. The plane Cu NMR, then, is mostly a probe of the Cu atom

local moments. We extract parameters for the local moment--its spin and orbital susceptibilty, the hyperfine coupling parameters, and the correlation time for fluctuations of the spin orientation. From transverse relaxation measurements we prove the existence of exchange coupling between the plane Cu atom local moments. We find an effective J for the coupling of approximately 1000cm^{-1}--the same approximate magnitude as that seen in the oxygen depleted $YBa_2Cu_3O_{6+x}$ (950cm^{-1}).

We then proceed with an analysis of the relaxation and magnetic shift of the chain Cu. In sharp contrast to the plane Cu, the chain Cu spin-lattice relaxation is dominated not by the interaction with a localized spin, but by a transferred hyperfine coupling to holes on oxygen p orbitals, giving an isotropic relaxation rate that varies linearly with temperature. We utilize the data of Barrett et al to find the Knight shift associated with this Korringa relaxation, and show that relaxation is not greatly enhanced, indicating that the oxygen holes giving relaxation in the chains may be viewed to a good approximation with a simple band theory picture neglecting electron-electron interactions.

We discuss briefly the nature of the oxygen holes giving relaxation in the chain Cu and explain why the relaxation behaviours of the plane and chain Cu are so very different. We argue that although we have less direct evidence for local moments on the chain Cu, we nonetheless believe that the chain, as well as the plane Cu, are near this limit. We discuss possible reasons for the suppression of spin-lattice relaxation resulting from interactions with the spin of the chain Cu local moments.

7.2 Local Moment vs. Band Theory

Many theories of the mechanism of pairing in high temperature superconductivity ascribe a role to magnetism. This is partly due to the fact that $La_{2-x}Sr_xCuO_4$, for example, is antiferromagnetic, with local moments on the Cu atoms, for the non-superconducting composition x=0; for higher values of x, such as .15 or .2, the material becomes a superconductor, with T_c approximately 30K, yet the local moments remain on the copper atoms [Shirane, 1987; Birgenau, 1988]. Magnetism is known to disrupt conventional BCS superconductivity, thus its presence in high temperature superconductors suggests that it may play a role in a new pairing

mechanism. It is also well known that $YBa_2Cu_3O_{6+x}$ with $x<<1$ is an antiferromagnet, with moments on the plane Cu atoms. The evidence is not as clear, however, for superconducting $YBa_2Cu_3O_7$ with T_c=90K, since neutron scattering studies of the tiny crystals available are not feasible. NMR has been a most important tool in the study of this question.

Descriptions based on the local moment picture and on the band theory picture are in fact just limits on a continuous spectrum. This spectrum of possibilities has been studied and described extensively by Moriya [1979].

7.21 Nuclear-Electron Spin Hamiltonian for Local Moment

Let us discuss, then, ways in which localized moments interact with nuclear spins. We shall consider a local moment in an $x^2 - y^2$ orbital. The coupling parameters for this situation have been described in detail by Bleaney et al [1955] in their studies of the electron spin resonance of the Cu^{2+} ion. The Hamiltonian describing the interaction of the nuclear spin **I** with the localized electron spin **S** is given by:

$$H = \sum_{\alpha} I_\alpha A_{\alpha\alpha} S_\alpha \qquad (7.1)$$

where the hyperfine coupling tensor is given by:

$$A_{zz}/\gamma_n\gamma_e\hbar^2 = \left\langle \frac{1}{r^3} \right\rangle \left[-\kappa - \frac{4}{7} - \lambda\left(\frac{6}{7}\frac{1}{E_{xz}} + \frac{8}{E_{xy}}\right)\right] \qquad (7.2a)$$

$$A_{xx}/\gamma_n\gamma_e\hbar^2 = A_{yy}/\gamma_n\gamma_e\hbar^2 = \left\langle \frac{1}{r^3} \right\rangle \left[-\kappa + \frac{2}{7} - \lambda\frac{11}{7}\frac{1}{E_{xz}}\right] \qquad (7.2b)$$

where λ is the spin orbit coupling parameter, κ the core polarization, $\left\langle \frac{1}{r^3} \right\rangle$ the expectation value of $\frac{1}{r^3}$, E_{xy} and E_{xz} the energies to the excited orbital states xy and xz (with $x^2 - y^2$ taken to be zero) , and γ_n and γ_e the nuclear and electron gyromagnetic ratios. In this thesis we will routinely refer to the hyperfine coupling tensor in units of magnetic field; this quantity is in fact $A_{\alpha\alpha}/\gamma_n\hbar$, but we will refer to it as $A_{\alpha\alpha}$ for convenience. The effective magnetic field h_α produced at the nuclear spin is then given by:

$$h_\alpha = A_{\alpha\alpha} S_\alpha \qquad (7.3)$$

Typical values of the parameters in these equations are known from the work of Bleaney et al [1955]. The core polarization κ ranges from approximately .25 to .32. λ is -826 cm^{-1} in the free atom, typically reduced

to -710cm^{-1} (0.088eV) in a solid. $\langle \frac{1}{r^3} \rangle$ takes on a value 7.5 atomic units in the free atom, typically reduced to 6.3 in a solid.

For typical values of the parameters, λ=-710cm^{-1}, κ=.28, $\langle \frac{1}{r^3} \rangle$=6.3, and $E_{xy}=E_{xz}=2eV$ one finds $A_{zz}/\gamma_n\hbar$ and $A_{xx}/\gamma_n\hbar$ equal to -364kG and 59kG respectively; the values are very anisotropic, with A_{zz} negative and much larger in magnitude.

In addition to the coupling of the nuclear spin to the electron at the same site there is also a coupling to the neighboring electron spins of the form:

$$H = B_1 I_\alpha \sum_{\alpha n} S_{\alpha n} \qquad (7.4)$$

where n indicates a sum over the nearest neighbor electron spins. Again, we will use a notation in which B_1 is equal the B_1 defined above divided by $\gamma_n\hbar$. Mila and Rice [1989] estimate B_1= 90kG. Our data indicate a value closer to 60kG--in reasonable agreement considering the type of calculation involved in Mila and Rice's estimate.

7.22 Spin-Lattice Relaxation from Nuclear-Electron Spin Interactions

In Appendix C we have calculated the spin-lattice relaxation rate W_1 from the interaction of the nuclear spin with a randomly fluctuating magnetic field $h_\alpha(t)$, where h_α fluctuates with an exponential correlation time τ_0.

We now apply this model to spin-lattice relaxation from nuclear spin interactions with localized electrons. One expects the electron spins to be exchange coupled to one another, causing the spin orientations to fluctuate rapidly. (Other mechanisms may contribute to the fluctuation rate of the electron spins.). We shall suppose that these fluctuations have an exponential correlation function with time constant τ_0.

With the z axis defined with respect to the $x^2 - y^2$ orbital we define $W_{1\parallel}$ to be the relaxation rate (defined in Appendix C) with the static magnetic field applied along the z-axis, and $W_{1\perp}$ the rate with the field along x or y. $h_{\parallel(\perp)}$ is then the component of the fluctuating field in the z (x) direction. We then have the relations:

$$W_{1\parallel} = \frac{3}{2}\gamma_n^2(2\overline{h_\perp^{\,2}})\tau_0 \qquad (7.5a)$$

$$W_{1\perp} = \frac{3}{2}\gamma_n^2(\overline{h_\perp^2} + \overline{h_\parallel^2})\tau_0 \qquad (7.5b)$$

If we assume that all electron spins are uncorrelated, and that there are four neighboring spins giving transferred hyperfine coupling, then we obtain:

$$\overline{h_{\parallel(\perp)}^2} = A_{\parallel(\perp)}^2\,\overline{S_{\parallel(\perp)}^2} + B_1^2\sum_n \overline{S_{\parallel(\perp)}^2} \qquad (7.6a)$$

$$\overline{S_{\parallel(\perp)}^2} = \frac{1}{4} \qquad (7.6b)$$

$$\overline{h_{\parallel(\perp)}^2} = \frac{A_{\parallel(\perp)}^2}{4} + B_1^2 \qquad (7.6c)$$

We may then write the spin lattice relaxation rate anisotropy $W_{1\perp} / W_{1\parallel}$:

$$W_{1\perp} / W_{1\parallel} = \frac{A_\parallel^2 + A_\perp^2 + 8B_1^2}{2A_\perp^2 + 8B_1^2} \qquad (7.7)$$

7.23 Magnetic Shift from Nuclear-Electron Spin Interactions

We have seen that the anisotropic hyperfine tensor describes very large effective magnetic fields per unit of electron spin at the Cu nucleus. Were these fields static they would drastically shift the NMR resonance frequency. Such is the case in an ferro or antiferromagnet, and in fact one may perform nuclear magnetic resonance experiments using the hyperfine field in place of an applied external field. In $YBa_2Cu_3O_7$, however, the fields are far from static, and fluctuate with characteristic times of 10^{-15} seconds, a time much faster than the precession time of the nucleus in the hyperfine field. This is precisely the condition for "motional narrowing," and thus, the large fields do not broaden the resonance. In the presence of a large external field, however, the electron spins do cause a shift in the resonance lines; this effect is due to their non-zero spin susceptibility. Each electron, though it may fluctuate rapidly between an up and down spin state, spends slightly more time with its magnetic moment parallel to the static field. As a result the NMR line position is shifted. The fractional shift is given by:

$$K^s_{\parallel(\perp)} = (A_{\parallel(\perp)} + 4B_1)\chi^s \qquad (7.8)$$

where we have assumed that the susceptibility of the localized spins is isotropic.

7.24 Magnetic Shift from Orbital Susceptibility

In addition to the NMR shift resulting from interactions of the nucleus with the spin of the local moment, the resonance line is also shifted from the

orbital susceptibility of the local hole--the so-called "chemical shift." This effect is described in detail in Appendix B. The shift is given by:

$$K^l_{\parallel} = 16\beta^2 \left\langle \frac{1}{r^3} \right\rangle \frac{1}{E_{xy}} = 2 \left\langle \frac{1}{r^3} \right\rangle \chi^l_{\parallel} \qquad (7.9a)$$

$$K^l_{\perp} = 4\beta^2 \left\langle \frac{1}{r^3} \right\rangle \frac{1}{E_{xz}} = 2 \left\langle \frac{1}{r^3} \right\rangle \chi^l_{\perp} \qquad (7.9b)$$

We note that the chemical shift, like the anisotropic hyperfine coupling tensor, contains the factor $\left\langle \frac{1}{r^3} \right\rangle$ and E_{xy}.

7.25 Antiferromagnetic Nuclear Magnetic Resonance

Valuable information on the hyperfine coupling tensor has been obtained by Yasuoka et al [1988]. They have made NMR measurements on antiferromagnetic $YBa_2Cu_3O_6$. In this material the Cu spins have an approximately static orientation pointing in the plane, with an effective magnetic moment of $.6\mu_B$, where μ_B is the Bohr magneton. The resonance frequency in zero applied field implied a hyperfine field of some 160kG. From equations (7.1) and (7.4) this implies:

$$(A_{\perp} - 4B_1)(\mu_{eff} / \mu_B) = -160kG \qquad (7.10)$$

7.3 The NMR Tensors of the Plane Cu

7.31 Analysis Using Only Data Obtained in the Normal State; Neglecting Effects of Oxygen Holes

We now analyze the magnetic shift tensor, the spin-lattice relaxation, and the transverse relaxation using the local moment picture for the holes in the Cu d-shell orbital. In a preliminary analysis we neglect the effects of oxygen holes.

The goal of this analysis is to explain the data self-consistently and with reasonable parameters, and with parameters consistent with the antiferromagnetic NMR data of Yasuoka et al (Equation 7.10).

In our analysis we ask both for typical values of numbers calculated and for the limits of error which are determined by experimental error and by the non-zero range of reasonable values for certain input parameters. For economy of presentation we have not included the variation with absolutely every parameter; we have, however, seen that these variations do not effect our major conclusions. To explore the limits that can be placed on the calculated quantities in this system is the idea of Hartmut Monien [1989]. He has shown from this kind of analysis that a small oxygen hole contribution is

needed in order to account for both our data and the measurements by Takigawa et al [1989] and Barrett et al [1989] of Knight shifts in the superconducting state. We have extended his analysis by a computerized iteration procedure in which we demand self-consistency of all numbers calculated.

We use Yasuoka's result as a convenient starting point, giving:

$$4B_1 = A_\perp + 160kG/\mu_{eff} \qquad (7.11)$$

This result can be substituted into equation (7.7), giving a second order equation involving the spin-lattice relaxation rate anisotropy $W_{1\perp} / W_{1\parallel}$, with a best experimental value of 4.48 (Table 5.3), $A_\parallel$, and $A_\perp$. We then choose reasonable values of κ, λ, E_{xy}, and E_{xz} and determine the ratio $A_\perp/A_\parallel$, defined as α. Equation 7.2 gives:

$$\alpha = \frac{-\kappa + \frac{2}{7} - \lambda \frac{1}{7}\frac{1}{E_{xz}}}{-\kappa - \frac{4}{7} - \lambda\left(\frac{6}{7}\frac{1}{Exz} + \frac{8}{E_{xy}}\right)} \qquad (7.12)$$

Using this relation one may then solve for $A_\parallel$, $A_\perp$, and B_1. With the assumed values of κ, λ, and E_{xy} and E_{xz} one may then find $\left\langle \frac{1}{r^3} \right\rangle$ from relation (7.2).

The anisotropy of the spin susceptibility contribution to the shift may be determined from equation (7.8). We have:

$$\frac{K^s_\parallel}{K^s_\perp} = \frac{(A_\parallel + 4B_1)}{(A_\perp + 4B_1)} \qquad (7.13)$$

If we assume a ratio E_{xz} / E_{xy}, then the chemical shift anisotropy is determined from (7.9):

$$K^l_\parallel / K^l_\perp = 4E_{xz} / E_{xy} \qquad (7.14)$$

Neglecting the effects of oxygen holes, the total magnetic shift (Table 5.2) K is given by:

$$K_\parallel = K^s_\parallel + K^l_\parallel \qquad (7.15a)$$

$$K_\perp = K^s_\perp + K^l_\perp \qquad (7.15b)$$

At this point one has calculated $\left\langle \frac{1}{r^3} \right\rangle$, and one knows the chemical shift.

Then from equation (7.9) one can calculate the energies $E_{xy,xz}$. They will presumably be different from those chosen initially. Thus, one repeats the calculation with the new energy values, and iterates until the energy converges.

With the new self-consistent values of the hyperfine tensors, $\langle \frac{1}{r^3} \rangle$, $E_{xy,yz}$, one may calculate various other quantities of interest. A schematic diagram for this method of self-consistent calculation is given in Figure 7.1.

We have carried through this calculation, using our experimental numbers $W_{1a}=4.7ms^{-1}$, $W_{1c}= 1.05ms^{-1}$, $K_{cc}=1.267\%$, $K_{aa}=0.59\%$, a typical spin-orbit interaction $\lambda=-0.088eV$, neutron scattering estimates of the magnetic moment of Cu in antiferromagneticYBa2Cu3O6+x, $\mu=.6\mu_B$, and an assumed ratio $E_{xz} / E_{xy}=1$. We have examined values of κ in the range of .22 to .36; typical values are between .25 and .32. Of course, one may also vary λ; we have tried several values of λ in a reasonable range. The λ variation changes the numbers slightly, but does not effect our major conclusions. Shown in Figure 7.2a are the values determined for $\langle \frac{1}{r^3} \rangle$ and $E_{xy,xz}$. The iteration procedure does not converge for $\kappa<.25$. We find for $.26<\kappa<.36$ that $\langle \frac{1}{r^3} \rangle$ varies between 5 and 7 atomic units, and E ranges between 2 and 3 eV. At $\kappa=.28$ we have $\langle \frac{1}{r^3} \rangle=6.1$ and $E_{xy}=2.6eV$. These numbers are quite reasonable; typical values of $\langle \frac{1}{r^3} \rangle$ in the Cu^{2+} ion in a solid are approximately 6. McMahan et al [1988] have estimated the energies E_{xy} and E_{xz} as 2 and 2.2 eV respectively. The anisotropic hyperfine tensor parameters are given in Figure 7.2b. A_{cc} (or $A_{\|}$) ranges from -500kG to -350kG. A_{aa} (or $A_{\perp}$) is much smaller in magnitude ranging from 60kG to -5kG, and B_1 stays in a narrow range from 65 to 82kG. Typical values at $\kappa=.28$ are $A_{\|}=-429kG$, $A_{\perp}=44kG$, and $B_1=78kG$.

From relation 7.9 we determine the orbital, or chemical shift contribution. This is plotted in Figure 7.2c; K^l_{cc} and K^l_{aa} stay nearly constant as a function of κ, taking on values of 1.35% and .34% at $\kappa=.28$. The spin shift (Figure 7.2d) also stays nearly constant, with $K^s_{cc}= -.084\%$ and $K^s_{aa}=.25\%$ at $\kappa=.28$.

From relations (7.8) and (7.9) one may obtain the orbital and spin contributions to susceptibility. Shown in Figure 7.2e are χ^l_{cc},χ^l_{aa},χ^s, and the powder average of these quantities, $\chi^{tot}=\frac{1}{3}(\chi^l_{cc}+2\chi^l_{aa})+\chi^s$. We find $\chi^{tot}=21\times10^{-29}$emu per plane Cu atom. One may compare this number with

measurements of Junod et al [1988] which indicate a total paramagnetic susceptiblility of 35×10^{-29}emu per Cu atom; their measurement, of course, averages over the plane and chain Cu atoms and includes contributions from oxygen holes.

Finally from relation 7.5 and 7.6 we may calculate τ_0, the exponential correlation time of the spin orientation of the Cu local moment. We find, for $\kappa = .28$, $\tau_0 = 1.1 \times 10^{-15}$sec. We may compare this time with that which would be given from precession of the electron spins in the exchange fields of their neighbors. Horvatic et al [1989] give this time as $\frac{\sqrt{\pi}}{4}\frac{\hbar}{J}$ where J is the exchange coupling. With J estimated at 950cm^{-1} from light scattering experiments on $YBa_2Cu_3O_{6+x}$ [Lyons et al, (1988)] we find $\tau_{exchange} = 2.4 \times 10^{-15}$. The time is of the same order of magnitude, though perhaps by coincidence, since our system is not in the antiferromagnetic state, and hence one expects to have something similar to a spin-wave excitation spectrum.

7.32 Analysis Using Both Normal and Superconducting State Data, with Inclusion of the Effects of Oxygen Holes

Infomation highly relevant to the analysis of the Cu NMR data has been obtained by Takigawa et al [1989] and in our laboratory by Sean Barrett et al [1989]. They have measured the magnetic shift tensor in oriented powder samples in the superconducting state. In BCS theory one expects electron spins to "pair," and thus the Knight shift should vanish. The orbital, or chemical shift does not depend on electron spin, and should thus remain constant in the superconducting state. At zero temperature, then, the magnetic shift will be equal to the chemical shift. The change in the magnetic shift as one increases temperature from zero to above the transition temperature will be equal to the Knight shift. Of course, in $YBa_2Cu_3O_{7-\delta}$ one has, in addition to a Knight shift and an orbital shift, the spin susceptibility shift due to local moments on the Cu atoms--the term referred to as $K^s_{\alpha\alpha}$ in the above analysis. In the normal state the copper spins are well described as local moments, yet in the superconducting state very convincing evidence exists that these local moments must pair in some

way. Indeed, the data of Barrett et al is most easily understood if one accepts this picture.

The aforementioned change in the magnetic shift as one increases temperature from zero into the normal state will be referred to as the "spin shift," and it is in fact the sum of the shift due to local moments on the Cu atoms and the Knight shift from itinerant holes in oxygen p-orbitals. The shift and relaxation rate mechanism from the oxygen p-orbitals, the so-called "transferred hyperfine coupling I," is discussed in Section [3.1221]. Both the shift and relaxation rate contributions from this mechanism are isotropic, and are related by the Korringa relation:

$$\frac{3T}{2W_1}(K_h)^2 = \left(\frac{\hbar}{4\pi k_B}\right)\frac{\gamma_e^2}{\gamma_n^2} \qquad (7.16)$$

if one may neglect electron-electron interactions for the oxygen hole band.

Using the data of Barrett et al we may then determine the quantities $K^l_{\alpha\alpha}$ and $K^s_{\alpha\alpha}$. They find for the plane Cu K^l_{cc}=1.267%, K^s_{cc}=0, K^l_{aa}=.24%, and K^s_{aa}=.36%. These values are qualitatively similar to the results found in Section 7.31, in which we neglect the effects of oxygen holes. Yet, the numbers disagree, and we may obtain detailed information by a re-analysis of the data with these facts incorporated.

We first note that the chemical shift anisotropy K^l_{cc}/K^l_{aa} is not equal to 4, as would be the case if $E_{xy}=E_{xz}$ (see Equation [7.9] and [7.14]). The anisotropy of the chemical shift indicates an energy ratio E_{xz}/E_{xy}=1.32. We shall incorporate this ratio into our new analysis; the analysis described in section 7.31 preserves this energy ratio, thus, we will merely use starting values of E_{xy} and E_{xz} which satisfy this condition.

Our new analysis then proceeds as follows. Again we select reasonable values of κ and λ(-0.088eV). We select initial energies E_{xy} and E_{xz} of 3 and 3.96eV respectively in order to satisfy the relation E_{xz}/E_{xy}=1.32. In our first of many iterations we use our experimental values of W_{1c} and W_{1a}, and of the total magnetic shift K_{cc} and K_{aa}. In a procedure identical to that described in Section 7.31 we find all relevant quantities, including the spin shift K^s_{cc}. In general this spin shift will not be zero, though it is measured to be zero by Barrett et al. We are forced to conclude that there is an isotropic oxygen hole contribution to the shift with a value that cancels the

shift calculated from the Cu local moments. We must then reanalyze the Cu local moment contribution; the analysis described in 7.31 is no longer correct unless one now replaces the input parameters K_{cc}, K_{aa},W_{1c}, and W_{1a} with new effective values which describe the local moment contribution only:

$$W_{1c\ effective}=W_{1c\ experimental} - W_{1\ oxygen\ hole}\ (W_{1h}) \quad (7.17a)$$

$$W_{1a\ effective}=W_{1a\ experimental} - W_{1\ oxygen\ hole} \quad (7.17b)$$

$$K_{aa\ effective}=K_{aa\ experimental} - K_{oxygen\ hole}\ (K_h) \quad (7.17c)$$

$$K_{cc\ effective}=K_{cc\ experimental} - K_{oxygen\ hole} \quad (7.17d)$$

At each iteration the oxygen hole contribution to the shift is required to cancel K^s_{cc} of the local moments. The quantity W_{1h} is calculated from K_h from equation (7.16). One iterates until the values W_{1h} and K_h converge; this occurs very rapidly, since W_{1h} is typically only a few percent of the total W_1. Of course within each iteration of this type one also iterates as in section 7.31. A schematic diagram for this procedure is given in Figure 7.3.

The results of this calculation are given in Figure 7.4a-x, with a format similar to that given in 7.2a-x. In 7.4a are given $\langle \frac{1}{r^3} \rangle$, E_{xy}, and E_{xz}. The calculation is done in such a way that these parameters automatically give the correct chemical shift. $\langle \frac{1}{r^3} \rangle$ takes on values between 5 and 7.5 atomic units, all quite reasonable, and E_{xy} ranges from 2.5 to 3.5eV. At κ=.28 we find $\langle \frac{1}{r^3} \rangle$=5.8 and E_{xy}=2.7eV. Figure 7.4b shows the hyperfine coupling tensor parameters; the curves are very similar to those found in earlier analysis. At κ=.28 we have $A_{||}$= -413kG, $A_{\perp}$=33kG, and B_1=75kG.

The spin suscepibility shift from the Cu local moments $K^s_{cc,aa}$ and the Knight shift from the oxygen holes are given in Figure 7.4c. The values $K^s_{cc,aa}$ are similar to those found in the earlier analysis; at κ=.28 we have K^s_{cc}= -0.09% and K^s_{aa}=.26%. K_h automatically cancels K^s_{cc} to give zero total spin shift along c.

Spin and orbital susceptibilities of the Cu local moments are shown in Figure 7.4d with a format identical to that in Figure 7.2e. We obtain for κ=.28 the values χ^s=14x10^{-29}emu per Cu atom, χ^l_{cc}=16x10^{-29}emu, χ^l_{aa}=3x10^{-29}emu, and χ^{tot}=22x10^{-29}emu. The susceptibility of the oxygen holes can not be determined from the data.

The fluctuation time of the Cu spins τ_o is given in Figure 7.4e. It, again, is not greatly affected by the refinements of our analysis, and takes on a value τ_o=1.3x10^{-15}sec at κ=.28.

Shown in Figure 7.4f is the deduced spin-lattice relaxation rate contribution from the oxygen holes, W_{1h}. We find that W_{1h} contributes a rate of only approximately .04ms^{-1}, almost completely negligible in comparison to W_{1c} (1.05ms^{-1}) and W_{1a} (4.7ms^{-1}).

7.33 Exchange Coupling of Plane Cu Atoms

We have described anomolous behaviour in the plane Cu transverse relaxation in Section 5.4. Knowing the complete spin-lattice relaxation rate tensor enables one to use Redfield theory to place a lower limit on transverse relaxation rates; the effect is the so-called "lifetime broadening" of the resonance line, and the calculation of the transverse relaxation from this effect is given in Appendix C. In addition to this relaxation we find an additional Gaussian decay when the static field is placed along the c-axis, and an additional exponential decay when the field is along the a or b axis. We have confirmed through a double resonance experiment that this extra relaxation for H_0 along c is due to a very strong Cu nuclear spin-spin interaction; the strength is approximately ten times greater than one expects from the well understood magnetic dipole-dipole coupling. The coupling, then, must be an electron mediated, or indirect coupling.

Such a coupling is well known in the case of an antiferromagnet, and is known as Suhl-Nakamura coupling [Nakamura, 1958]. The existence of the coupling in $YBa_2Cu_3O_7$, which is not an antiferromagnet, indicates that exchange coupling of plane Cu atoms is present in this material. We have used the strength of the nuclear spin-spin coupling to deduce the exchange coupling between neighboring plane Cu local moments. We find an effective exchange coupling, J_{eff}, equal to approximately 2400cm^{-1}, a value that is of the same order of magnitude as the nearest neighbor exchange coupling J, 950cm^{-1}, measured with light scattering experiments in the oxygen deficient antiferromagnet $YBa_2Cu_3O_{6+x}$. The analysis that we present here has appeared in print [Pennington (b); 1989]; we now have a somewhat higher value for the hyperfine coupling tensor term A_{cc}, and this results in our finding a somewhat higher J_{eff}.

In this analysis we shall neglect the somewhat smaller B_1 transferred hyperfine coupling.

We can derive an expression for the indirect nuclear coupling assuming that the coupling is the net effect of the hyperfine coupling between each plane Cu nucleus and local moments, and antiferromagnetic exchange coupling between neighboring plane Cu local moments. We write the spin Hamiltonian for two nearest neighbor Cu sites, site 1 and 2, where I is a nuclear and S an electron spin:

$$H = \sum_{\alpha=x\ y\ z} I_{1\alpha}A_{\alpha\alpha}S_{1\alpha} + \sum_{\alpha=x\ y\ z} I_{2\alpha}A_{\alpha\alpha}S_{2\alpha}$$

$$+ \sum_{i\ j;i<j} J_{ij}\mathbf{S}_i\cdot\mathbf{S}_j \qquad (7.18)$$

The last term contains exchange coupling between the entire lattice of electron spins. Denoting the excited eigenstates of the exchange coupled electron spins as $|n\rangle$ and the ground state as $|0\rangle$ we treat the electron-nuclear interaction in second order perturbation theory to get an effective Hamiltonian involving only nuclear spin coordinates. We find:

$$H_{eff} = \hbar \sum_{\alpha=x\ y\ z} a_{\alpha\alpha 12} I_{1\alpha} I_{2\alpha} \qquad (7.19)$$

where

$$\hbar a_{\alpha\alpha 12} = (A_{\alpha\alpha})^2 \left(\sum_n \frac{\langle 0|S_{1\alpha}|n\rangle\langle n|S_{2\alpha}|0\rangle}{E_0 - E_n} + c.c \right) \qquad (7.20)$$

Evaluation of the sum requires a complete theory of the system of exchange coupled Cu electron spins. The physical content of the expression can be found by considering an isolated pair of Cu atoms with an antiferromagnetic exchange coupling for which $|0\rangle$ is the electronic singlet ground state and $|n\rangle$ are the excited triplet states. One finds

$$\hbar a_{\alpha\alpha 12} = \left(\frac{A_{\alpha\alpha}{}^2}{2J} \right) \qquad (7.21)$$

We then define, again for nearest neighbor Cu atoms, a quantity J_{eff} by the equation:

$$\frac{1}{2J_{eff}} = \sum_n \frac{\langle 0|S_{1\alpha}|n\rangle\langle n|S_{2\alpha}|0\rangle}{E_0 - E_n} + c.c \qquad (7.22)$$

giving

$$\hbar a_{\alpha\alpha 12} = \frac{(A_{\alpha\alpha})^2}{2J_{eff}} \qquad (7.23)$$

Since one expects the states |n> to be something like spin waves, we expect J_{eff} to be somewhat different from J. For simplicity we keep only nearest-neighbor nuclear spin-spin coupling.

We know the range of A_{cc}, A_{aa}, and from the analysis of section 7.32. For κ=.28 we have A_{cc}= -412kG and A_{aa}= -33kG. In the presence of a strong magnetic field H_0 parallel to c, since $(A_{cc})^2>>(A_{aa})^2$ we can keep only terms involving a_{cc} in the total nuclear Hamiltonian for i and j nearest neighbors:

$$H = -\gamma_n \hbar H_0 \sum_i I_{iz} + \hbar a_{cc} \sum_{i\ j;i<j} I_{iz} I_{jz} \qquad (7.24)$$

This Hamiltonian can be solved exactly [see, e.g. Ansermet et al (1988)] since the individual I_{iz}'s commute with it. Consider an experiment in which we are tuned to the 1/2,-1/2 Cu transition and the neighboring plane Cu is the same isotope. Following a 90° pulse an ensemble of nuclei each coupled to one spin-3/2 neighbor would have a normalized free induction decay given by $\frac{1}{2}[\cos(at/2)+\cos(3at/2)]$, where the first term results from neighboring nuclei with spin 1/2 or -1/2 and the second 3/2 or -3/2. One then applies a 180° pulse to the 1/2,-1/2 transition, and if the neighboring spin is a "like" nucleus, then its 1/2,-1/2 transition will be flipped as well. The first term, then, will not refocus. The neighbor spin's m_I=3/2 and -3/2 levels will, however, be unaffected, so the second term will be refocused in the spin-echo. Thus, the spin-echo decay will be $\frac{1}{2}[\cos(at/2)+1]$, where t is the time at which the echo forms.

If instead there are two "like" neighbors, then one has a 25% probability that both neighbors will be either plus or minus 3/2, a 50% probability that one will be plus or minus 3/2 and the other plus or minus 1/2, and a 25% probability that they will both be plus or minus 1/2. The echo, then, from two "like" neighbors will be $\frac{1}{4}[\cos(at)+2\cos(at/2)+1]$ where the first term is from nuclei having two neighbors with m=±1/2. Generalizing for N "like" neighbors one has an echo given by :

$$\frac{1}{2^N}\sum_{n=0}^{N}\frac{N!}{n!(N-n)!}\cos(nat/2) \qquad (7.25)$$

In $YBa_2Cu_3O_{7-\delta}$ the plane Cu's are each bordered by four nearest neighbor plane Cu's. One must consider, of course, the probability that a neighbor will be an "unlike" nucleus, i.e. Cu^{65} rather than Cu^{63}. A Cu^{63} with four spin 3/2 neighbors may have as many as four and as few as zero neighbors that are "like." To obtain the echo one must then weight the sum (7.25) with the a factor Pr(N), the probability of having N like neighbors. For Cu^{63} these probabilities are:

$$Pr(4) = P_{63}^4 \qquad (7.26)$$
$$Pr(3) = 4P_{63}^3P_{65}$$
$$Pr(2) = 6P_{63}^2P_{65}^2$$
$$Pr(1) = 4P_{63}P_{65}^3$$
$$Pr(0) = P_{65}^4$$

where P_{63} is the fractional abundance of the Cu^{63} isotope. The weighted sum is of the form:

$$A + B\cos(at/2) + C\cos(at) + D\cos(3at/2) + E\cos(2at) \qquad (7.27)$$

Using abudances for Cu^{63} 69% and Cu^{65} 31% we obtain:

$$A=.3422 \qquad (7.28a)$$
$$B=.4684 \qquad (7.28b)$$
$$C=.1605 \qquad (7.28c)$$
$$D=.0270 \qquad (7.28d)$$
$$E=.0018 \qquad (7.28e)$$

For short times this decay can be approximated by a Gaussian whose τ is related to the coefficient of the t^2 term in the series expansion of the decay. For Cu^{63} we get $(1/\tau)^2=.346a_{cc}^2$, or:

$$a_{cc}=\frac{1}{\hbar}\left(\frac{A_{cc}^2}{2J}\right)= 1.70/\tau \qquad (7.29)$$

Using τ=117μs from the spin-echo experiment we find a_{cc}=14.5x10^3rad/sec. There is also a direct magnetic dipolar interaction between plane Cu nuclei of like isotopes for which the coefficient of $I_{iz}I_{jz}$ is 1x10^3rad/sec, much less than the measured a_{cc}. Correcting the measured a_{cc} for the dipolar coupling and using values of A_{cc} and B_1 corresponding to κ equal to .28 (Section 7.32) we find J_{eff}=1700cm^{-1}. This value is significantly higher than the value

$J=950cm^{-1}$ deduced from light scattering experiments. We note that a nearest neighbor J exchange coupling and our defined quantity J_{eff} should presumably be of the same approximate magnitude but not exactly the same.

Next consider the decay of the plane Cu^{63} nuclei when H_0 is along the a-axis. Defining z as the direction of H_0 and x as lying along the c-axis, the secular part of the Hamiltonian can be written as:

$$H = \hbar[a_{aa}I_{1z}I_{2z} + \frac{1}{4}(a_{aa} + a_{cc})(I_1^+I_2^- + I_1^-I_2^+)] \qquad (7.30)$$

Recognizing that $a_{cc}>>a_{aa}$ (since for the values of the hyperfine coupling tensor we have seen that typically $(A_{cc})^2>>(A_{aa})^2$), we see that the second term, which leads to mutual spin flips, is much larger than the first term, which leads to a static frequency splitting. In this limit, the transverse relaxation will be exponential with T_2' being determined by the size of the frequency jumps and the jumping rate, leading to the relationship $T_2^{65}=(\gamma_{63}/\gamma_{65})^2T_2^{63}$. This relation is shown to work in Figure 5.4b.

7.34 Conclusions from Plane Cu NMR

We have given a self-consistent description, using quite reasonable parameters, of the spin-lattice relaxation, the magnetic shift, and the transverse relaxation of the plane Cu using a "permanent moment" model for the holes in the plane Cu 3d shell. The normal state NMR data thus indicates that these holes are very near to that end of the spectrum.

We find in the transverse relaxation an indication that the exchange coupling that one finds in oxygen depleted $YBa_2Cu_3O_{7-\delta}$ persists in the fully oxygenated 90K superconductor, as one would expect in a permanent moment picture.

Surprisingly, however, the measurements of Takigawa et al and of Barrett et al of magnetic shifts in the superconducting state indicate that these localized moments must in fact "pair." Taken together these results suggest a picture in which we have essentially local moments but are not completely in that limit; the small amount of itinerant character results in a pairing of the spins. The theoretical description of this state will be a most important step in the understanding of high temperature superconductivity.

7.4 The NMR Tensors of the Chain Cu

7.41 The Spin-Lattice Relaxation and Magnetic Shift Tensors

For the chain Cu the permanent moment model of the holes in the Cu d-shell would predict a $W_{1\alpha}$ anisotropy which would be very similar to that for the planes, except that the a-axis would be the axis of symmetry. Clearly we have a problem since the chain $W_{1\alpha}$'s are isotropic to within experimental error.

One then wonders if the chain Cu also have permanent electron spin magnetic moments (or more precisely are close to the permanent moment end of the fluctuation spectrum). There are two things which make us think they probably do. The first is the very great similarity of the plane Cu and chain Cu magnetic shift tensor; the shift tensors of the plane and chain Cu are identical except that the symmetry axis of the chain Cu is along the a-axis for the chain and along c for the plane. The shift tensor reflects the electronic structure of the plane and chain Cu atoms. The second is that the existence of the permanent moment arises fundamentally because of the relative size of certain matrix elements involving Cu^{2+} ions and oxygens. It is hard then to see why the plane Cu meet the criteria for permanent moments and the chain Cu do not.

Analysis of the chain relaxation and shift is somewhat simpler than that of the plane. The first important clue is the fact that the chain Cu spin-lattice relaxation rate is linear with temperature (Table 5.3). The temperature dependence of the relaxation rate of the chains has been measured using zero field nuclear quadrupole resonance as well, and all workers report a dependence close to linear. The linear temperature dependence of the relaxation rate is the signature of relaxation from a conduction band.

A candidate mechanism for the hyperfine coupling of the nucleus to holes in a conduction band is the "Transferred Hyperfine Coupling I" mechanism (Section 3.51221) in which a hole that is essentially in a band of oxygen 2p orbitals has some overlap with the Cu 4s state. The coupling, the Knight shift contribution, and the spin-lattice relaxation are isotropic. We measure, within experimental error, a relaxation rate that is isotropic (Table 5.3). The magnetic shift, however, is anisotropic, with the same degree of

anisotropy as the plane Cu. We propose, then that the magnetic shift of the chain Cu is the sum of an isotropic Knight shift and an anisotropic chemical shift. We have seen from our electric field gradient data that the chain Cu has a single hole in the y^2-z^2 orbital, where we associate x,y, and z with the crystal axes a,b, and c as we have done for the plane Cu. The anisotropy of the chain chemical shift is given by:

$$K^l_{aa}/K^l_{cc} = 4E_{yx}/E_{yz} \qquad (7.31)$$

We shall suppose that $E_{yx}=E_{yz}$ and thus we have a 4 to 1 chemical shift anisotropy. We now use the total magnetic shift to deduce the contributions from the isotropic Knight shift and the anisotropic chemical shift. Using the magnetic shifts in Table 5.2 we find:

$$K_{aa}= 1.38\% = 4K^l_{cc} + K_h \qquad (7.32a)$$

$$K_{cc}= 0.58\% = K^l_{cc} + K_h \qquad (7.32b)$$

where K_h is the Knight shift from oxygen holes, and we have used for K_{cc} the average of K_{cc} and K_{bb}, since we expect these quantities to be equal, and since they are equal within experimental error. We solve to find:

$$K_h = .31\%$$
$$K^l_{cc}=.27\%$$
$$K^l_{aa}=1.08\%$$

The measurements of Barrett et al in our laboratory of the magnetic shift tensor in the superconducting state give the Knight shift contribution and chemical shift contribution to the magnetic shift. They find:

$$K_h = .33\%$$
$$K^l_{cc}=.25\%$$
$$K^l_{aa}=1.05\%$$

The agreement is remarkable and confirms our picture of a hole in the chain Cu y^2-z^2 orbital, with excited states E_{yz} and E_{yx} almost degenerate. Using a value of 6 atomic units for $\left\langle \frac{1}{r^3} \right\rangle$ we estimate from K^l_{aa} and equation (7.9) the value of E_{yz}. We find E_{yz}=3.5eV, a reasonable value. From equation (7.9), and again using $\left\langle \frac{1}{r^3} \right\rangle$=6 atomic units we obtain $\chi^l_{aa}=12.9\times10^{-29}$emu and $\chi^l_{cc}=3.1\times10^{-29}$emu per chain Cu atom.

We now turn our attention to K_h and $W_{1\alpha}$. As discussed in 3.131, K_h and W_1 are related by the Korringa law, giving:

$$\frac{3T}{2W_1}(K_h)^2 = \left(\frac{\hbar}{4\pi k_B}\right)\frac{\gamma_e^2}{\gamma_n^2}B \qquad (7.33)$$

where B is a factor introduced by Pines [1955] to take account of electron-electron interactions. B=1 for non-interacting electrons. Using an average of the $W_{1\alpha}$ (which are equal to within experimental error) at 100K of $1ms^{-1}$ we find B=.43; this is essentially the non-interacting electron limit. We conclude that the chain Cu nuclei are relaxed by weakly interacting holes in the conduction band.

We can understand this coupling if we consider states at the top of the oxygen hole band. These states are made up of O(1) and O(4) (chain and bridge) 2pσ orbitals, whose symmetry relative to the chain Cu sites is nearly the same as that of the y^2-z^2 orbitals. Such states are shown in Figure 7.5. The overlap of the oxygen 2pσ orbitals with the chain Cu atoms is larger for the bridge oxygens [O(4)] than for the chain oxygens, since the Cu-O interatomic distance is less for the bridge oxygens. Thus the oxygen band wave function acting at the chain Cu nucleus contains some Cu 4s wavefunction. (This mixing should be much less at the plane Cu site; the four plane oxygens surrounding the plane Cu are nearly equidistant from the Cu atom.)

In the above picture since the chemical shift and electric field gradient tensors of the chain Cu atoms correspond to a single hole in the $3dy^2$-z^2 orbital we still assume that chain Cu atoms have a permanent moment. The fact that it does not contribute a spin susceptibility shift or a contribution to spin-lattice relaxation requires that the spin be rapidly scattered by the oxygen holes, and its spin susceptibility must be very small. This situation has been seen experimentally by Narath for Co in W.

7.42 The Transverse Relaxation Tensor

The component of chain Cu transverse relaxation due to spin-lattice relaxation, which we have labeled "T_{2R}," is well understood from Redfield theory.

Additionally one finds extra relaxation in the chains, which for H_0 along c and b adds a Gaussian component, and along a an extra exponential component. We were able to understand plane Cu relaxation completely because the form of the transverse relaxation can be calculated

in two limiting cases: (1) Gaussian relaxation in the static limit where mutual spin flips are negligible, as is the case for H_0 along c, and (2) exponential relaxation in the motionally narrowed limit where mutual spin flips dominate (when H_0 parallel to a). We know, however, that the chain Cu relaxation for H_0 parallel to c is not described by either limit because though the decay is Gaussian, indicating absence of motional narrowing, the isotope dependence is $\tau_{65}=(\gamma_{63}/\gamma_{65})^2\tau_{63}$, not that of Equation (5.6). In this case there is no exact solution of the relaxation process, so we do not know whether or not an indirect nuclear spin-spin coupling is present.

7.43 Conclusions from the Chain Cu NMR

Whereas the plane Cu NMR probes the local moments on the plane Cu atoms, the chain Cu, in contrast, is apparently much less sensitive to the moments which we believe must exist on the chain Cu atoms. The chain Cu NMR instead probes the oxygen holes in the chain. We find that these holes are in a weakly interacting limit, and thus a band theory picture may serve to describe these holes.

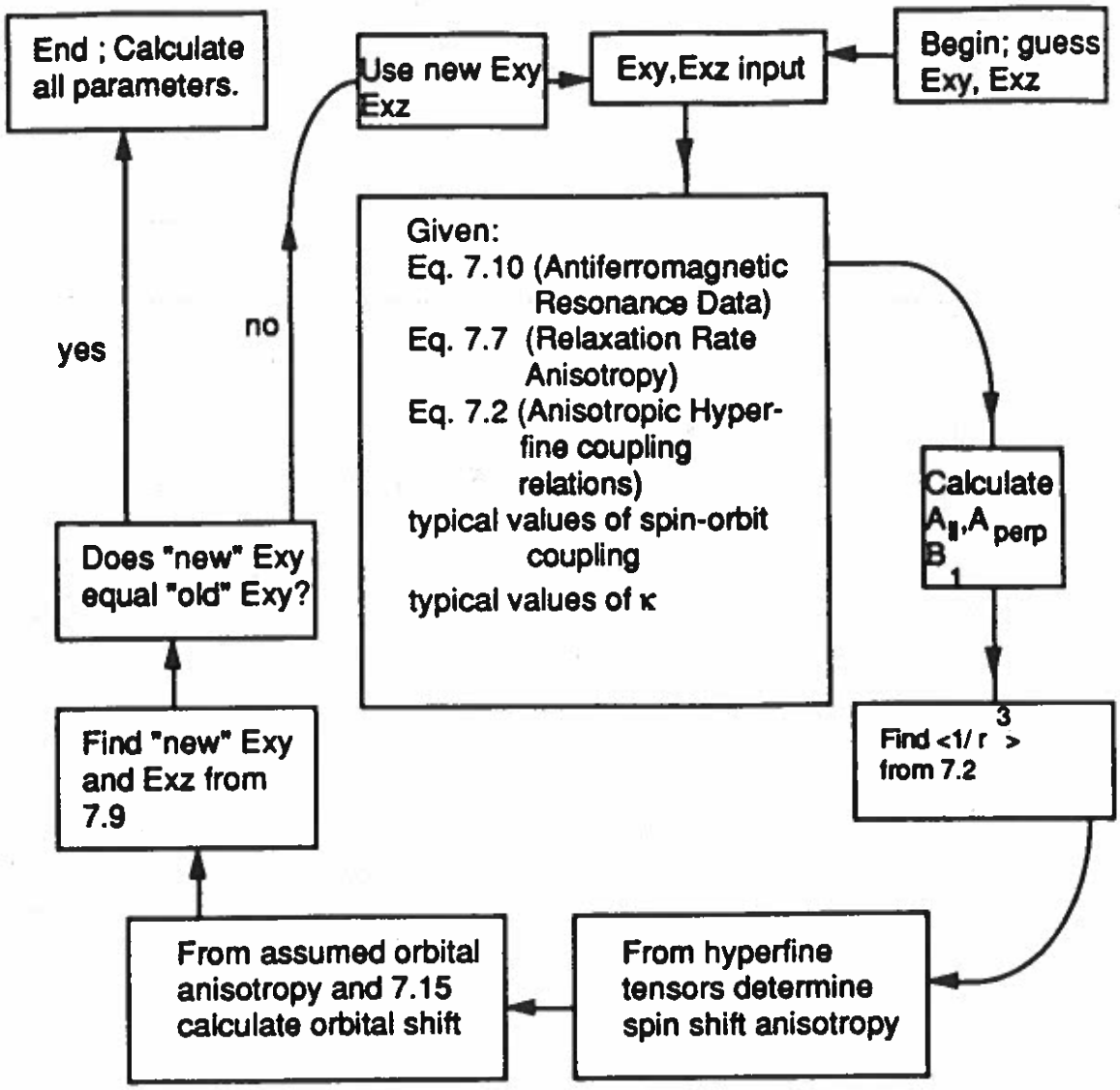


Figure 7.1 Schematic of iteration procedure used to calculate the quantities plotted in Figures 7.2.

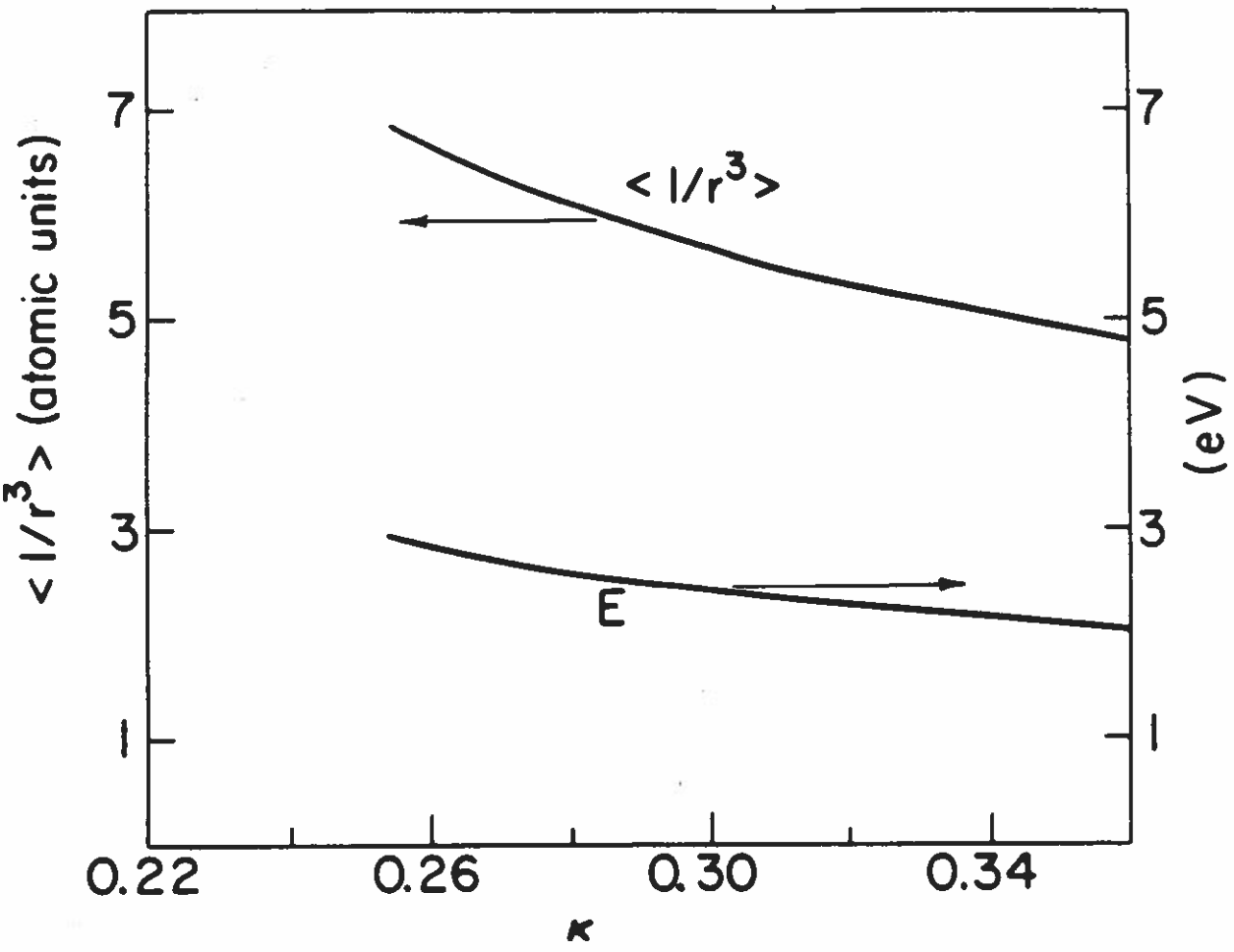


Figure 7.2a $<1/r^3>$ in atomic units and $E_{xy,yz}$ in eV calculated as described in text.

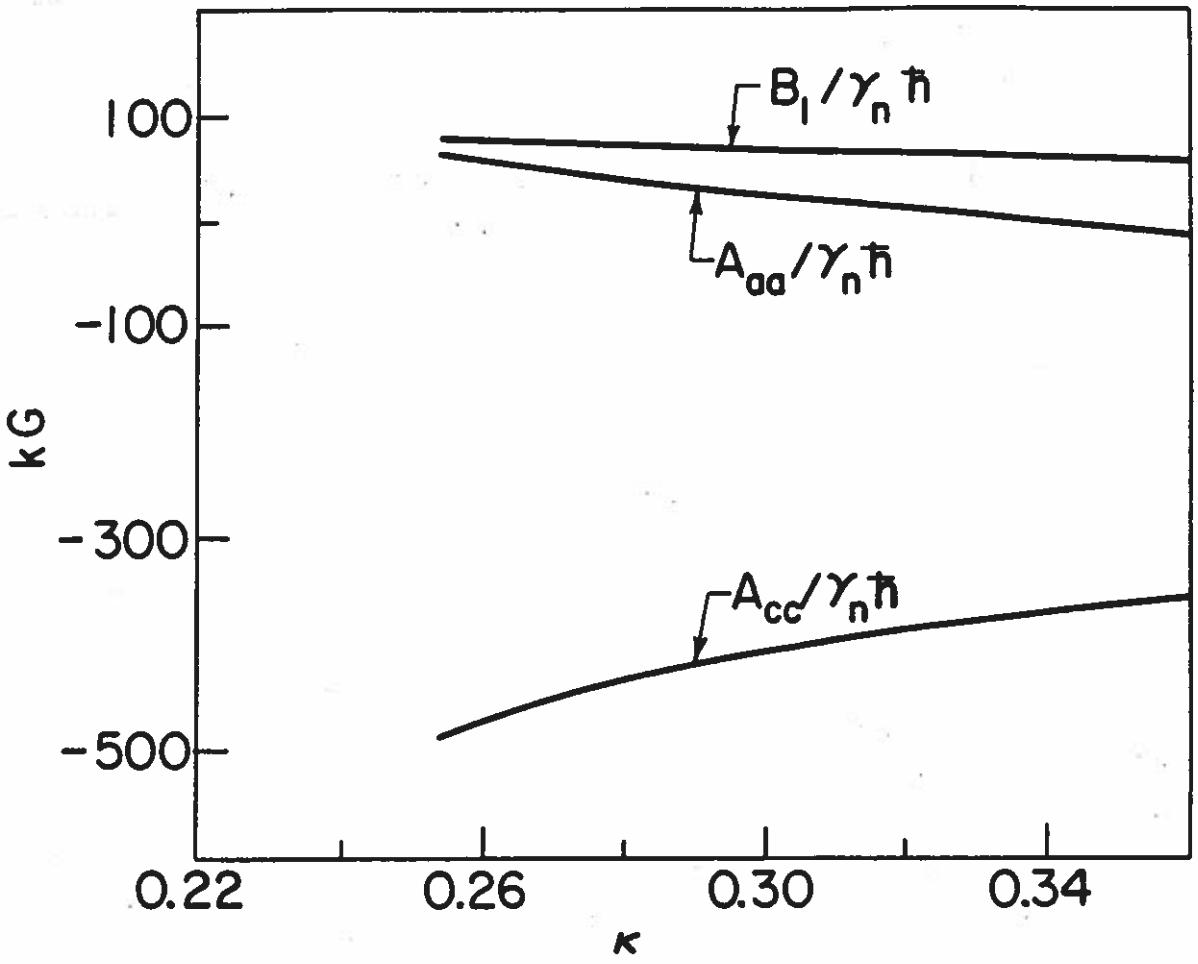


Figure 7.2b A_{cc}, A_{aa}, and B_1 in kG calculated as described in text.

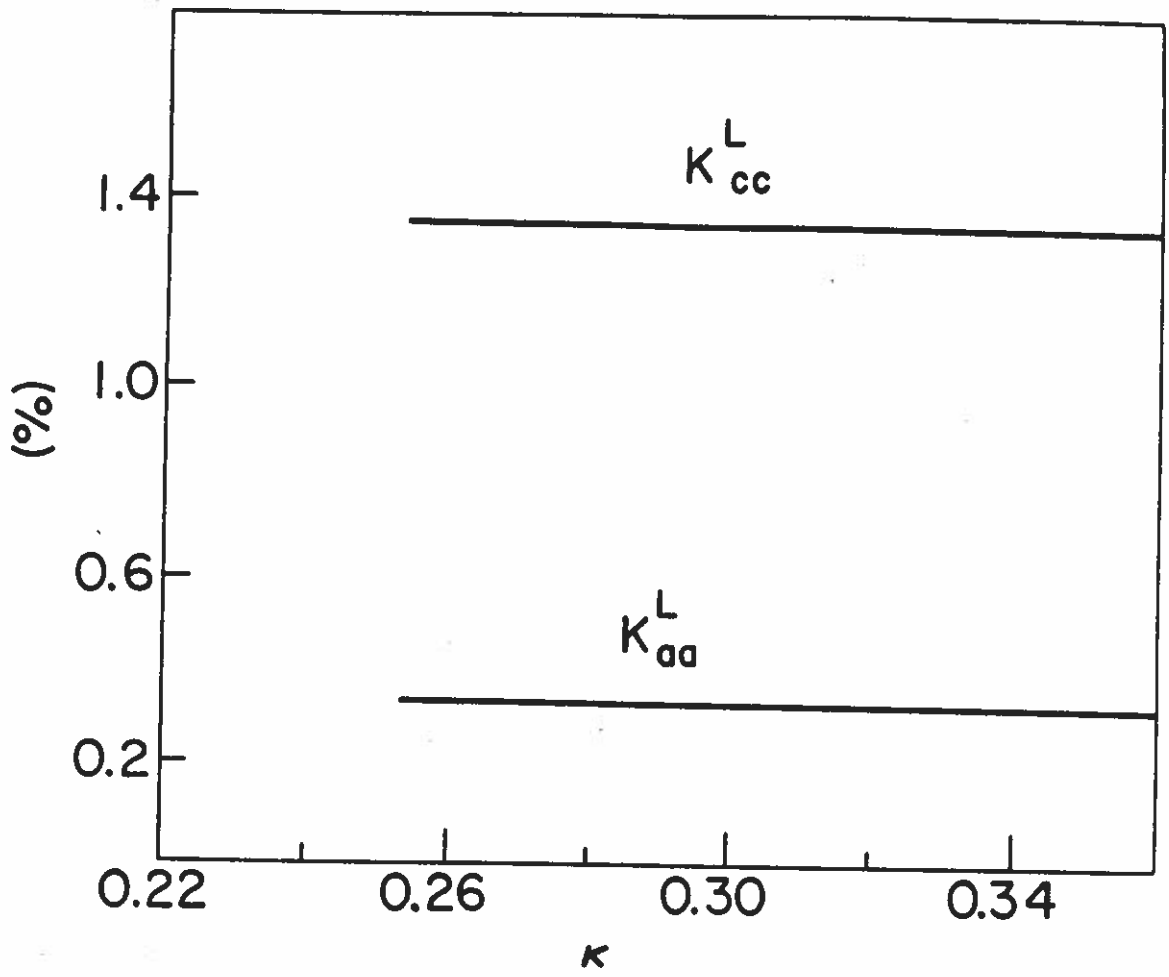


Figure 7.2c K^l_{cc}, K^l_{aa}, the orbital shifts in percent calculated as described in text.

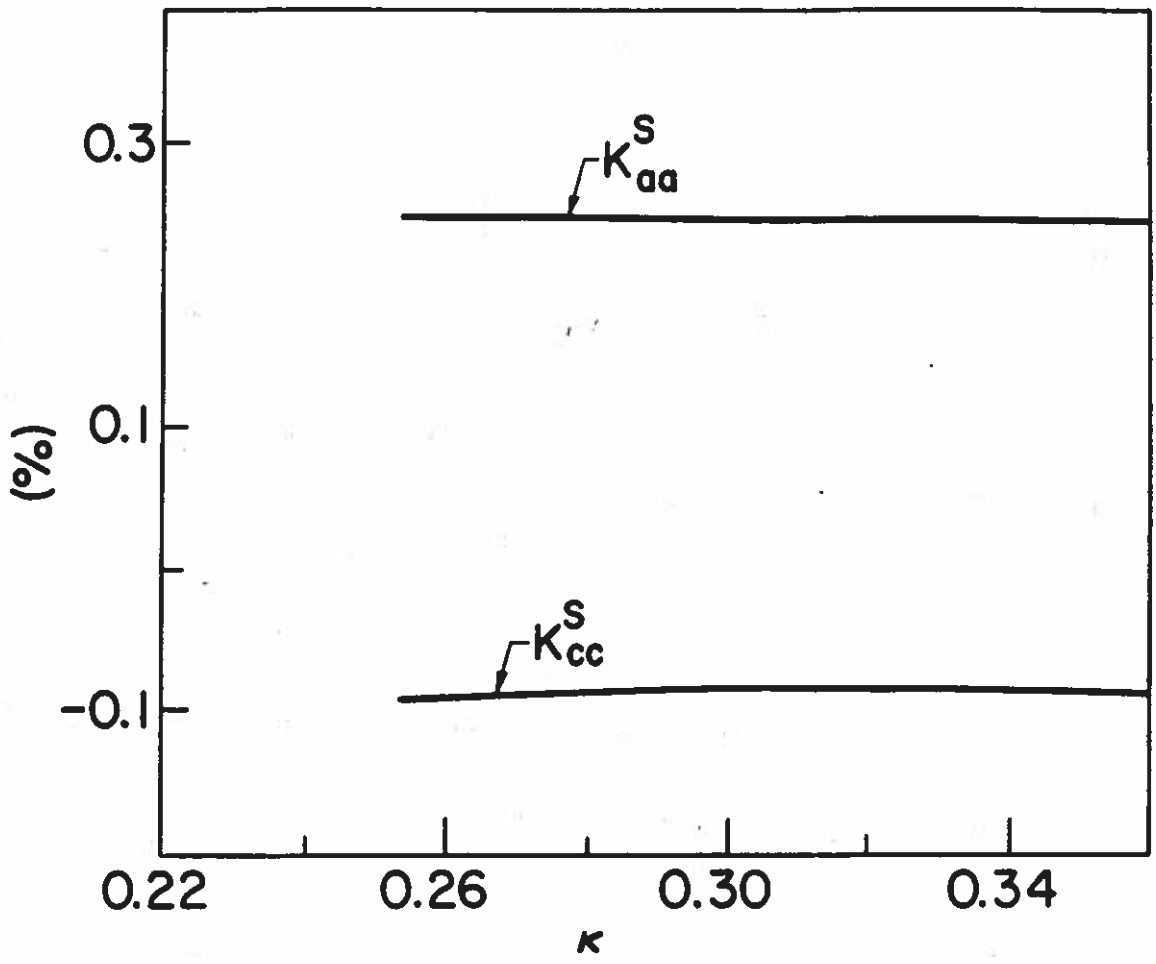


Figure 7.2d K^s_{cc}, K^s_{aa}, the spin shifts in percent calculated as described in text.

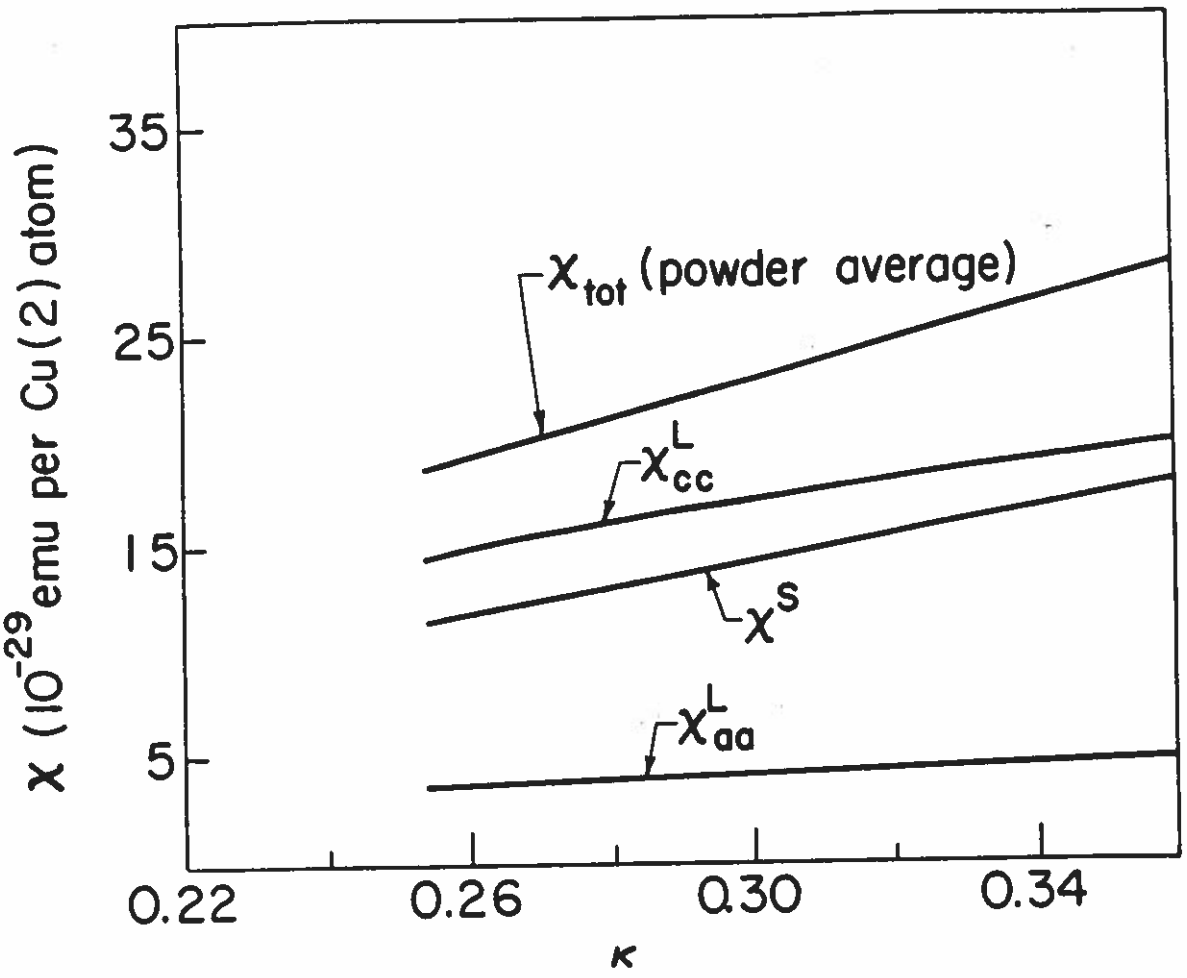


Figure 7.2e χ^s, the spin susceptibility, χ^l_{cc}, χ^l_{aa}, and $\chi^{total}_{powder\ average}$ in units of 10^{-29}emu per Cu atom calculated as described in text.

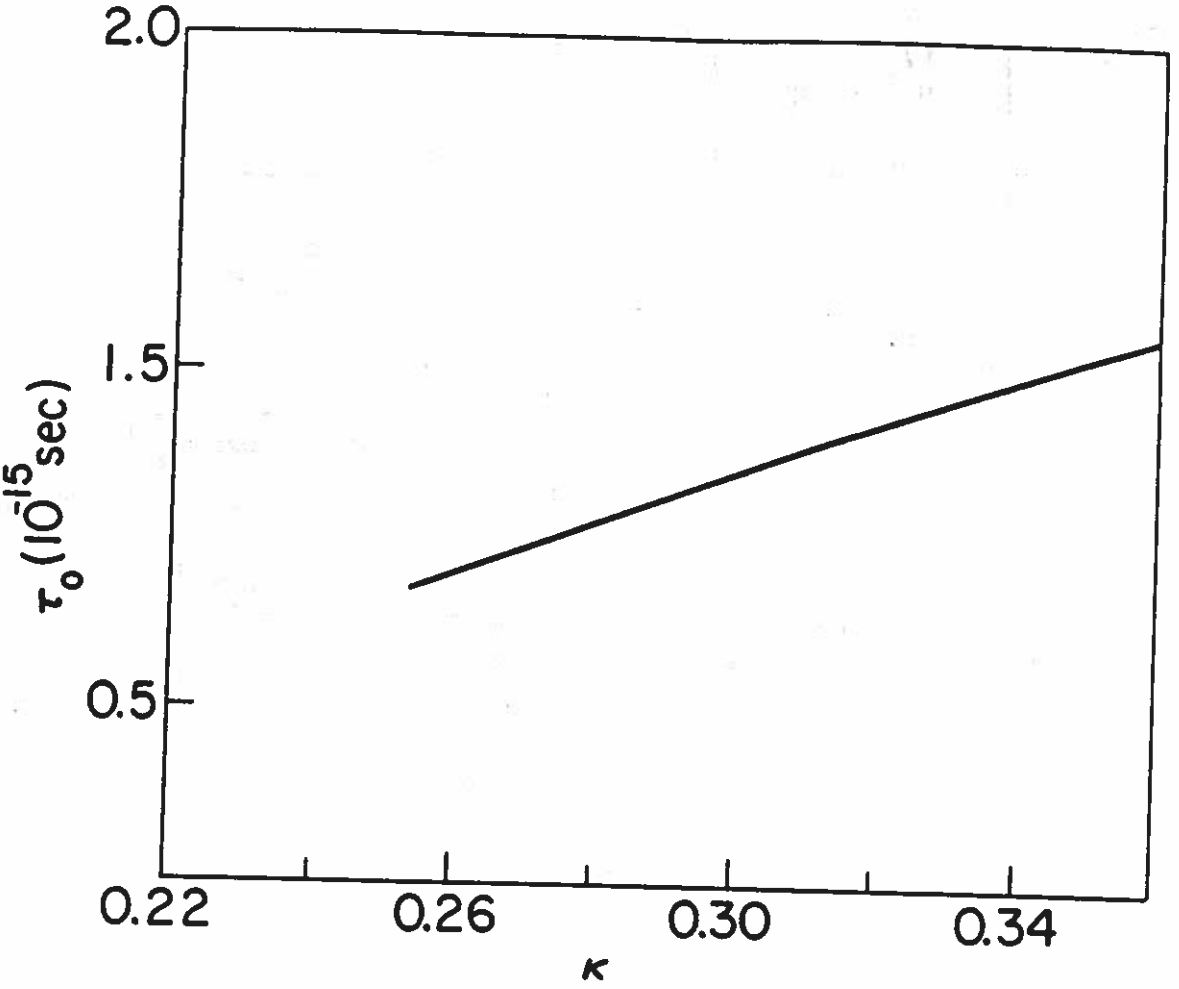


Figure 7.2f τ_o, the correlation time for fluctuations of the moment in the Cu d-shell, in units of 10^{-15}sec, calculated as described in text.

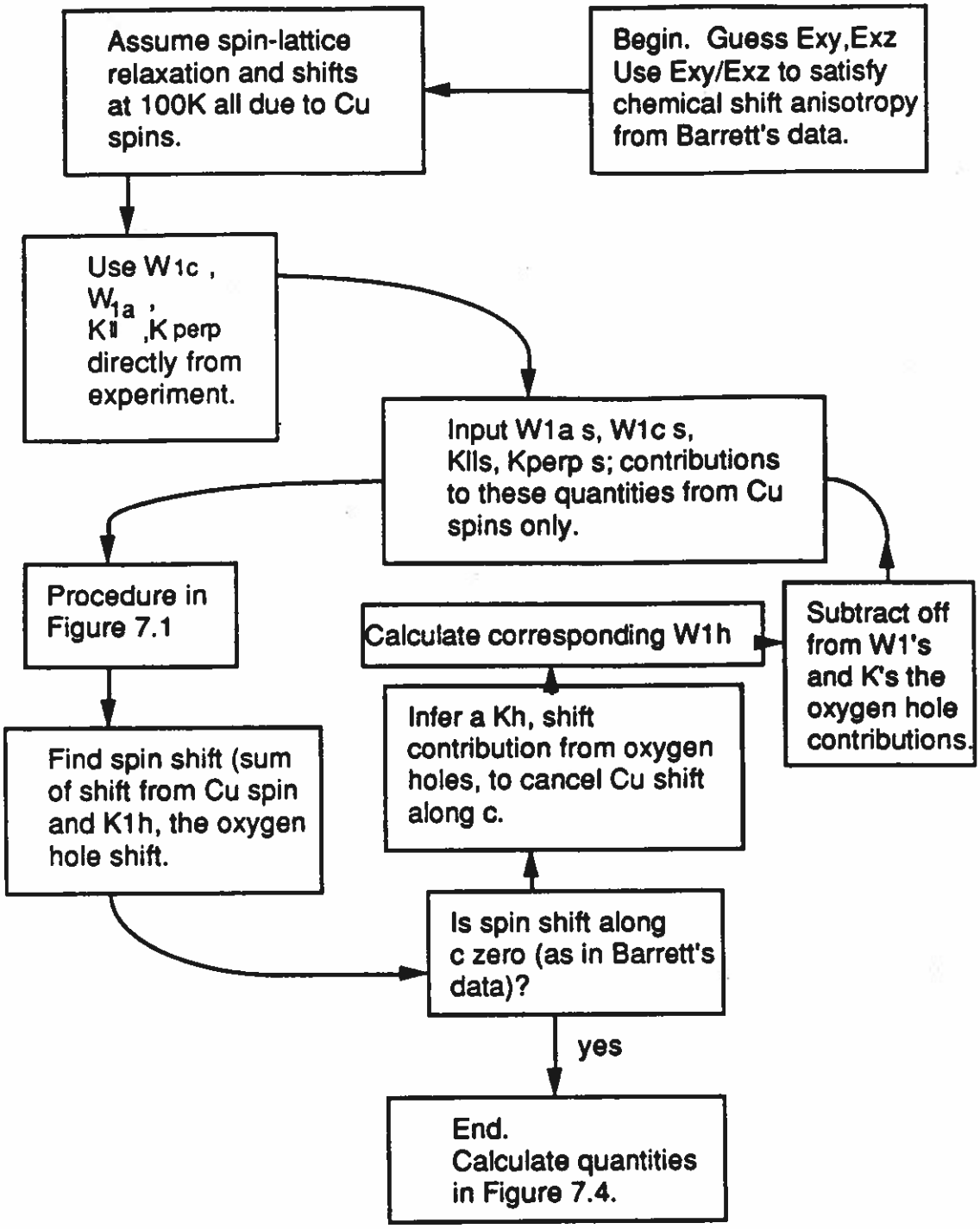


Figure 7.3 Schematic diagram showing the procedure used to calculate the quantities in Figure 7.4

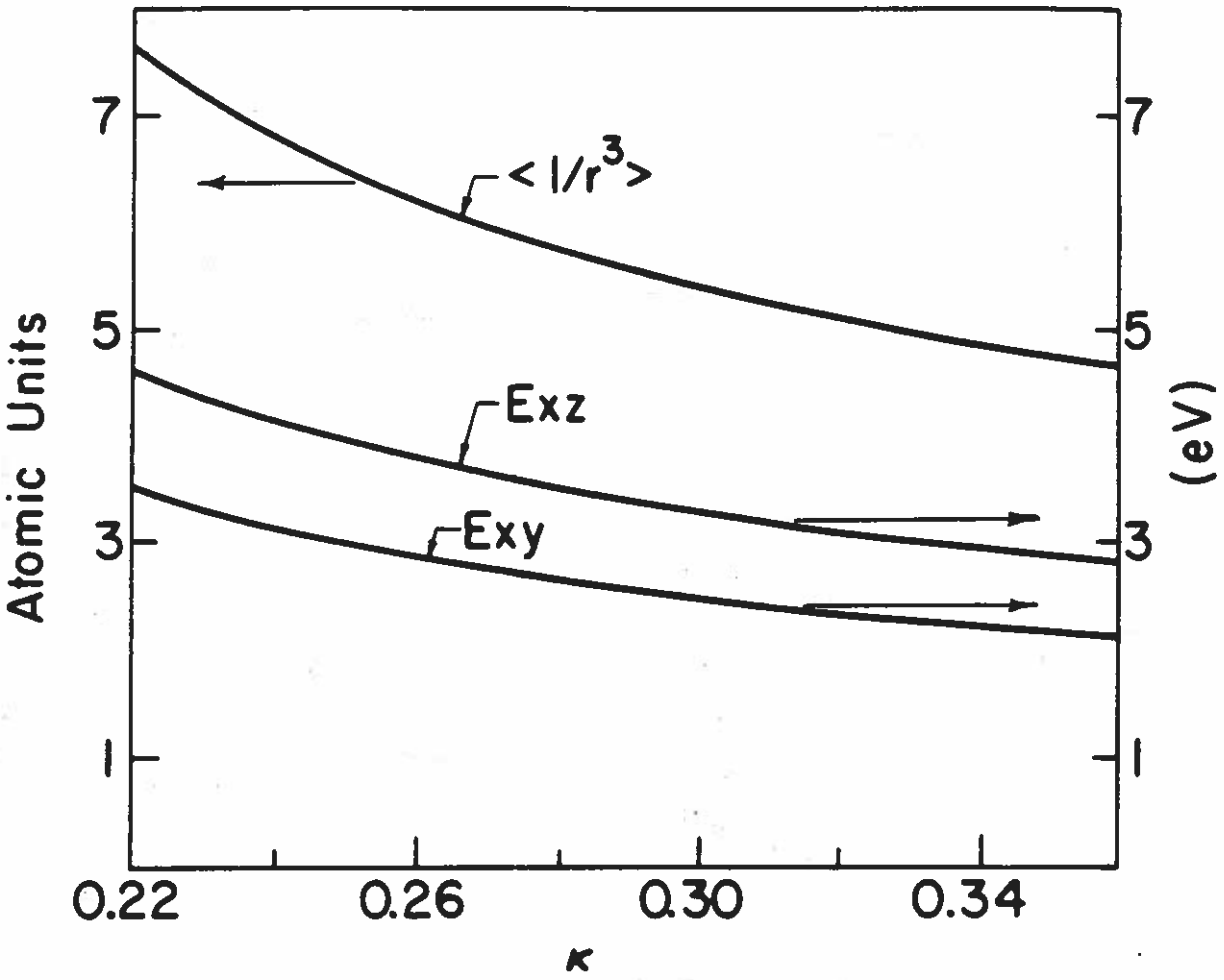


Figure 7.4a $<1/r^3>$ in atomic units and E_{xy}, E_{yz} in eV calculated as described in text.

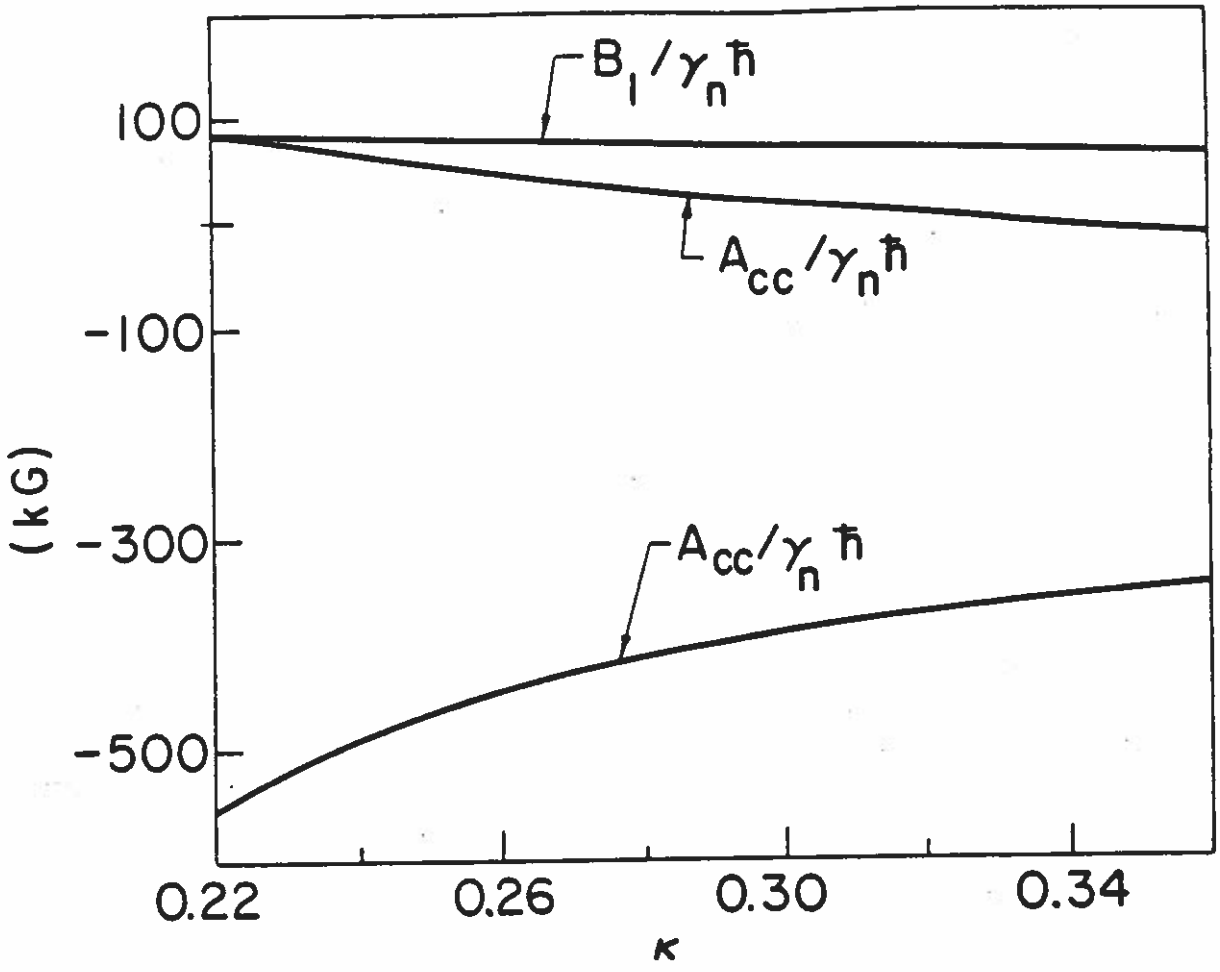


Figure 7.4b A_{cc}, A_{aa}, and B_1 in kG calculated as described in text.

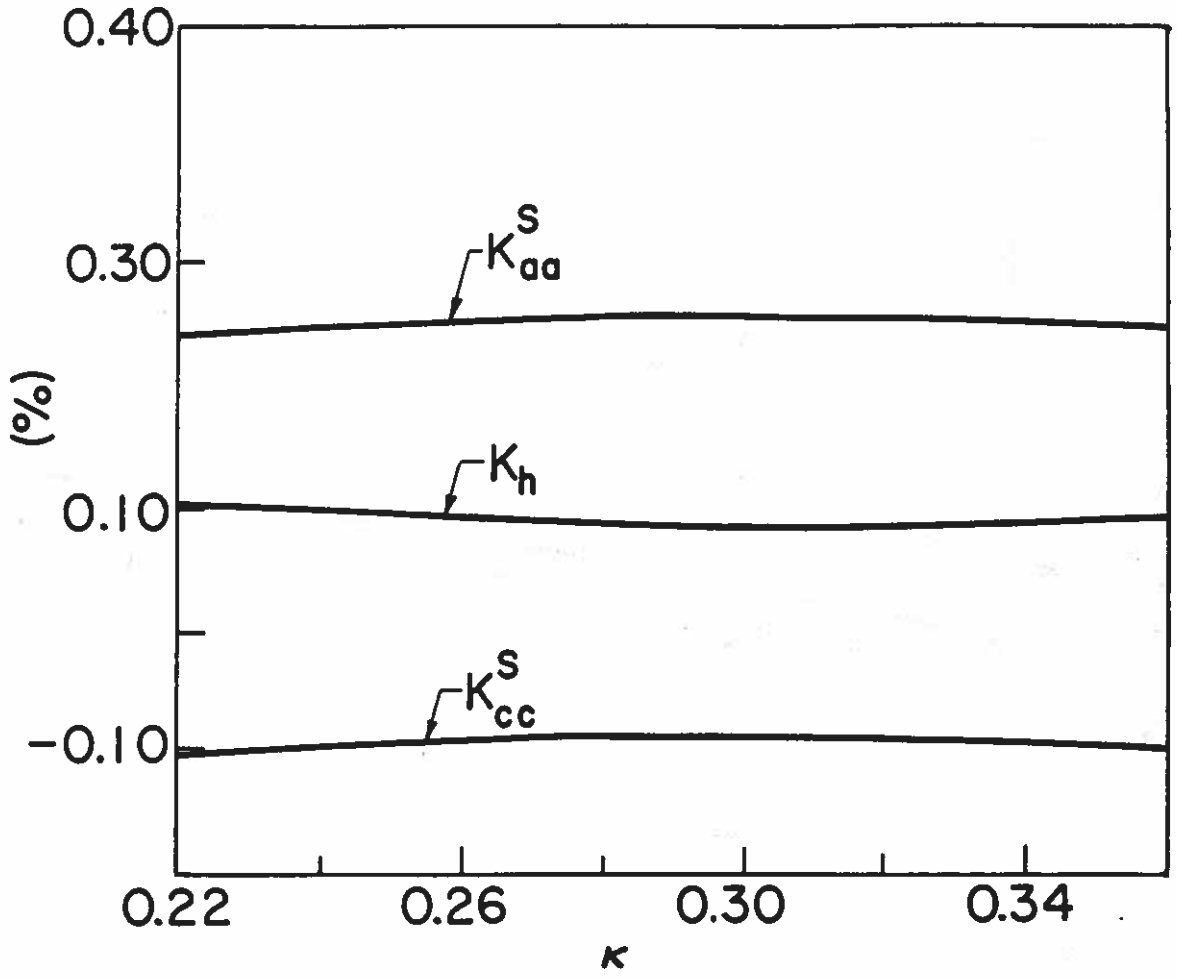


Figure 7.4c K^s_{cc}, K^s_{aa}, the Cu spin susceptibility shifts, and K_h, the contribution of oxygen holes, each given in percent and calculated as described in text.

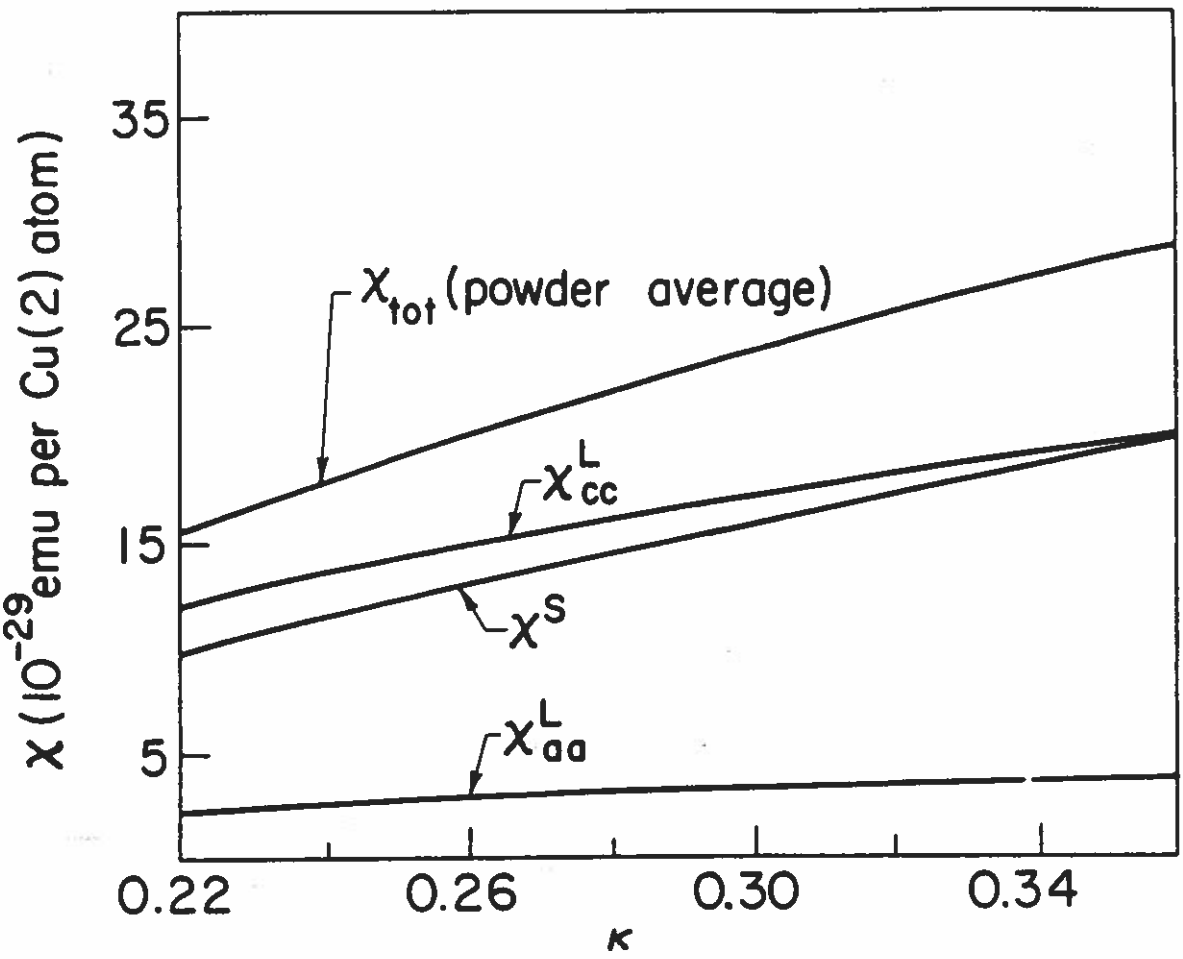


Figure 7.4d χ^s, the spin susceptibility, χ^l_{cc}, χ^l_{aa}, and $\chi^{total}_{powder\ average}$ in units of 10^{-29}emu per Cu atom calculated as described in text.

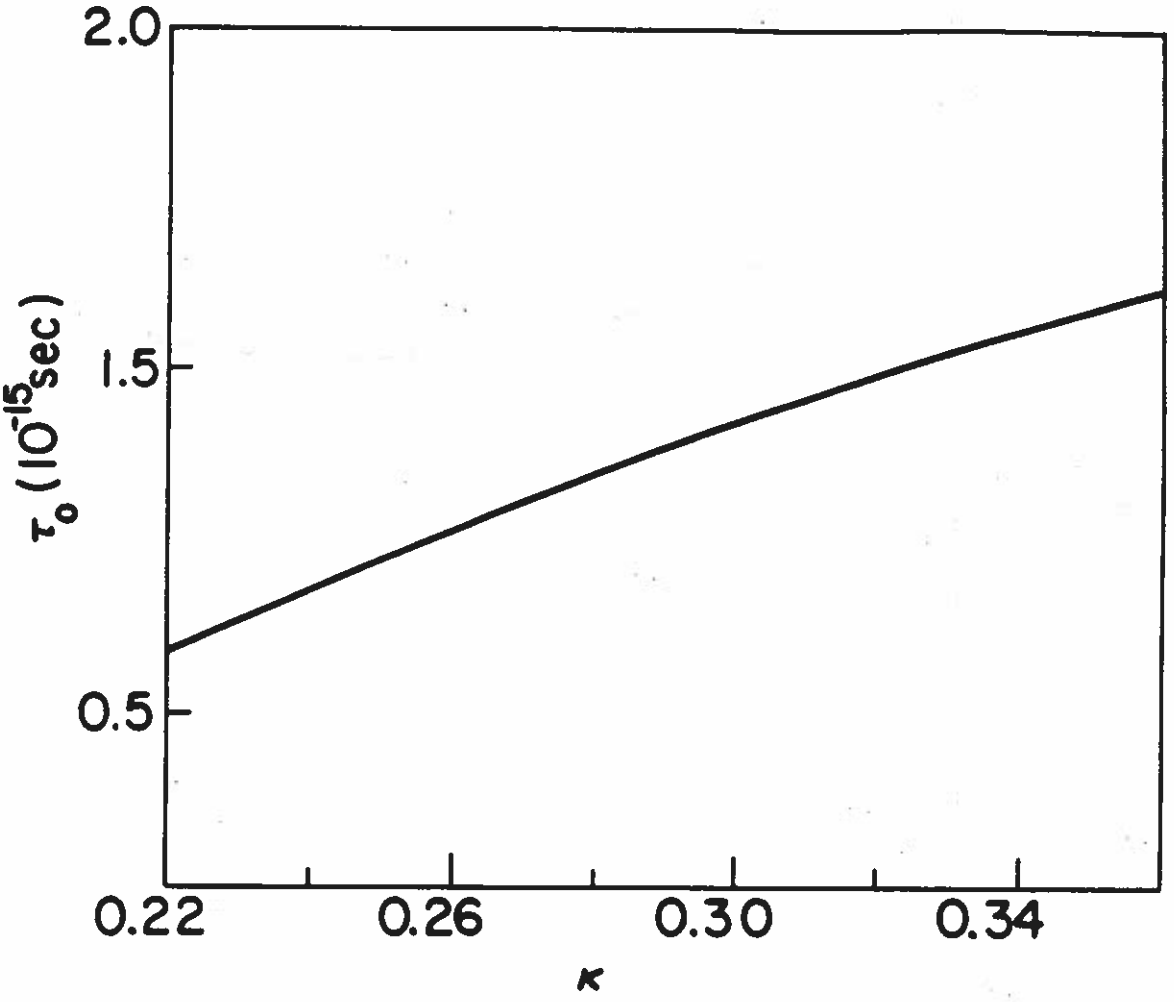


Figure 7.4e τ_o, the correlation time for fluctuations of the moment in the Cu d-shell, in units of 10^{-15}sec, calculated as described in text.

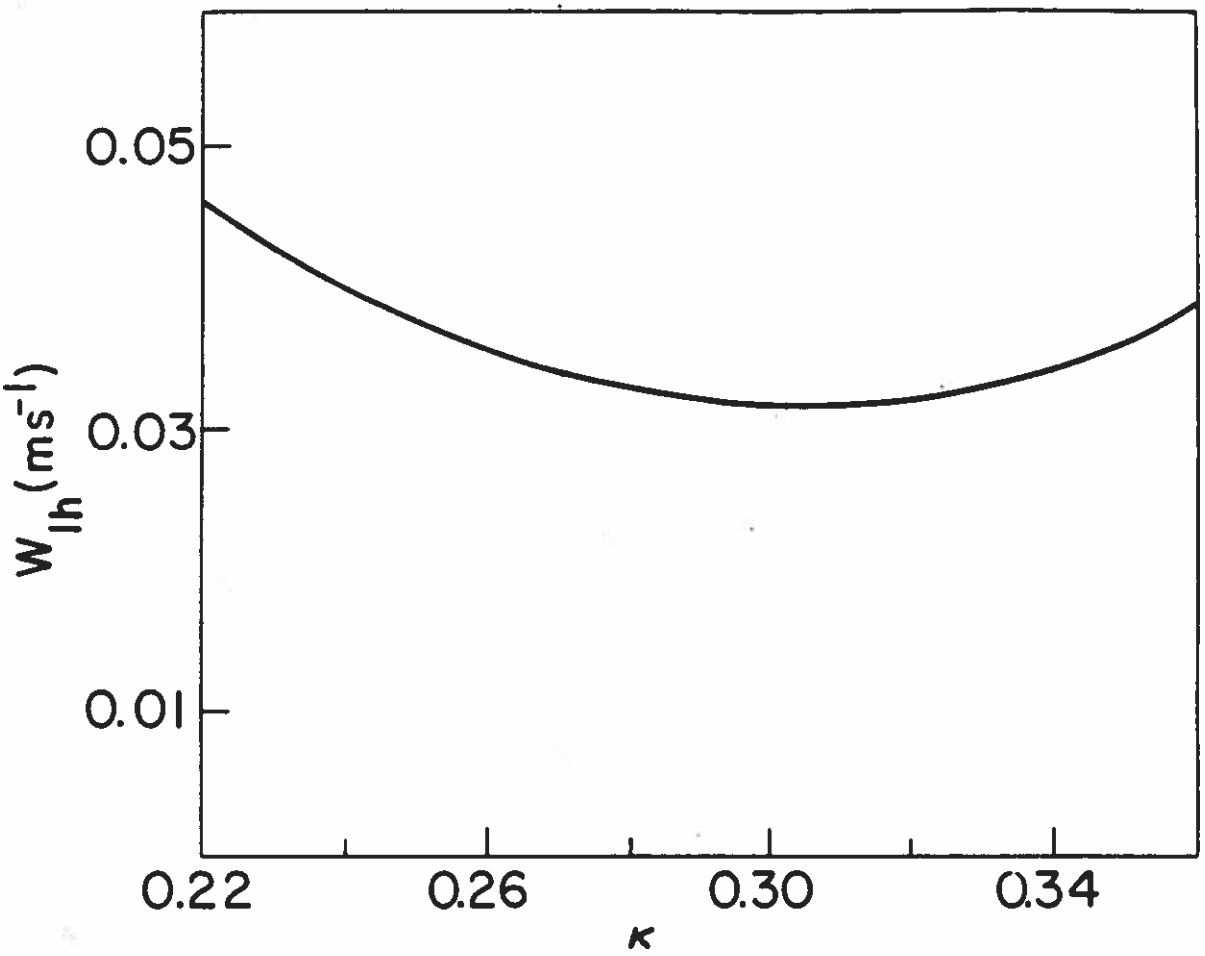


Figure 7.4f W_{1h}, the isotropic contribution of the oxygen holes to the spin-lattice relaxation rate, given in ms^{-1}. W_{1h} is shown to be almost negligible compared to the total W_1's, which range from 1 to $5ms^{-1}$. Calculated as described in text.

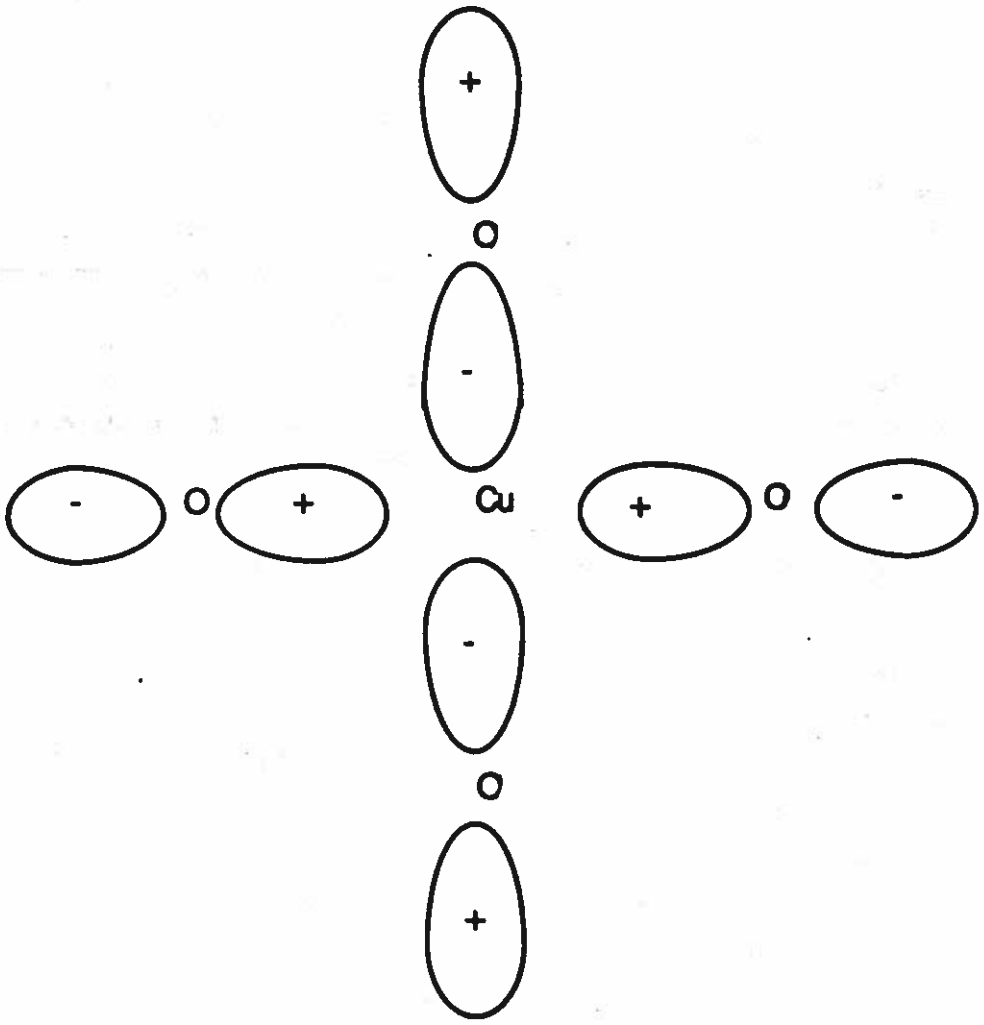


Figure 7.5 2p Oxygen hole orbitals. For equal Cu-O bond lengths along x and y the above orbital has zero overlap with Cu 4s states. In the chain, however, these bond lengths differ considerably and give a non-negligible overlap with Cu 4s.

Chapter 8: Conclusions

Our single crystal NMR study of the anisotropic properties of the high temperature superconductor $YBa_2Cu_3O_{7-\delta}$ has answered several questions. Our body of data consists of the measurement of the complete NMR tensors of four quantities--the electric field gradient, the magnetic shift, the spin-lattice relaxation, and the transverse relaxation at temperatures of 100K and 200K. We contribute information that ranges from the simple chemical assignment of valence charges and the site assignment of resonance lines to the description of the Cu atoms on a fluctuation spectrum and the nature of oxygen holes in the chain.

Analysis of the static NMR tensors--the electric field gradient and the magnetic shift tensor--allows us to assign our Cu NMR resonances to their respective crystallographic sites. We also obtain from these tensors the valence charges of the Cu atoms, Cu^{2+}. These valence assignments indicate an extra hole, which we must then assign to the oxygen atoms in the material, giving the oxygen atoms a valence that is less negative than 2-. It is likely that these "oxygen holes" are primarily involved in the electrical conduction in the material.

A self-consistent analysis of each of the plane Cu NMR tensors indicates a picture in which the plane Cu atoms behave in the normal state very much like atoms with permanent magnetic moments--much like a Cu^{2+} ion in a material such as copper sulfate. Local moments are observed using neutron scattering techniques in the related high temperature superconductor $La_{2-x}Sr_xCuO_4$, with a transition temperature of 30K; however, their existence in the 90K superconductor $YBa_2Cu_3O_{7-\delta}$ has not been detected with neutron scattering due to the necessity of using very large single crystals. Within this model of local moments on the Cu atoms one expects the existence of exchange coupling, just as one has in the oxygen depleted material $YBa_2Cu_3O_{6+x}$, which is an antiferromagnet. We find convincing evidence for the existence of electron exchange coupling in our measurement of the transverse relaxation of the plane Cu. This measurement uncovered an anomoly--one that can not be detected in a powder NMR experiment, where intensities from Cu with varied orientation and crystallographic site overlap--a very strong electron mediated nuclear

spin-spin coupling. We describe quantitatively how this nuclear spin-spin coupling is mediated through an electron exchange coupling with J_{eff}=1700cm^{-1}, the same approximate energy found in the oxygen deficient antiferromagnet.

Whereas the NMR properties of the plane Cu give information about the localized moments on the plane Cu atoms, the chain Cu is a probe of oxygen holes. We find that the chain Cu spin-lattice relaxation rate is isotropic, and that, consistent with previous NQR relaxation measurements, it is linear with temperature. We have identified the source of the relaxation as the transferred hyperfine coupling of the copper nucleus to oxygen holes in the chain, with the oxygen holes hybridized with the Cu 4s states. Combining our results with the superconducting state Knight shift measurements of Barrett et al in our lab we find that the relaxation obeys the Korringa relation between the Knight shift and the spin-lattice relaxation. This demonstrates that the oxygen holes that are involved in relaxation are in a band in which the effects of electron-electron interactions give only small corrections.

APPENDIX A

The Cu Spin Hamiltonian

The Cu nuclear spin Hamiltonian is the sum of the Zeeman Hamiltonian and the electric quadrupolar Hamiltonian. In our experiments we have applied the static magnetic field H_0 along the crystal axes a,b, and c. These axes coincide with the principle axes of the electric field gradient tensor.

We denote the electric potential at the origin as V and define:

$$V_{\alpha\beta} = \frac{\partial^2 V}{\partial\alpha\partial\beta} \qquad (A1)$$

Using the principle axis system we then define:

$$\nu_{\alpha\alpha} = \frac{e\,Q}{2h} V_{\alpha\beta} \qquad (A2)$$

Denoting the nuclear spin as **I**, the static field H_0, the nuclear gyromagnetic ratio γ, and the magnetic shift K we have the following Hamiltonian [Slichter, Chapter 9, 1980]:

$$H = -\gamma_n \hbar H_0(1+K_{zz})I_z + \frac{h}{6}\left[\nu_{zz}(3I_z^2 - \frac{15}{4}) + \frac{(\nu_{xx} - \nu_{yy})}{2}(I_+^2 + I_-^2)\right] \qquad (A3)$$

In the field of 81kG the first term is dominant, but the second term is also important. To obtain a physical feel for the energy eigenvalues of this Hamiltonian we treat the second term as a perturbation and show in Figure A1 its first and second order energy shifts.

For purposes of calculation we have treated the Hamiltonian exactly. We write the Hamiltonian in matrix form, $H_{m,m'}$, setting the quantity $\gamma_n\hbar(H_0(1+K_{zz}))$ equal to 1:

$$H = \begin{pmatrix} (-3/2) + \frac{h}{2}\nu_{zz} & 0 & \sqrt{3}(\nu_{xx}-\nu_{yy}) & 0 \\ 0 & (-1/2) - \frac{h}{2}\nu_{zz} & 0 & \sqrt{3}(\nu_{xx}-\nu_{yy}) \\ \sqrt{3}(\nu_{xx}-\nu_{yy}) & 0 & (1/2) - \frac{h}{2}\nu_{zz} & 0 \\ 0 & \sqrt{3}(\nu_{xx}-\nu_{yy}) & 0 & (3/2) + \frac{h}{2}\nu_{zz} \end{pmatrix} \qquad (A4)$$

This Hamiltonian has been diagonalized exactly by Durand and Barrett [1989]; we present their results for the frequencies of the three transitions in terms of ν_0, defined as $\gamma_n H_0(1+K_{zz})$, and $\underline{\nu}\{\pm\}$, where:

$$\underline{\nu}\{\pm\} = \sqrt{\nu_0^2 + \left(\frac{\nu_{zz}}{2}\right)^2 \pm \nu_{zz}\nu_0 + \frac{1}{3}\left(\frac{\nu_{xx} - \nu_{yy}}{2}\right)^2} \qquad \text{(A5))}$$

The three frequencies are:

$$\nu_{up} = \underline{\nu}\{+\} - \underline{\nu}\{-\} + \nu_0 \qquad \text{(A6)}$$

$$\nu_{middle} = \underline{\nu}\{-\} + \underline{\nu}\{+\} - \nu_0$$

$$\nu_{down} = -\underline{\nu}\{+\} + \underline{\nu}\{-\} + \nu_0$$

In published work we have diagonalized the Hamiltonian by computer, trying different values of magnetic shift and electric field gradient until we found values which fit all three NMR resonance lines.

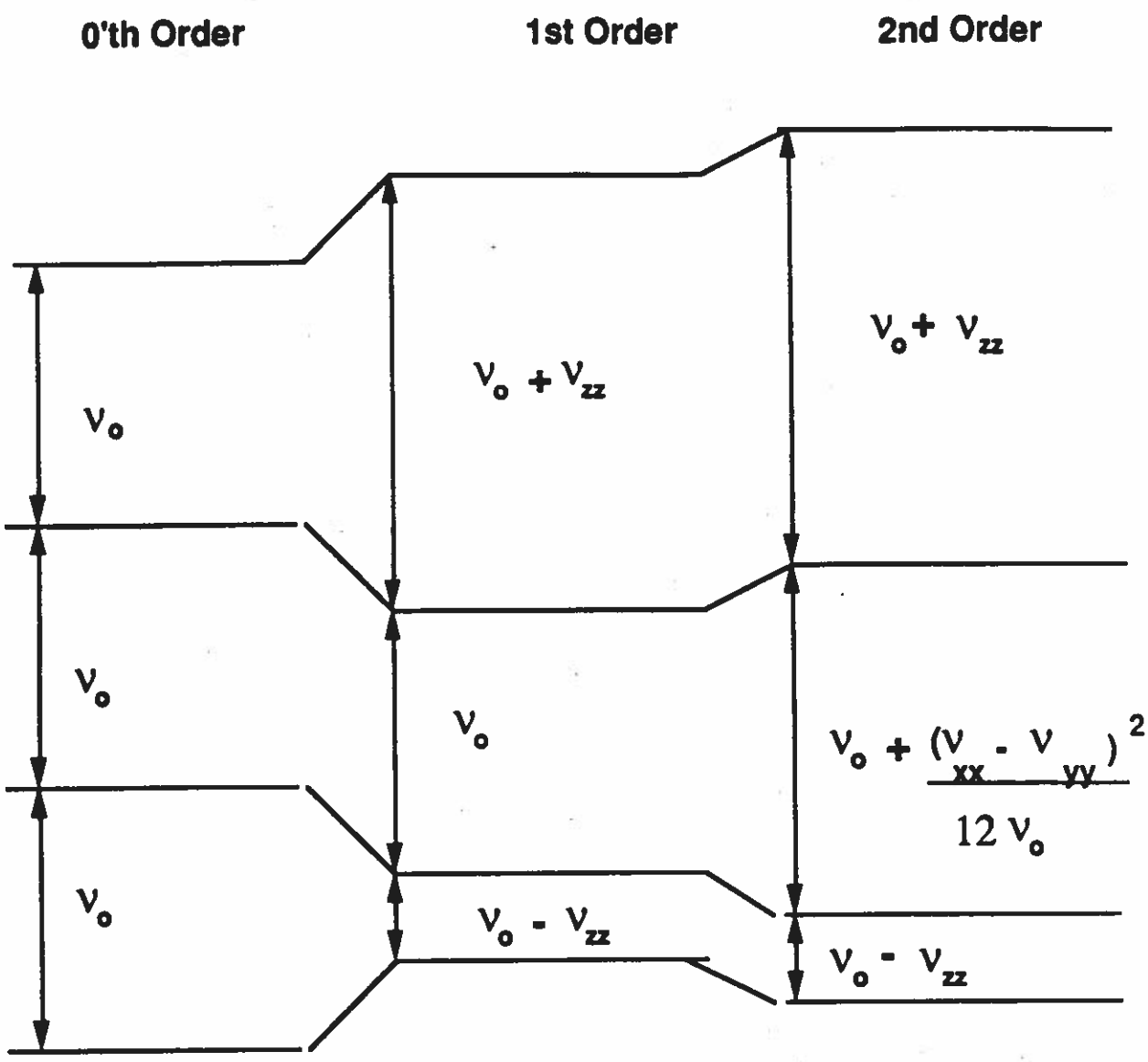


Figure A1 Energy level diagram showing first and second order splittings of resonance lines due to electric quadrupole effects.

APPENDIX B

Chemical Shift Due to a Single Hole in a $3dx^2$-y^2 orbital

We have seen that the Cu atoms in $YBa_2Cu_3O_{7-\delta}$ have a single hole in the $3d(x^2-y^2)$ orbital. The atoms are well described by the energy level diagram shown in Fig. B1.

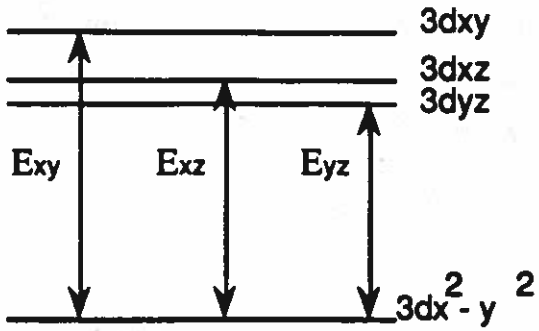


Fig. B1. Cu 3d hole energy levels involved in chemical shift

We wish to calculate the NMR chemical shift for this configuration. First, let us calculate some matrix elements which will be needed. First, we note that the 3d wavefunctions normalized relative to one another are x^2-y^2, 2xy,2xz,2yz, and $1/\sqrt{3}(3z^2-r^2)$. We calculate, then some matrix elements of angular momentum operators between these states:

$$L_z(x^2 - y^2) = i(x\partial y - y\partial x)(x^2 - y^2) = -4ixy \qquad (B1)$$

$$\longrightarrow (2xy \mid L_z \mid x^2 - y^2) = -2i$$

$$L_x(x^2 - y^2) = i(y\partial_z - z\partial_y)(x^2 - y^2) = 2iyz$$

$$\longrightarrow (2yz \mid L_x \mid x^2 - y^2) = i$$

$$L_y(x^2 - y^2) = i(z\partial_x - x\partial_z)(x^2 - y^2) = 2ixz$$

$$\longrightarrow (2xz \mid L_y \mid x^2 - y^2) = i.$$

The so-called "chemical shift" in NMR is the shift of the resonance frequency which results from magnetic fields produced by orbiting electrons. The chemical shift is conveniently divided into two contributions:

1. The Diamagnetic Shift: The diamagnetic shift results from orbital currents which "shield," or reduce, the applied magnetic field. These "shielding" currents can be understood as the result of the modification of the quantum mechanical expression for current density which applies in the presence of a magnetic field:

$$j(r) = \frac{q}{2mi}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) - \frac{q^2}{mc}A\Psi^*\Psi \qquad (B2)$$

If the applied field does not change the wave function, then the first term in the expression is just the current in the absence of a magnetic field--typically zero in a solid, where orbital currents are "quenched."

The second term gives the diamagnetic screening current. For a spatially uniform magnetic field the expression reduces to:

$$j_{dia}(r) = -\frac{q^2}{2mc}(H \times r)\Psi^*\Psi \qquad (B3)$$

The direction of the current flow is such that the magnetic field at the site of the nucleus is reduced. The diamagnetic contribution to the chemical shift is dominant in NMR of protons in organic compounds.

2. The Paramagnetic Shift: Referring back to (B2), let us consider the possibility that the presence of the magnetic field not only changes the form of the quantum mechanical current density operator, but also changes the ground state wavefunction. In that event the chemical shift is due predominantly to the first term in (B1); thus in what follows we omit the second term:

$$j_{para} = \frac{q}{2mi}(\Psi^*\nabla\Psi - \Psi\nabla\Psi^*) \qquad (B4)$$

We refer to the wavefunction in the absence of the field as Ψ_o, and to the perturbed wavefunction as Ψ.

We now treat the effects of the magnetic field upon the wavefunction using first order perturbation theory. The field, described by a vector potential **A**, will add a perturbation to the Hamiltonian:

$$H_{pert} = -\frac{e}{2m}[(\mathbf{A}\cdot\mathbf{p}) + (\mathbf{p}\cdot\mathbf{A})] \qquad (B5)$$

where we have neglected terms which area second order in **A**.

For a field constant in space we write:

$$\mathbf{A} = \frac{1}{2}\mathbf{H} \times \mathbf{r} \qquad (B6)$$

and obtain:

$$H_{pert} = (-e/2m)H_\alpha L_\alpha \qquad (B7)$$

where we consider fields in the α = x, y, or z direction.

The perturbation will change the wavefunction Ψ_o (in our case the $3d(x^2-y^2)$ state. To first order, the perturbed Ψ_o is given by:

$$\Psi_o' = \Psi_o + \frac{(n|H_{pert}|\,0)}{(E_0 - E_n)}\Psi_n$$

$$= \Psi_o + \varepsilon_{n,0}\Psi_n \qquad (B8)$$

The current is given, to first order, as:

$$\mathbf{j}(r) = \frac{e}{2mi}\sum_n(\varepsilon_{n,0} - \varepsilon_{n,0}^*)(\Psi_o\nabla\Psi_n - \Psi_n\nabla\Psi_o) \qquad (B9)$$

The matrix elements $\varepsilon_{n,0}$ are purely imaginary, as shown from (B7) and the calculations given earlier of matrix elements of **L**. Thus,

$$\mathbf{j}(r) = \frac{e}{m}\sum_n \varepsilon_{n,0}(\Psi_o\frac{1}{i}\nabla\Psi_n - \Psi_n\frac{1}{i}\nabla\Psi_0) \qquad (B10)$$

The magnetic field produced by the hole current at the nuclear site (the origin) is given by:

$$\Delta H = \int \frac{r \times j(r)}{r^3} d^3r \qquad \text{(B11)}$$

And the integrand is given by

$$\frac{r \times j(r)}{r^3} = \frac{e}{m}\sum_n \varepsilon_{n,0}[\Psi_o \frac{r}{r^3} \times \frac{1}{i}\nabla\Psi_n - \Psi_n \frac{r}{r^3} \times \frac{1}{i}\nabla\Psi_0] \qquad \text{(B12)}$$

$$= \frac{e}{m}\sum_n \varepsilon_{n,0}(\Psi_o \frac{L}{r^3}\Psi_n - \Psi_n \frac{L}{r^3}\Psi_0)$$

The field produced by the hole, ΔH is then:

$$\Delta H = \frac{-2e}{m}\sum_n \varepsilon_{n,0}(n|\frac{L}{r^3}|0) \qquad \text{(B13)}$$

where again we have used the fact that the matrix elements we consider are pure imaginary.

Let us calculate ΔH explicitly, first for H_0 along z. We see from our earlier matrix elements calculation that L_z couples $x^2 - y^2$ to 2xy. We find:

$$\varepsilon_{2xy,x^2-y^2} = \frac{2i}{E_{xy}}(\frac{-e}{2m})H_0 \qquad \text{(B14)}$$

$$(2xy|\frac{L}{r^3}|0) = -2i\langle\frac{1}{r^3}\rangle$$

with $\langle\frac{1}{r^3}\rangle$ being the expectation value of $\frac{1}{r^3}$.

thus:

$$\Delta H_z = \frac{16\beta^2}{E_{xy}}\langle\frac{1}{r^3}\rangle H_0 \qquad \text{(B15)}$$

where β is the Bohr magneton. On the other hand, a static field along the x-axis connects $x^2 - y^2$ with 2yz, and the matrix element squared is 1/4 as large, as shown. Thus, for a field along the x axis:

$$\Delta H_x = \frac{4\beta^2}{E_{yz}}\langle\frac{1}{r^3}\rangle H_0 \qquad \text{(B16)}$$

similarly for H_0 along y:

$$\Delta H_y = \frac{4\beta^2}{E_{xz}}\langle\frac{1}{r^3}\rangle H_0. \qquad \text{(B17)}$$

In the case that $E_{xy} = E_{yz} = E_{xz}$ the chemical shift for H_0 along z is four times that for the field along x or y.

APPENDIX C

Applying Redfield Theory to Calculate Rates of Transition Between Density Matrix Elements for Relaxation from Fluctuating Magnetic Fields

We consider a spin-3/2 nucleus, such as the Cu^{63} and Cu^{65} studied in this thesis. In a strong magnetic field the eigenstates of the nuclear spin Hamiltonian are the four eigenstates of the operator I_z, m=3/2, 1/2,-1/2,-3/2. The energy level diagram is shown in Figure C1:

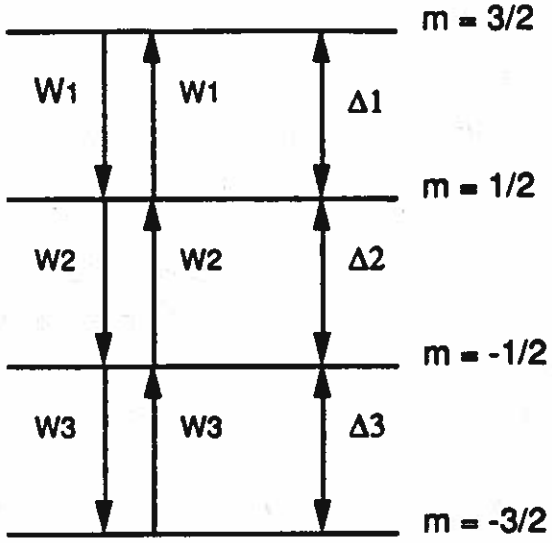


Figure C1

The energy splittings Δ_i are equal in the absence of electric quadrupole effects, and given by $\gamma_n H_0$. In $YBa_2Cu_3O_{7-\delta}$, however, lattice anisotropy results in quadrupole effects that make the energy splittings significantly different.

We will use the density matrix formalism to evaluate the transition probabilities W_i and the NMR T_2. The NMR spin-lattice relaxation is described in terms of the W_i in Appendix D.

We shall refer to $\rho_{\alpha\beta}$ as the "density matrix," although in fact we define $\rho_{\alpha\beta}$ as the actual density matrix in the rotating frame less the density matrix at thermal equilibrium. We shall use the notation given in Slichter Section 5.4-5.7. When using equations from the Slichter text we list equation numbers as well.

The Redfield equation describing density matrix transitions is given by:

$$\frac{d\rho_{\alpha\alpha'}}{dt} = \sum_{\beta,\beta'} R_{\alpha\alpha',\beta\beta'}\rho_{\beta\beta'} \qquad \text{(C1)} \quad (5.151)$$

where the energies of the states are labeled by the subscripts α. It is further stipulated that $R_{\alpha\alpha',\beta\beta'} = 0$ unless $\alpha - \alpha' = \beta - \beta'$.

We calculate the various $R_{\alpha\alpha',\beta\beta'}$ using the relations:

$$R_{\alpha\alpha',\beta\beta'} = \frac{1}{2}[2J_{\alpha\beta,\alpha'\beta'}(\alpha - \beta) - \delta_{\alpha'\beta'}\sum_{\gamma} J_{\gamma\beta\gamma\alpha}(\gamma - \beta) - \delta_{\alpha\beta}\sum_{\gamma} J_{\gamma\beta'\gamma\alpha'}(\gamma - \beta')] \quad \text{(C2)} \quad (5.177)$$

where:

$$J_{\alpha\alpha'\beta\beta'}(\omega) = \int_{-\infty}^{+\infty} \overline{(\alpha|H_1(t)|\alpha')(\beta|H_1(t+\tau)|\beta')}\, e^{-i\omega\tau} d\tau. \qquad \text{(C3)}$$

(5.175)

$H_1(t)$ is the time dependent part of the spin Hamiltonian.

We now use these results to calculate the R's in terms of the J's. We will use the fact that our Hamiltonian H_1 results from a fluctuating magnetic field and gives terms with the spin operators I_α where α = x,y, and z; thus it connects states with $\Delta m = 0$, +1, or -1 only.

The following R's are directly related to the transition rates between states of differing m:

$$R_{3/2\ 3/2,1/2,1/2} = W_1 = J_{3/2\ 1/2,3/2\ 1/2}(\omega_{3/2,1/2}) \qquad \text{(C4)}$$

$$R_{1/2\ 1/2,-1/2,-1/2} = W_2 = J_{1/2\ -1/2,1/2\ -1/2}(\omega_{1/2,-1/2})$$

$$R_{-1/2\ -1/2,-3/2\ -3/2} = W_3 = J_{-1/2\ -3/2,-1/2\ -3/2}(\omega_{-1/2,-3/2})$$

By using the relation $\langle o \rangle = Tr(o\rho)$ one can show that an R of the form $R_{\alpha\ \alpha-1,\alpha\ \alpha-1}$ is equal to $\frac{1}{T_2}$ for the α,α-1 transition. We give these R's in terms of the J's. In some cases we have omitted the argument for $J_{\alpha\alpha'\beta\beta'}(\omega)$; in each case it is given by $\omega_{\alpha\alpha'}$.

$$R_{3/2\ 1/2,3/2\ 1/2} = (1/2)[2J_{3/2\ 3/2\ 1/2\ 1/2}(\omega_{3/2\ 1/2}) - \sum_{\gamma} J_{\gamma 3/2\gamma 3/2}(\omega_{\gamma 3/2}) - \sum_{\gamma} J_{\gamma 1/2\gamma 1/2}(\omega_{\gamma 1/2})] \qquad \text{(C5)}$$

$$= (1/2)[2J_{3/2\ 3/2,1/2\ 1/2} - (J_{3/2\ 3/2,3/2\ 3/2} + J_{1/2\ 3/2,1/2\ 3/2}) - (J_{3/2\ 1/2,3/2\ 1/2} + J_{1/2\ 1/2,1/2\ 1/2} + J_{-1/2\ 1/2,-1/2\ 1/2})]$$

$$= -W_1 - W_2/2 + (1/2)[2J_{3/2\ 3/2,1/2\ 1/2} - J_{1/2\ 1/2,1/2\ 1/2} - J_{3/2\ 3/2,3/2\ 3/2}]$$

For $R_{1/2\ -1/2,1/2\ -1/2}$ we find:

$$R_{1/2\ -1/2,1/2\ -1/2} = -W_2 - W_1/2 - W_3/2 + (1/2)[2J_{1/2\ 1/2,-1/2\ -1/2} - J_{1/2\ 1/2,1/2\ 1/2} - J_{-1/2\ -1/2,-1/2\ -1/2}] \qquad \text{(C6)}$$

Clearly $R_{-1/2\ -3/2,-1/2\ -3/2}$ can be determined by inspection of $R_{3/2\ 1/2,3/2\ 1/2}$.

Now we need to explicitly work out these results for the fluctuating field $H_\alpha(t)$. We define $h_\alpha(t) = -\gamma_n H_\alpha(t)$. We then assume a simple model for the fluctuating fields: each component of h_α fluctuates independently and randomly with an exponential correlation time τ_0 (See Slichter 5.141 p.166 and Appendix C) so that:

$$\overline{h_\alpha(0)h_\alpha(\tau)} = \overline{h_{\alpha 0}^2}\exp(-2|\tau| / \tau_0) \qquad \text{(C7)}$$

We then find for W_1 and W_2:

$$W_1 = (3/2)(\overline{h_{x0}^2} + \overline{h_{y0}^2})\frac{\tau_0}{1+(\omega_{3/2,1/2})^2\tau_0^2} \qquad \text{(C8a)}$$

$$W_2 = (2)(\overline{h_{x0}^2} + \overline{h_{y0}^2})\frac{\tau_0}{1+(\omega_{1/2,-1/2})^2\tau_0^2} \qquad \text{(C8b)}$$

W_3 can be obtained from W_1 by replacing 3/2,1/2 by -3/2,-1/2. We note briefly that in the short correlation time limit, when $\omega\tau_0 << 1$, we have $W_2 = (4/3)W_1$. This limit applies well to the work in this thesis, where ω is of the order of $2\pi \times 90\times10^6 sec^{-1}$, and τ_0 is about 2×10^{-15}sec.

We now give the results of calculations of the J's needed to find T_2 for the various transitions:

$$J_{3/2\ 3/2,3/2\ 3/2} = \frac{9}{4}(\overline{h_{z0}^2})(2\tau_0) \qquad \text{(C9a)}$$

$$J_{3/2\ 3/2,1/2\ 1/2} = \frac{3}{4}(\overline{h_{z0}^2})(2\tau_0) \qquad \text{(C9b)}$$

$$J_{1/2\ 1/2,1/2\ 1/2} = \frac{1}{4}(\overline{h_{z0}^2})(2\tau_0) \qquad \text{(C9c)}$$

$$J_{1/2\ 1/2,-1/2\ -1/2} = \frac{-1}{4}(\overline{h_{z0}^2})(2\tau_0) \qquad \text{(C9d)}$$

Thus, we give T_2's:

$$\left(\frac{1}{T_2}\right)_{3/2,1/2} = W_1 + W_2/2 + \overline{h_{z0}^2}\,\tau_0 \qquad \text{(C10a)}$$

$$\left(\frac{1}{T_2}\right)_{1/2,-1/2} = W_2 + W_1/2 + W_3/2 + \overline{h_{z0}^2}\,\tau_0 \qquad \text{(C10b)}$$

We now give these results in the limit of $\omega\tau_0$ approaching zero, the short correlation time limit:

$$W_1 = (3/2)(\overline{h_{x0}^2} + \overline{h_{y0}^2})\tau_0 \qquad \text{(C11a)}$$

$$W_2 = (4/3)W_1 \qquad \text{(C11b)}$$

$$\left(\frac{1}{T_2}\right)_{3/2,1/2} = (5/3)W_1 + \overline{h_{z0}^2}\,\tau_0 \qquad \text{(C12a)}$$

$$\left(\frac{1}{T_2}\right)_{1/2,-1/2} = (7/3)W_1 + \overline{h_{z0}^2}\,\tau_0 \qquad \text{(C12b)}$$

APPENDIX D

Cu NMR Spin-Lattice Relaxation in the Presence of Quadrupolar Splittings

Cu^{63} and Cu^{65} are spin-3/2 nuclei with four eigenstates of the spin Hamiltonian. In a strong magnetic field along z the eigenstates of the Hamiltonian are the four eigenstates of the operator I_z, m=3/2,1/2,-1/2,-3/2. The energy level diagram is shown in Figure D1:

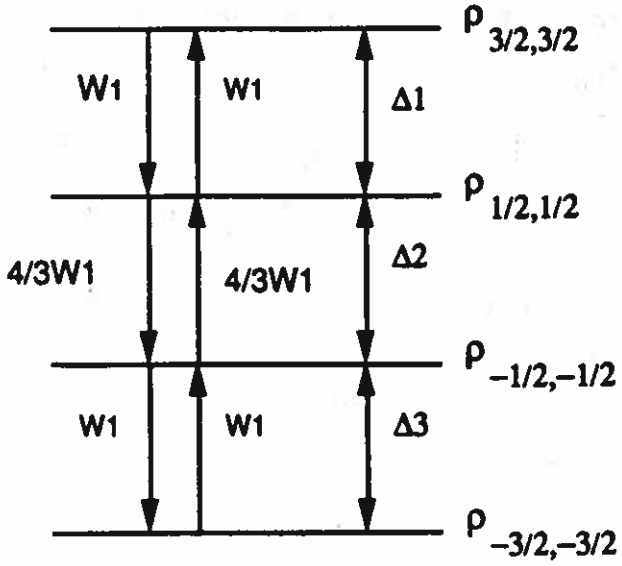


Figure D1

The levels for m=α are labeled by the corresponding density matrix element $\rho_{\alpha\alpha}$. We shall refer to $\rho_{\alpha\beta}$ as the "density matrix" although in fact we define it as the actual density matrix $P_{\alpha\beta}$ in the rotating frame less the density matrix at thermal equilibrium. The element $\rho_{\alpha\alpha}$, then, refers to the dev'ation of the population level of the m=α state from its thermal equilibrium population.

The assignment shown in Figure D1of the relative relaxation rates for the different transitions is based on the assumption that the relaxation results from a fluctuating magnetic field rather than from a fluctuating electric field gradient, and on the assumption that the time scale for the fluctuations of this magnetic field is much faster than the NMR resonance frequency (See Appendix C.). We have shown experimentally that these assumptions are correct.

With the four-level system with three possible transitions the relaxation to thermal equilibrium will not in general be characterized by a

single exponential (T_1), but by as many as three exponentials. A non-equilibrium state of the system can be described as a superposition of three "normal modes" of relaxation, each of which relaxes with a single exponential.

We will now write the differential equation described by Figure D1 and find the normal modes of relaxation. We then consider various NMR pulse sequences which excite non-equilibrium states and examine the functional form of their relaxation back to equilibrium.

The differential equation described by Figure D1 is given by:

$$\frac{d}{dt}\begin{pmatrix} \rho_{3/2,3/2} \\ \rho_{1/2,1/2} \\ \rho_{-1/2,-1/2} \\ \rho_{-3/2,-3/2} \end{pmatrix} = W_1 \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -7/3 & 4/3 & 0 \\ 0 & 4/3 & -7/3 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} \rho_{3/2,3/2} \\ \rho_{1/2,1/2} \\ \rho_{-1/2,-1/2} \\ \rho_{-3/2,-3/2} \end{pmatrix} \qquad \text{(D1)}$$

Diagonalization of the above matrix gives the three relaxation normal modes (and a trivial solution, corresponding to thermal equilibrium). The normalized eigen-modes, together with the corresponding relaxation rate of each mode, are given:

$$\text{(1)} \qquad \frac{1}{2}\begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix} \quad : \quad 2W_1 \qquad \text{(D2)}$$

$$\text{(2)} \qquad \frac{1}{\sqrt{5}}\begin{pmatrix} 3/2 \\ 1/2 \\ -1/2 \\ -3/2 \end{pmatrix} \quad : \quad 2/3W_1$$

$$\text{(3)} \qquad \frac{1}{\sqrt{5}}\begin{pmatrix} 1/2 \\ -3/2 \\ 3/2 \\ -1/2 \end{pmatrix} \quad : \quad 4W_1$$

Now let us consider the excitations produced by various NMR pulses. We consider first the simplest situation. We suppose that quadrupole splittings of the resonance lines are absent. This situation arises, for example, in Cu metal, which has a cubic lattice. In this situation $\Delta_1=\Delta_2=\Delta_3$. An NMR pulse applied at the frequency of these transitions will "flip," or invert the population of the entire density matrix. The static Hamiltonian is given by:

$$H_{stat} = -\gamma_n H_0 I_z \qquad \text{(D3)}$$

Thus the thermal equilibrium density matrix $P_{\alpha\beta}$ will be given by:

$$(P_{\alpha\beta})_{eq} = \exp(-H_{stat}/k_BT)_{\alpha\beta} \approx 1 + \gamma_n H_0 I_z/k_BT \qquad (D4)$$

For the purposes of our experiment the constant one is unimportant, so we have:

$$(P_{\alpha\beta})_{eq} \sim (I_z)_{\alpha\beta} \qquad (D5)$$

A so-called 180° pulse will invert I_z to minus I_z. The density matrix $\rho_{\alpha\beta}$, defined as $P_{\alpha\beta} - (P_{\alpha\beta})_{eq}$ will then be proportional to I_z. The diagonal elements, however, of the operator I_z, however, are the same as eigen-mode (2)! Thus, we conclude that such a system will relax with a single exponential rate $2/3W_1$. In Cu metal, for example, the single exponential relaxation is expressed in terms of $1/T_1$, where $1/T_1=2/3W_1$.

We next consider another simple case in which only a single exponential is involved, namely, a Nuclear Quadrupole Resonance (NQR) experiment, with zero magnetic field. We suppose that there is an electric field gradient (EFG) tensor which is axially symmetric about the z axis, as is the case for the plane Cu in $YBa_2Cu_3O_{7-\delta}$, where we associate the crystal c-axis with z. (Note: The chain Cu electric field gradient is not axially symmetric about c; in fact it is close to antisymmetric, with $v_{xx} = -v_{yy}$. In a zero field NQR experiment the chain Cu is not adequately described by Figure D1. We will treat the chain NQR as a seperate case.) In this situation the static Hamiltonian is given by:

$$H_{stat} = \frac{\omega_Q}{2} I_z^2 + \text{const.} \qquad (D6)$$

The m=1/2 and -1/2 levels are degenerate, as are m=3/2 and -3/2. A pulse which flips the (3/2,1/2) transition must also flip the (-1/2,-3/2) transition. The resulting non-equilibrium state will be proportional to eigen-mode (1). Thus, in an NQR experiment with an EFG tensor axially symmetric about z the relaxation rate is a single exponential rate $2W_1$.

The NMR spectrum of Cu in $YBa_2Cu_3O_{7-\delta}$, referring to Figure D1, has $\Delta_1 \neq \Delta_2 \neq \Delta_3$, and an rf pulse at the frequency of Δ_2 will only excite the (1/2,-1/2) transition. In order to measure spin-lattice relaxation one "flips" the (1/2,-1/2) transition, i.e. one very quickly inverts the populations $\rho_{1/2,1/2}$ and $\rho_{-1/2,-1/2}$, and then, after a parameter time τ, one observes by means of a spin-echo the population difference $\rho_{1/2,1/2} - \rho_{-1/2,-1/2}$. (In principle one

could observe a transition different from the one flipped.) The initial state following the population inversion is given by:

$$\Psi(\tau=0) = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} \qquad \text{(D7)}$$

We can easily expand this state in terms of eigen-modes (2) and (3) to obtain the time dependence:

$$\Psi(\tau) = \begin{pmatrix} .3 \\ .1 \\ -.1 \\ -.3 \end{pmatrix} \exp(-\tfrac{2}{3}W_1\tau) + \begin{pmatrix} -.3 \\ .9 \\ -.9 \\ .3 \end{pmatrix} \exp(-4W_1\tau) \qquad \text{(D8)}$$

The measured quantity $\rho_{1/2,1/2} - \rho_{-1/2,-1/2}$ is given by:

$$\rho_{1/2,1/2} - \rho_{-1/2,-1/2}(\tau) = .2\exp(-2/3W_1\tau) + 1.8\exp(-4W_1\tau)$$

(D9)

giving a 10 percent component at $2/3W_1$ and a 90 percent component at $4W_1$. Note that Ψ gives the population differences of all transitions; one could, for example, flip the (1/2,-1/2) transition and then observe the (3/2,1/2) transition.

If one tries a similar experiment on the (3/2,1/2) transition, i.e. one flips this transtion and then observes $\rho_{3/2,3/2} - \rho_{1/2,1/2}(\tau)$, similar analysis gives:

$$\rho_{3/2,3/2} - \rho_{1/2,1/2}(\tau) = \exp(-2W_1\tau) + .8\exp(-4W_1\tau) + .2\exp(-\tfrac{2}{3}W_1\tau)$$

(D10)

giving a 50 percent $2W_1$ component, 40 percent $4W_1$, and 10 percent $2/3W_1$.

We have observed a quantitative agreement with these functional forms in our study of Cu in $YBa_2Cu_3O_{7-\delta}$. Shown in Figure 5.2 is the spin-lattice relaxation of the plane Cu with H_0 parallel to the crystal c axis. The (1/2,-1/2) transition at 92.79MHz, together with the (3/2,1/2) and (-3/2,-1/2) transitions at 124 and 61MHz. The fits shown are using the functional forms derived above with the **same** W_1 for each transition.

We now consider the spin-lattice relaxation of the chain Cu in a nuclear quadrupole resonance experiment; as mentioned earlier, the chain Cu NQR is not described by Figure D1. The chain Cu electric field gradient is not axially symmetric; in fact, to a good approximation, we have $\nu_{xx} = -\nu_{yy}$ and $\nu_{zz} = 0$. In that limit the eigenstates are not eigenstates of

I_z with quantum numbers m. Instead they are linear combinations of the states |m>. We label the states a, b, c, and d:

$$|a\rangle = \frac{1}{\sqrt{2}}(|3/2\rangle + |-1/2\rangle) \qquad E = \frac{1}{2}\omega_Q \qquad \text{(D11)}$$

$$|b\rangle = \frac{1}{\sqrt{2}}(|3/2\rangle - |-1/2\rangle) \qquad E = -\frac{1}{2}\omega_Q$$

$$|c\rangle = \frac{1}{\sqrt{2}}(|-3/2\rangle + |1/2\rangle) \qquad E = \frac{1}{2}\omega_Q$$

$$|d\rangle = \frac{1}{\sqrt{2}}(|-3/2\rangle - |1/2\rangle) \qquad E = -\frac{1}{2}\omega_Q$$

The relaxation diagram is shown in Figure D2:

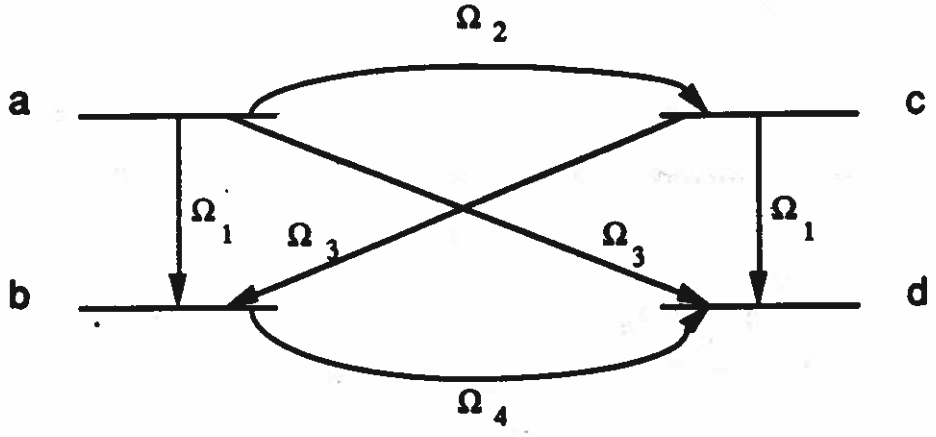


Figure D2: Relaxation diagram for the chain Cu NQR.

We can calculate the rates Ω using Redfield theory as in Appendix C, and again assuming that relaxation is from a fluctuating magnetic field with a correlation time fast compared to the NQR frequency. We obtain:

$$\Omega_1 = 2\gamma_n^2 h_z^2 \tau_0 \qquad \text{(D12)}$$

$$\Omega_2 = \frac{(\sqrt{3}+1)^2}{2}\gamma_n^2 h_x^2 \tau_0 + \frac{(\sqrt{3}-1)^2}{2}\gamma_n^2 h_y^2 \tau_0$$

$$\Omega_3 = \frac{1}{2}\gamma_n^2 (h_x^2 + h_y^2)\tau_0$$

$$\Omega_4 = \frac{(\sqrt{3}-1)^2}{2}\gamma_n^2 h_x^2 \tau_0 + \frac{(\sqrt{3}+1)^2}{2}\gamma_n^2 h_y^2 \tau_0$$

The differential equations described by Figure D2 are given by:

$$\frac{d}{dt}\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} -(\Omega_1+\Omega_2+\Omega_3) & \Omega_1 & \Omega_2 & \Omega_3 \\ \Omega_1 & -(\Omega_1+\Omega_3+\Omega_4) & \Omega_3 & \Omega_4 \\ \Omega_2 & \Omega_3 & -(\Omega_1+\Omega_2+\Omega_3) & \Omega_1 \\ \Omega_3 & \Omega_4 & \Omega_1 & -(\Omega_1+\Omega_3+\Omega_4) \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$$

(D13)

In an Inversion Recovery NQR measurement of $1/T_1$ the quantities $\rho_{aa}-\rho_{cc}$ and $\rho_{bb}-\rho_{dd}$ remain equal, since in the inversion pulse the c to a and d to b transitions are excited equally. The relaxation mode, then, must be:

$$\begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} .5 \\ -.5 \\ .5 \\ -.5 \end{pmatrix} \qquad \text{(D14)}$$

Indeed, (D14) is easily shown to be an eigenvector of the relaxation matrix with eigenvalue $-2(\Omega_1 + \Omega_3)$. Thus, we have:

$$\left(\frac{1}{T_1}\right)_{NQR} = 2(\Omega_1 + \Omega_3) \qquad \text{(D15)}$$

or, using Relations (D12),

$$\left(\frac{1}{T_1}\right)_{NQR} = \frac{3}{2}\gamma_n^2\left(\frac{8}{3}h_z^2 + \frac{2}{3}(h_x^2 + h_y^2)\right)\tau_0 \qquad \text{(D16)}$$

Using the experimental values of $(3/2)h_\alpha^2\tau_0$ from Table 5.3 we obtain:

$$\left(\frac{1}{T_1}\right)_{NQR} = 2.17\text{ms}^{-1}$$

This result is in good agreement with our measurements (Figure 2.3) and with those of other groups.

APPENDIX E

The Pulse Sequence Generator

The function of the Pulse Sequence Generator (PSG) is the timed control of exeriments. A PSG of some form is a necessary part of any pulsed NMR rig; it essentially controls the entire experiment through its complete control of the timing and phases of rf pulses and the triggering of the signal digitizer.

The PSG exerts control through 24 TTL outputs. We refer to the state of the 24 TTL outputs as the "output state" of the PSG. To control an experiment one typically wishes to put out a sequence of N output states S_1, S_2,...S_N for duration times t_1, t_2,...t_N. We shall refer to a single output state, S_n, together with its corresponding duration time t_n, as a "line." Among the desired features of the PSG are that the times t_n may be varied over a very wide range and with high precision, that there is an adequate number of outputs, and that the PSG can accomodate enough lines to perform the experiment contemplated.

Our lab has previously used a PSG designed by Zhiyue Wang [1986]. The new PSG incorporates many of the same design features; however, it has been improved to accomodate as many as 8192 lines, whereas the previous PSG accomodated 256. The number of lines that are necessary is a function of the complexity of the experiment. In Section 4.4 we describe various pulse sequences that have been used in this thesis in order to accomplish various objectives. Most of the sequences involve an extra factor of 2, 4, 8, or even more in the number of PSG lines required. Prior to the construction of the new PSG we often had to simplify our pulse sequences in order to maintain fewer than 256 lines. At present we routinely use sequences of 800-1000 lines. With this new ability to use the more complex sequences we are able to see much tinier signals that might otherwise be obscured by the coil ringdown (Section 4.43). Using the Alternating Phase Inversion Recovery we can measure the extremely fast relaxation rates encountered, for example, with the plane Cu with H_0 parallel to a (Section 4.45). We may perform all of these experiments using the technique described in Section 4.47 to eliminate the effects of drift--all parameter times are put on one pulse sequence, so each parameter time

will be signal averaged over the same time period, and each time will experience the same "drift," if present.

The PSG holds the pulse sequence (consisting of the PSG "lines" in the proper order) in a random access memory (RAM). The operator can "load" the PSG with a pulse sequence sequence using the IBM PC-XT. The GPIB system is used for the interface of the computer with the PSG; the interfacing is accomplished with a circuit designed by Joe Ross and Chris Klug [Ross, 1986], the "GPIB to PSG Interface." The software for the interfacing has been written by Durand [1989].

A schematic of the PSG is given in Figure E1. The schematic does not describe completely the loading of the PSG but describes the PSG in the running state. The RAM memory is shown with 8k lines and 64 bits per line. The RAM consists of 8 AMC9988 chips by Advanced Micro Devices, each with 8 bits by 8k lines. For a given line 40 of the 64 bits are used to specify the line's duration time in units of the clock pulse. Our 10MHz clock gives a unit of .1μs. The minimum time that can be used is, however, .2μs, due to finite chip switching speeds. The maximum time is 2^{40}x.1μs = 30.5 hours. The remaining 24 bits per line give the 24 TTL outputs.

The memory is addressed by a 13 bit counter (note that 2^{13}=8192, the full number of lines in the memory). Let us suppose that the n'th line in the memory is being addressed. At that time the (n-1)'th line will be at the outputs of the PSG, and will remain there for a duration t_{n-1}. The time is counted down by a 40 bit count-down counter/timer; at each clock pulse the time is counted down by one unit. As the (n-1)'th line finishes, the count-down counter/timer will reach a count of zero. At that time it will trigger several events. First, the count-down counter/timer loads a new forty bit time, time t_n, which is waiting at the output RAM memory on a 40 bit bus. Simultaneously a flip-flop is triggered and the output state of the PSG switches from that of the (n-1)'th line to that of the n'th line. Finally, with the new time loaded the memory address counter is advanced by one to address the (n+1)'th line.

The lines are counted through until the end of the sequence is reached. Let us suppose, then that the n'th line is the last line of the sequence. Now, in an NMR experiment one typically wishes to repeat a

sequence many times and signal average. If the sequence contains the full 8192 lines (which it usually does not), then the 13 bit count-up memory address will start over at zero after it reaches 8192. If there are fewer than 8192 lines, then a dummy line, which will never appear at the outputs of the PSG, must be inserted. The dummy line is tagged by having all of its final seven output bits high (this is then a forbidden state for an ordinary PSG line). With the dummy (n+1)'th line at the ouput of the RAM memory the n'th (final) line will be at the output of the PSG. The (n+1)'th line is immediately identified as the dummy line, and the 13 bit count-up memory address is set to zero, putting the zero'th line on the memory address. At the conclusion of the n'th (final) line the PSG restarts the sequence. This solution is different from Wang's PSG. Wang's PSG holds a maximum of 256 lines. For sequences of less than 256 lines the final line is partitioned into many lines of shorter time duration, enough lines to give 256 total lines. This solution is appropriate for only 256 lines. With 8192 lines, however, the final line would have to be divided into up into approximately 8000 lines. With a minimum of .2μs duration for each line the sequence would require a total duration of at least 1.6ms; we sometimes repeat sequences more quickly, as fast as once per ms.

Copies of the PSG board as well as the GPIB to PSG Interface are available from Pentaflex, 11314 Main St., Huntley IL; (312)-669-3350.

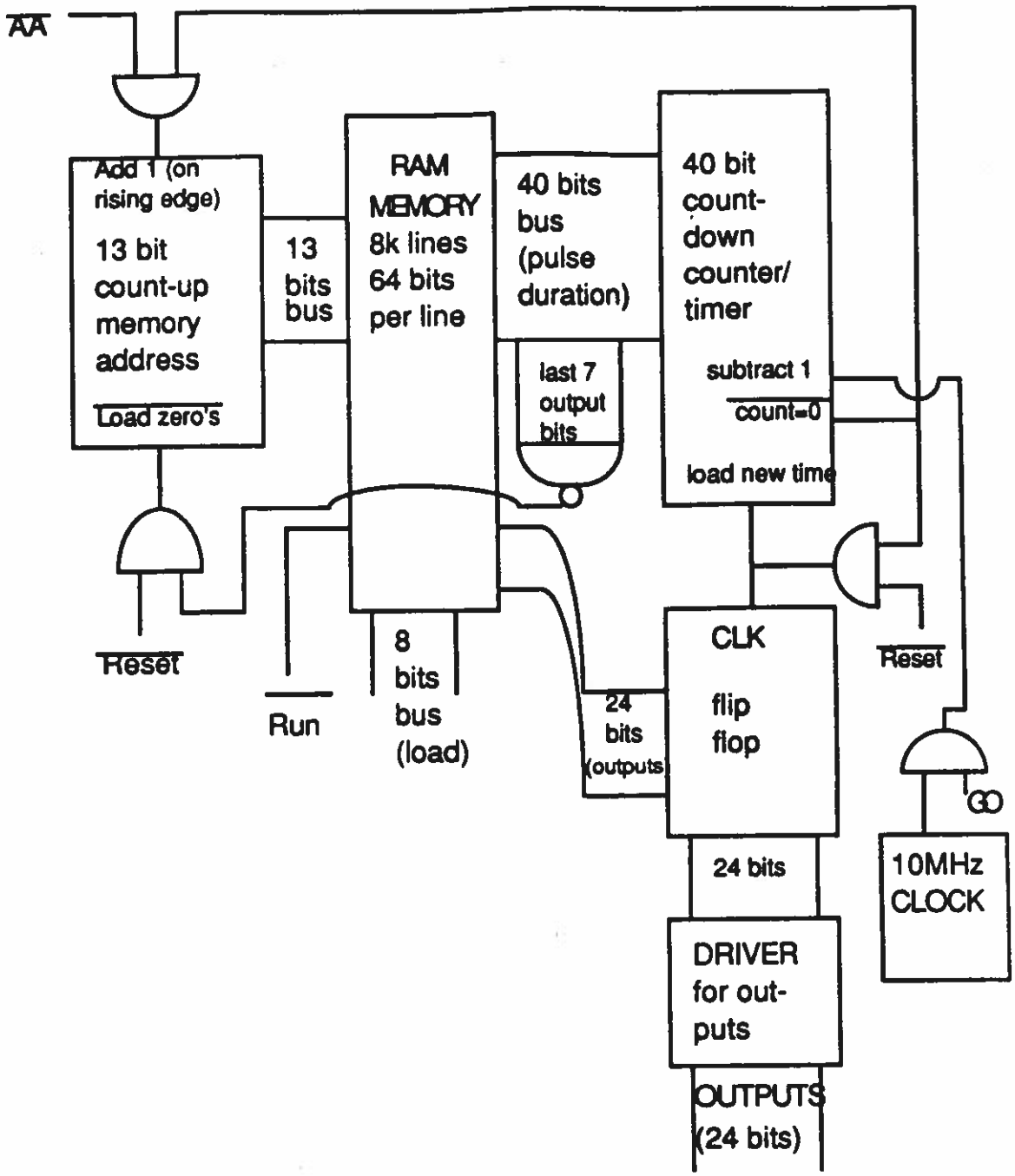


Figure E1: Schematic diagram of the Pulse Sequence Generator (PSG). The schematic describes the operation of the PSG when it is running a sequence. Details of the process of loading the PSG are not shown.

APPENDIX F

The Alternating Phase Inversion Recovery

The Alternating Phase Inversion Recovery Sequence is described in Section 4.45. The purpose of using an alternating phase inversion pulse in an inversion recovery sequence is to cancel the so-called "stimulated echoes" which occur when one applies three pulses. There may be as many as four stimulated echoes; they are described by Hahn [1950].

We now look briefly at how one may cancel the stimulated echoes by alternating the phase of the inversion pulse. The inversion recovery pulse is shown in Figure F1 (stimulated echoes not pictured):

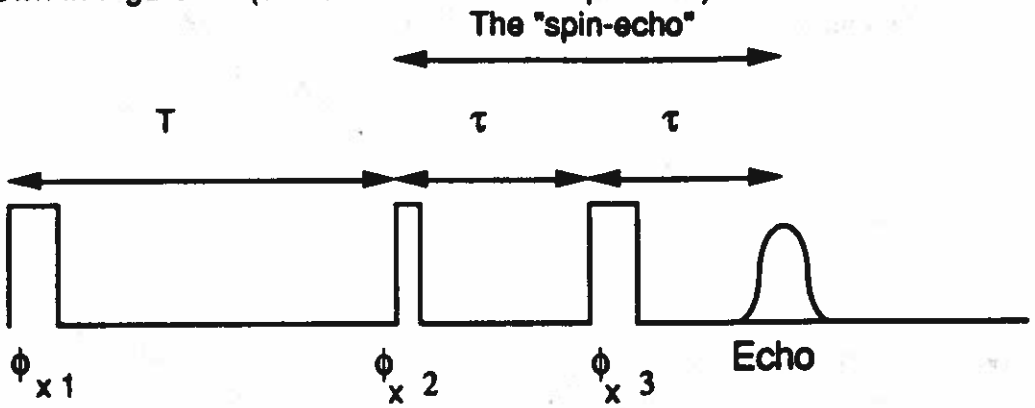


Figure F1: The Inversion-recovery sequence

To analyze this sequence we will make use of the Pictorial Product Operator Formalism described by Wang and Slichter [1986], and of some of the simple results derived in this reference.

We first give an intuitive argument about why this sequence works. We note that the result of a spin-echo experiment on magnetization initially along z is well understood; it gives a single echo proportional to the initial magnetization at a time following the 180° pulse that is equal to the time between the 90° and 180° pulses. The stimulated echoes in a three pulse sequence described by Hahn must then result from transverse magnetization (or, more precisely, coherence between levels differing by $\Delta m = \pm 1$) present just prior to the time of the second pulse. This transverse magnetization results from the imperfection of the 180° inversion pulse. If the pulse is not exactly 180°, then immediately following it transverse magnetization will be present. Intuitively, one knows that this transverse magnetization must be the source of the stimulated echoes. If, however, we

alternate the phase of the 180° inversion pulse, alternately pulsing about the +x axis and the -x axis, then we will alternate the sign of this initial transverse magnetization. One expects intuitively that the stimulated echoes will also have alternating signs and will cancel if one averages over the two cases.

We now give a complete argument, using the techniques of the Product Operator Formalism, to prove the result.

Immediately prior to the initial inversion pulse one has a density matrix I_z. Immediately following the pulse, which we assume to be a pulse of ϕ_1 about the x axis in the rotating frame, the density matrix is given by $I_z\cos\phi_1 + I_y\sin\phi_1$.

We work in the rotating frame which is on resonance for the magnetization considered. Then the time evolution of the density matrix during the period T before the second pulse will not display a precession, but will only relax, following the Bloch equations. After the evolution of time T the new density matrix is:

$$I_z[1+(\cos\phi_1 - 1)e^{-T/T1}] + I_y(\sin\phi_1)e^{-T/T2} \quad (F1)$$
$$\equiv pI_z + qI_y$$

where we have defined $p \equiv [1+(\cos\phi_1 - 1)e^{-T/T1}]$ and $q \equiv \sin\phi_1 e^{-T/T2}$.

After time T one applies a second pulse of angle ϕ_2 about the x axis. This results in a new density matrix:

$$p(I_z\cos\phi_2 + I_y\sin\phi_2) + q(I_y\cos\phi_2 - I_z\sin\phi_2) \quad (F2)$$
$$= (p\cos\phi_2 - q\sin\phi_2)I_z + (p\sin\phi_2 + q\cos\phi_2)I_y$$

Again, the new density matrix propagates, now for a time τ until the next pulse. The new density matrix immediately preceding the third pulse is:

$$[1 + \{(p\cos\phi_2 - q\sin\phi_2)I_z - 1\}e^{-\tau/T1}]I_z + [(p\sin\phi_2 + q\cos\phi_2)e^{-\tau/T2}]I_y \quad (F3)$$

Immediately following the third pulse, a pulse of angle ϕ_3 about the x axis, the density matrix will again have a I_y and an I_z term. At this point we may discard the I_z term, since the final pulse has been applied; the I_z term corresponds to static magnetization which is not observed in NMR measurements. We need only write the I_y term:

$$[\sin\phi_3[1 + \{(p\cos\phi_2 - q\sin\phi_2)I_z - 1\}e^{-\tau/T1}] + \cos\phi_3(p\sin\phi_2 + q\cos\phi_2)e^{-\tau/T2}]I_y \quad (F4)$$

We now wait a time t following the final pulse to observe the signal. (To observe the spin echo one waits a time τ). The density matrix at the time t will be given by:

$$\{[\sin\phi_3[1 + \{(p\cos\phi_2 - q\sin\phi_2)I_z - 1\}e^{-\tau/T1}] + \cos\phi_3(p\sin\phi_2 + q\cos\phi_2)e^{-\tau/T2}]e^{-t/T2}\}I_y \quad (F5)$$

The NMR signal is in fact a measurement of the expectation value of I_y, which is given by $Tr[I_y\rho]$, where ρ is the density matrix. The signal, then, will be proportional to the coefficient of I_y in the density matrix. Thus, we have:

$$\text{Signal } \alpha \{[\sin\phi_3[1 + \{(p\cos\phi_2 - q\sin\phi_2)I_z - 1\}e^{-\tau/T1}] + \cos\phi_3(p\sin\phi_2 + q\cos\phi_2)e^{-\tau/T2}]e^{-t/T2} \quad (F6)$$

In the alternating phase inversion recovery sequence one alternates the sign of the angle ϕ_1. We note from equation (F1) that q is an odd function of this angle, and p is an even function. As we average over plus and minus angles ϕ_1 we obtain:

$$\text{Signal (average) } \alpha \{\sin\phi_3(1+(p\cos\phi_2 - 1)e^{-\tau/T1}+p\cos\phi_3\sin\phi_2e^{-\tau/T2}\}e^{-t/T2} \quad (F7)$$

This, then, is the signal obtained. In an inversion recovery experiment the time T is the variable parameter. One measures the above signal (Eq. [F6]) using an "infinite" time T_∞ very long compared to T_1, and then measures the signal for a shorter time T and subtracts the result. We note that for $T=\infty$ we have p=1. The result of the subtraction is then:

$$(\text{Signal}(T = \infty) - \text{Signal}(T)) = (p - 1)\{\sin\phi_3\cos\phi_2e^{-\tau/T1} + \cos\phi_3\sin\phi_2e^{-\tau/T2}\}e^{-t/T2} \quad (F8)$$

Holding ϕ_2, ϕ_3, τ, and t constant we see that the above quantity is proportional to p-1. But p-1 is given by:

$$p - 1 = (\cos\phi_1 - 1)e^{-T/T1} \quad (F9)$$

In the pulse sequence one alternates the sign of ϕ_1 but does not vary the magnitude. Since the cosine is an even function we see that the quantity measured, (Signal($T = \infty$) - Signal(T)), is proportional to $e^{-T/T1}$, as desired.

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He began graduate study in physics at the University of Illinois at Urbana-Champaign in June, 1985. During that summer he received a University Fellowship and was a Research Assistant working with Prof. Peter Debrunner. He held a Teaching Assistantship during the academic year of 1985. Subsequently he became a Research Assistant in the laboratory of Prof. Charles P. Slichter.